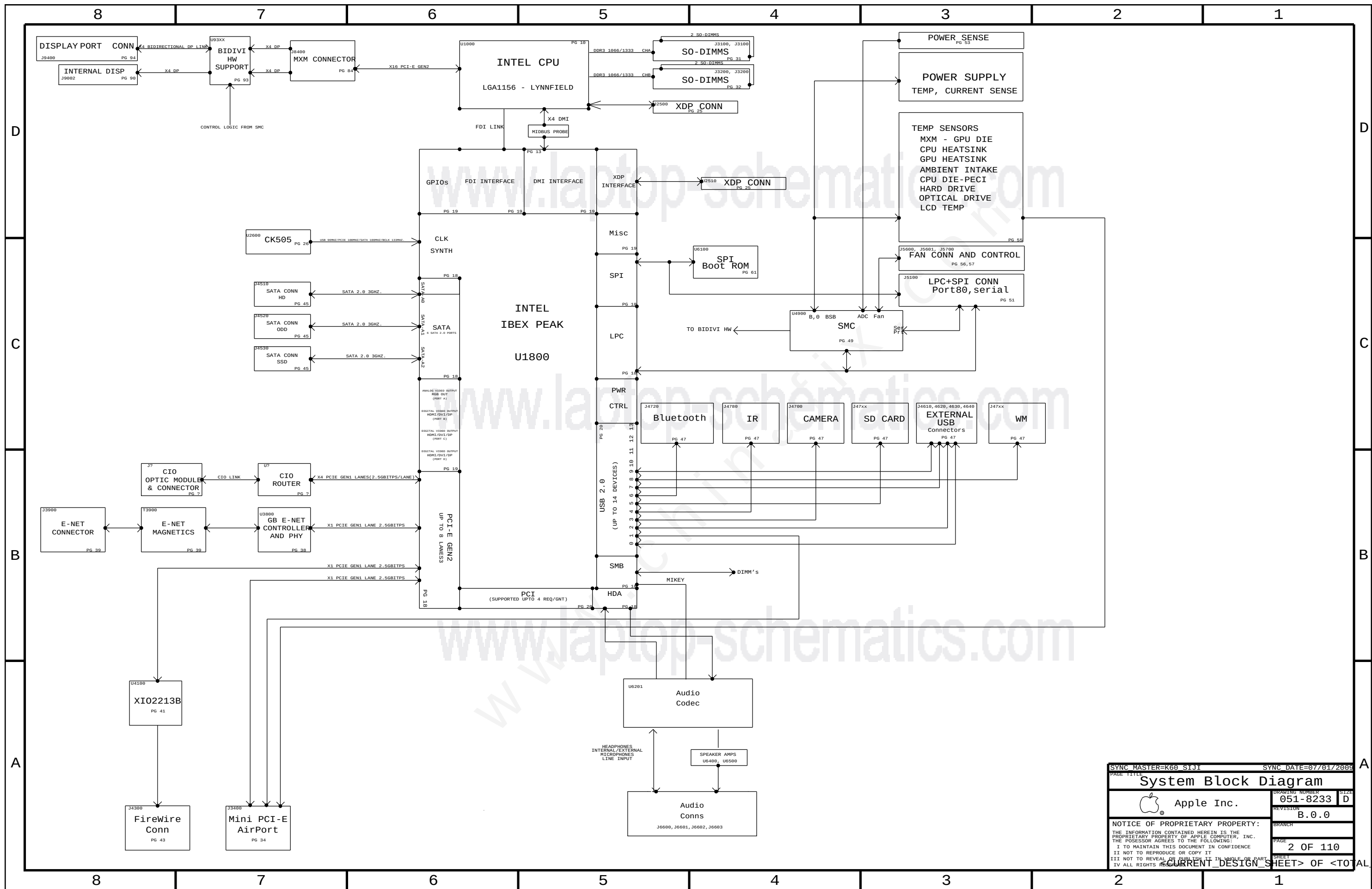
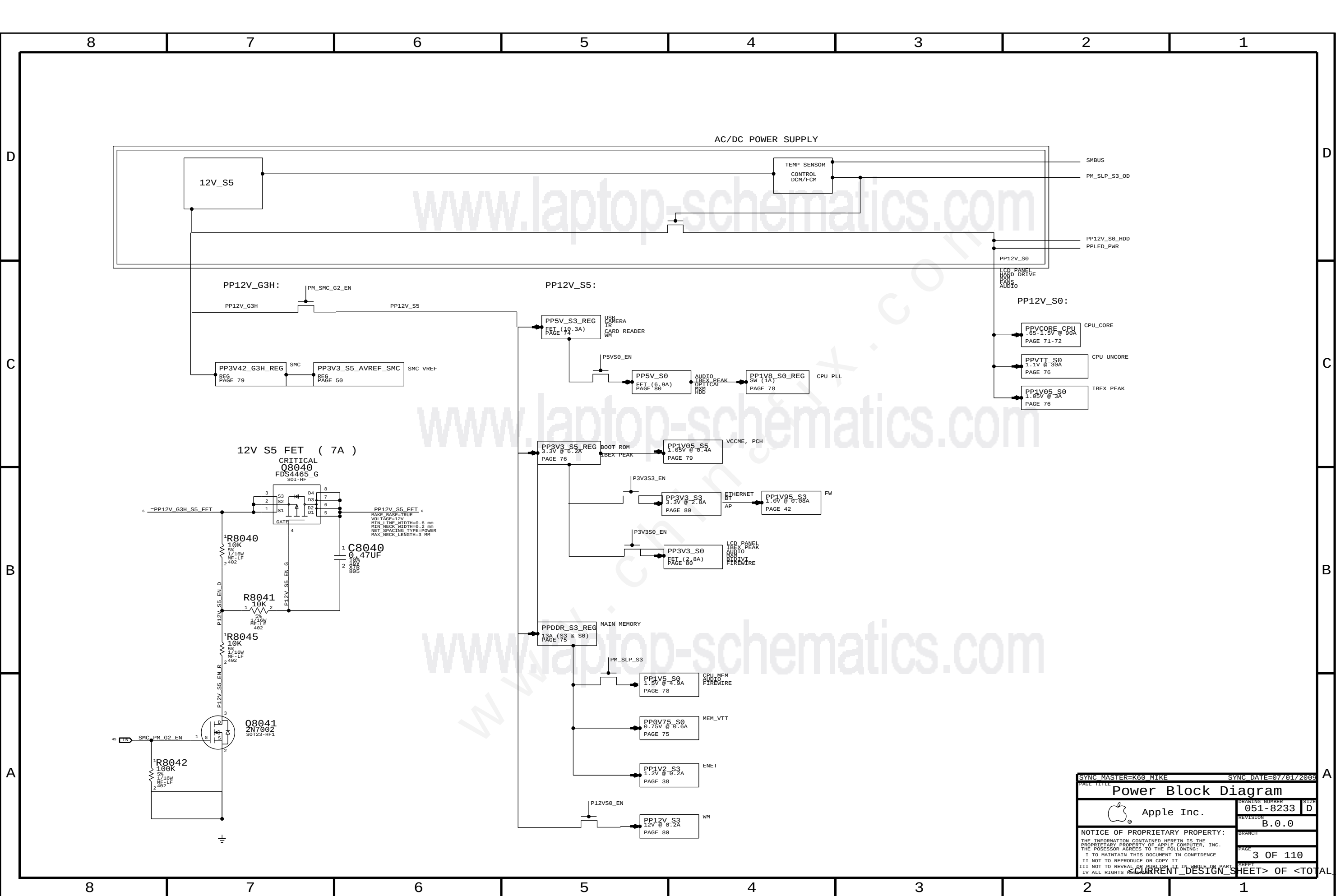


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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.												REV	ECN	DESCRIPTION OF REVISION	CK APPD
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.												B	0000803841	PRODUCTION RELEASED	2009-10-09
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.															
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-0439	PCBA, MLB, K23F, 2.66GHZ	K23F, 2P66GHZ_CPU, BASIC, CPUPOC_IMAX_100_120
639-0440	PCBA, MLB, K23F, 2.80GHZ	K23F, 2P80GHZ_CPU, BASIC, CPUPOC_IMAX_100_120
085-1023	PCBA, MLB, DEV, K23F	DEVELOPMENT, DEV_GROUP

BOM GROUPS

BOM_GROUP	BOM_OPTIONS
BASIC	COMMON, ALTERNATE, XDP, BETTER, MXM, XDP_CPU_BPM, INT_VREF, PCH_VRM, BUF_CLK, PRODUCTION
DEV_GROUP	XDP_CONN, LPCPLUS, MOJOMUX, CPU_TDIODE, MEM_RESET_HW

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3823	1	IC, IBEX PEAK B3, DESKTOP, FCBGA, PCH, P425	U1800	CRITICAL	
359S0157	1	IC, SLG62AP100, CLK GEN, KCS05, QFN3	U2600	CRITICAL	BUF_CLK
341T0211	1	IC, EFI BOOTROM, K23F	U6100	CRITICAL	
338S0765	1	IC, XI02211ZAY, 1394B_PCIE, PHY/LINK	U4100	CRITICAL	
343S0485	1	IC, BCM5764M, 68PIN QFN	U3700	CRITICAL	
341T0213	1	IC, FLASH, 450B0011D, SOIC-8S1	U3701	CRITICAL	
511S0003	1	SOCKET, LGA1150, CPU- L F	U1000	CRITICAL	
825-7122	1	MLB LABEL, 48.0X4.8	X14	CRITICAL	

CPUS

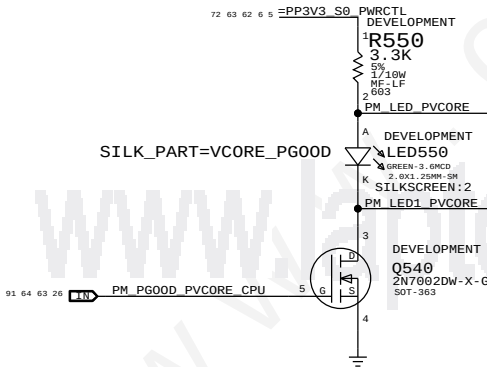
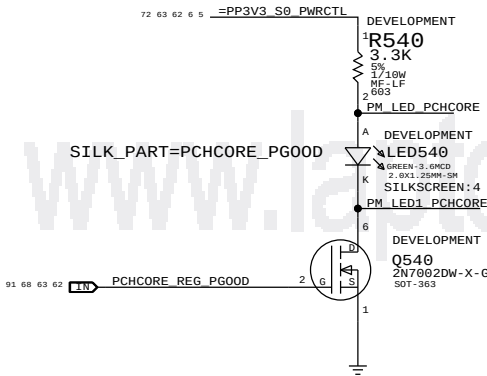
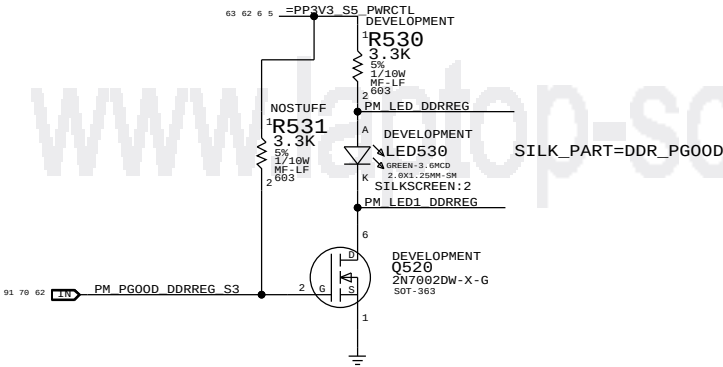
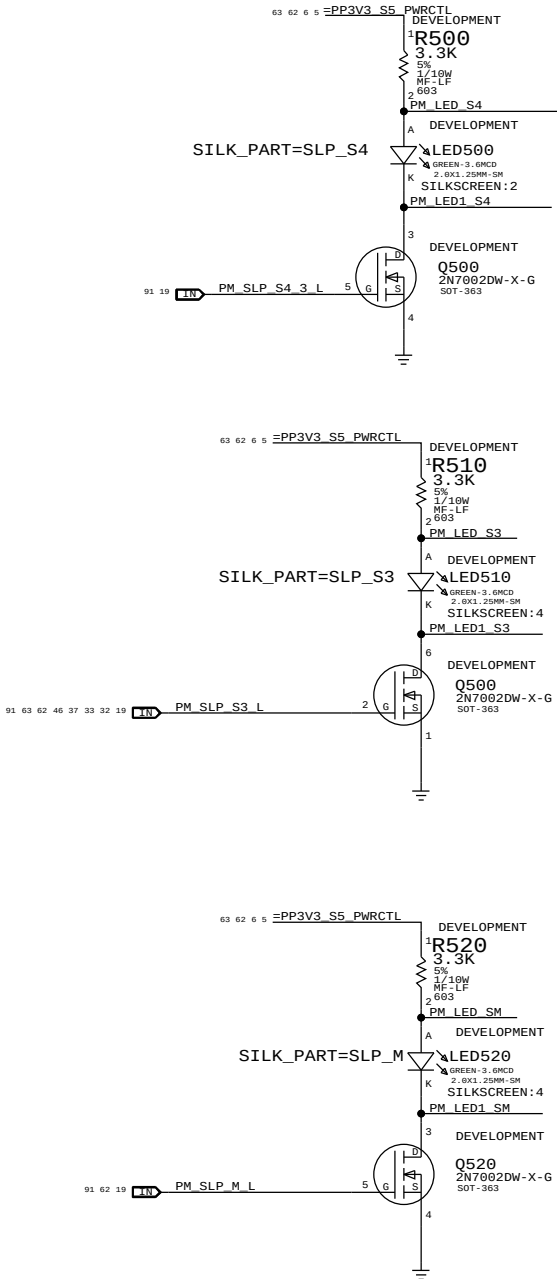
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337S3810	1	LFD, SLBLC, PRQ, 2.66, 95W, 1333, B1, 8M, LGA	CPU	CRITICAL	2P66GHZ_CPU
337S3811	1	LFD, SLBJJ, PRQ, 2.80, 95W, 1333, B1, 8M, LGA	CPU	CRITICAL	2P80GHZ_CPU

K23F PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-8233	1	SCH, K23F, MLB	SCH1		K23F
828-2733	1	PCBF, K23F, MLB	MLB1		K23F
341T0212	1	IC, SMC, K23F	U4900	CRITICAL	K23F

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State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SYNC MASTER=K60 MIKE

SYNC DATE=07/01/2009

PROTO 0 DEBUG LEDS

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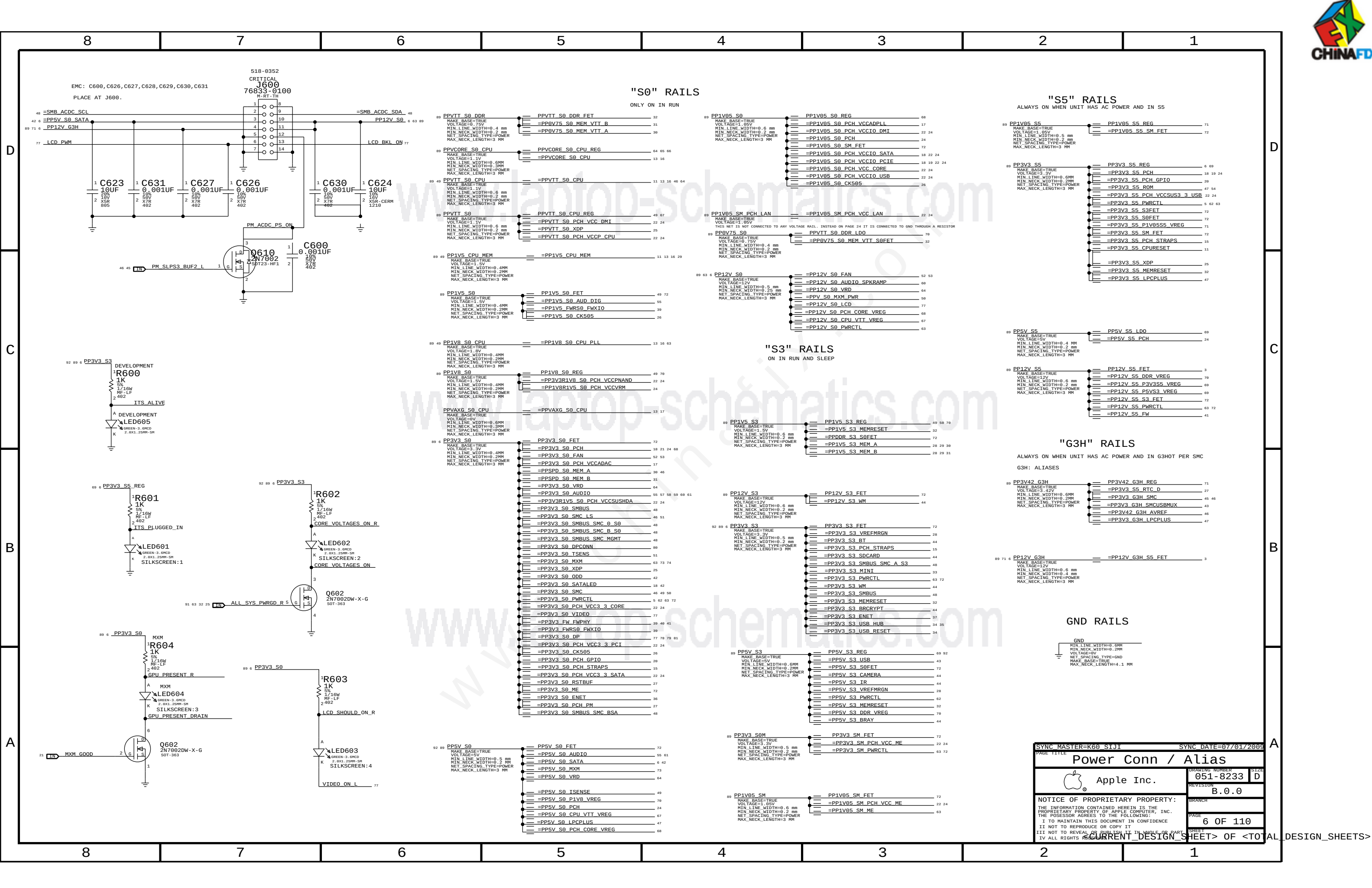
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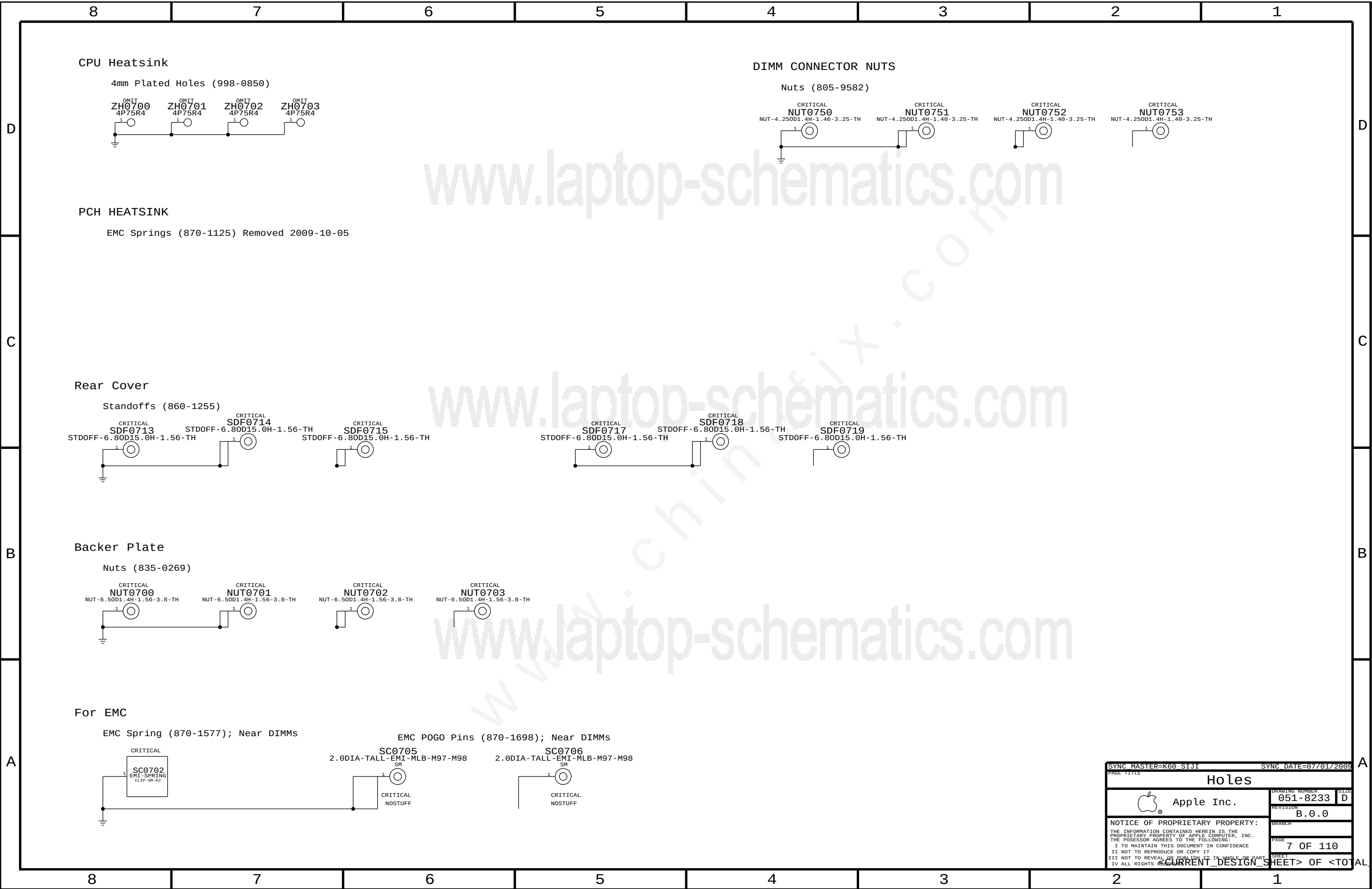
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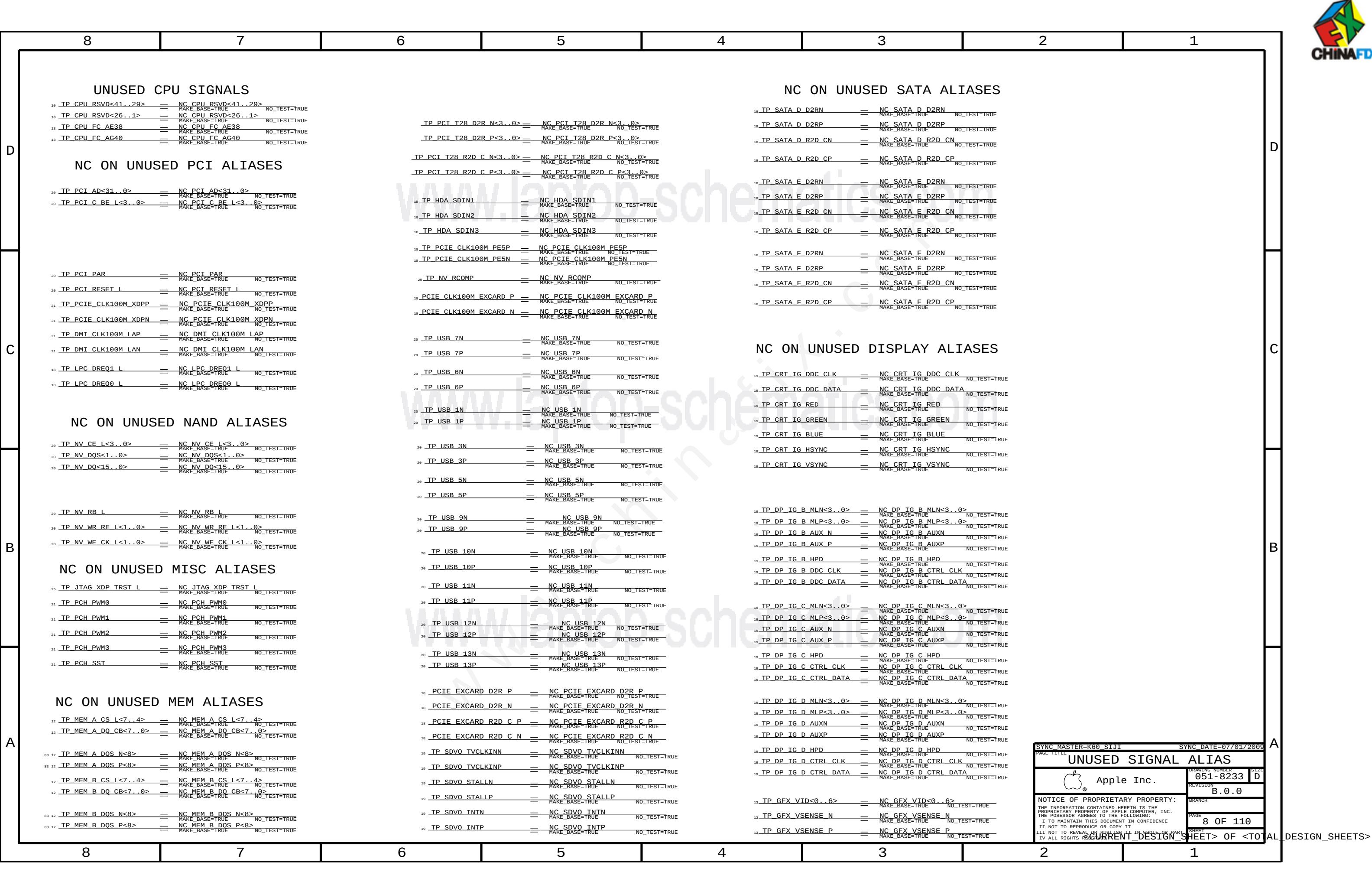
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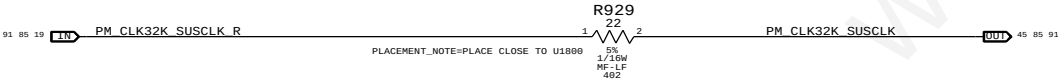
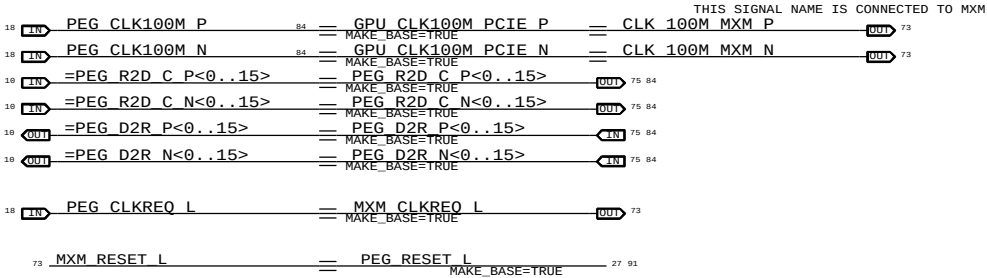
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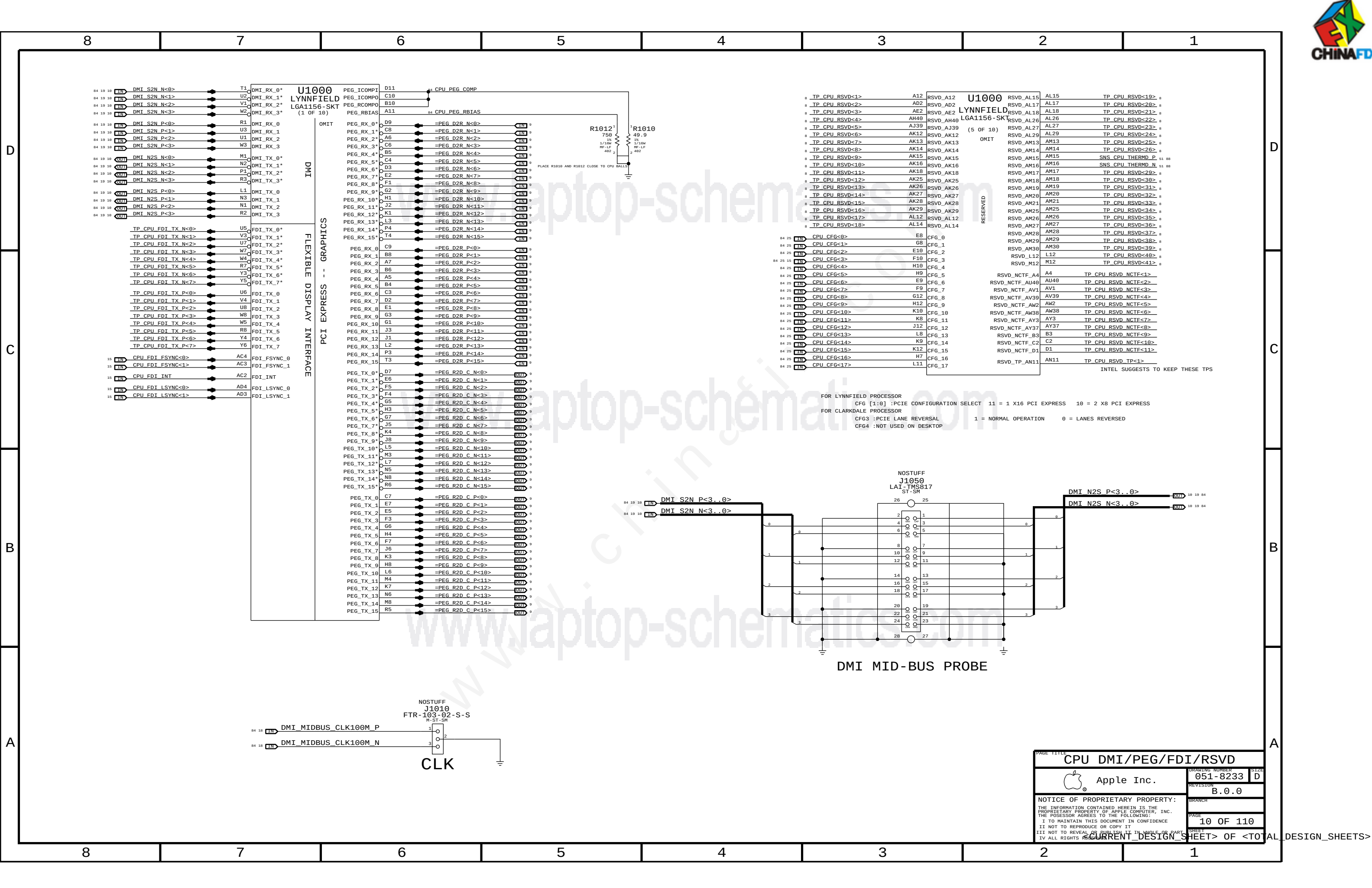
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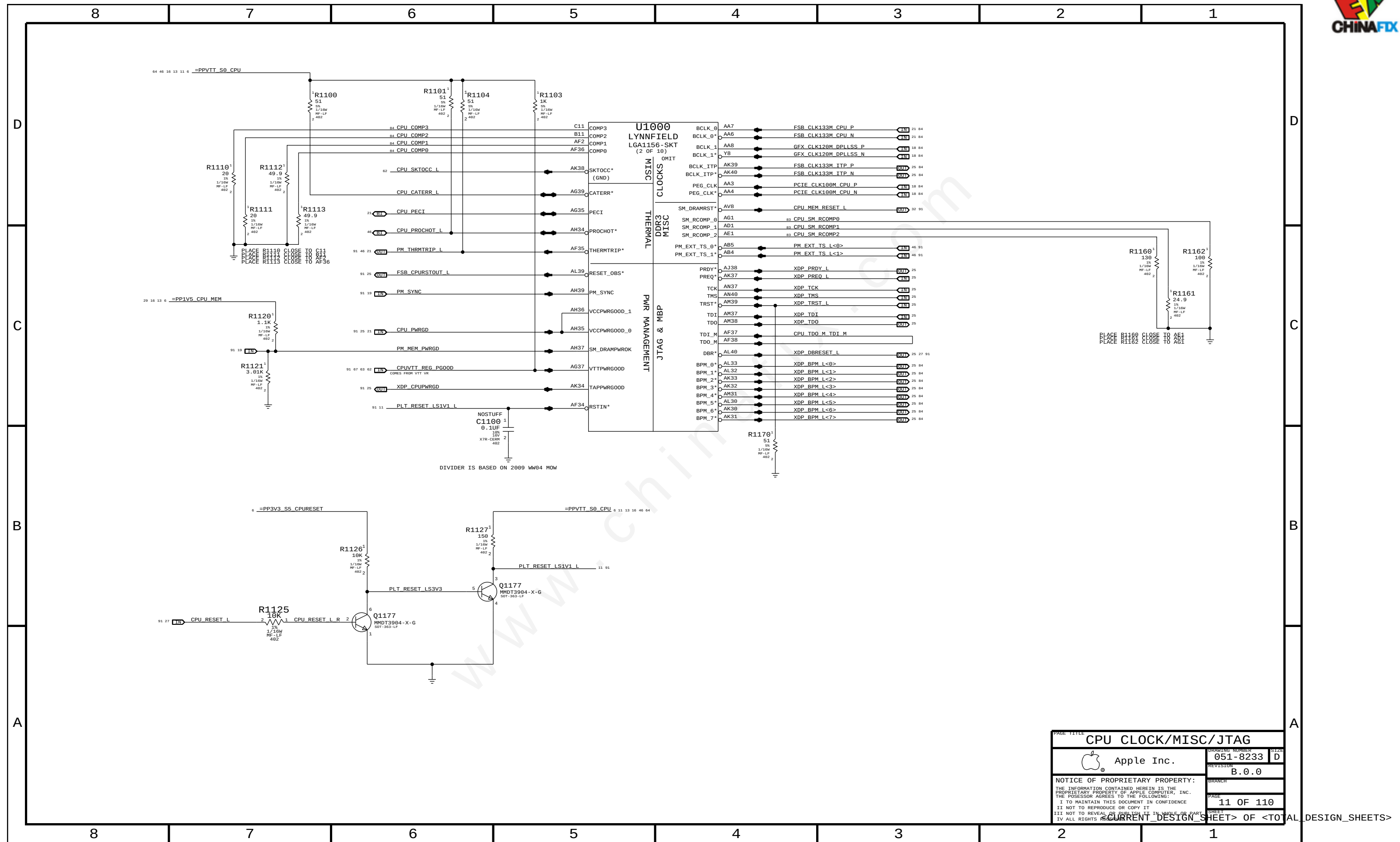
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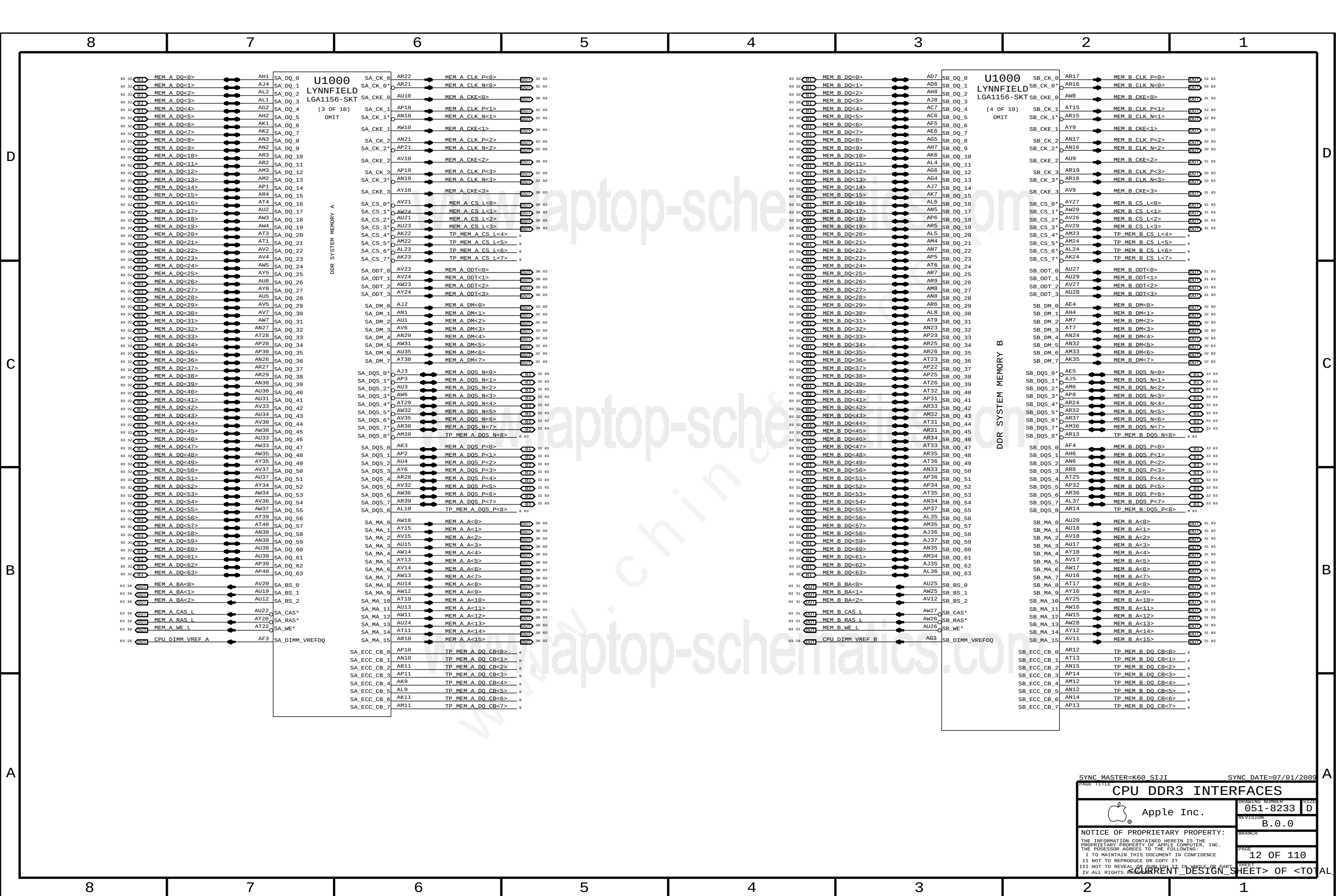
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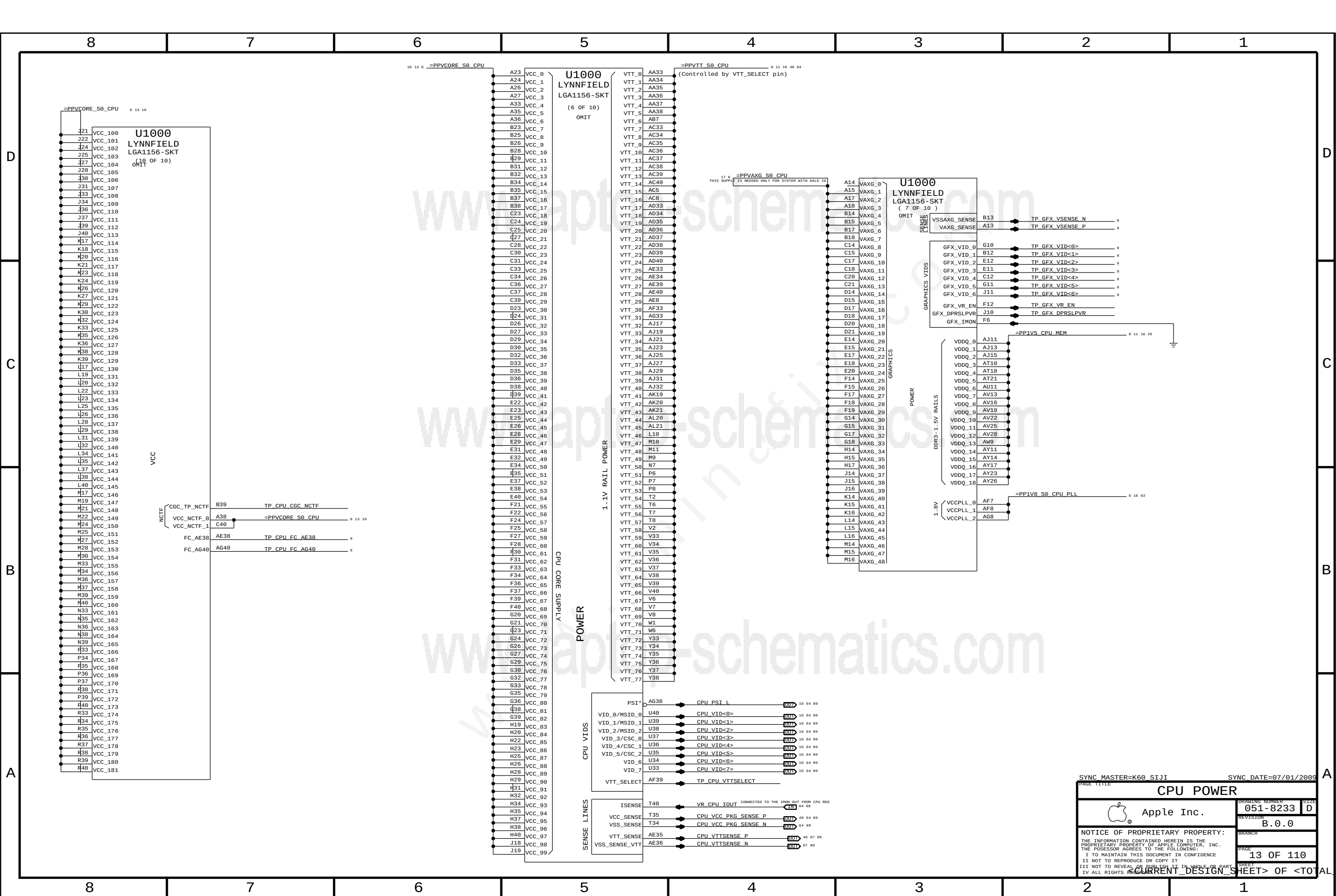


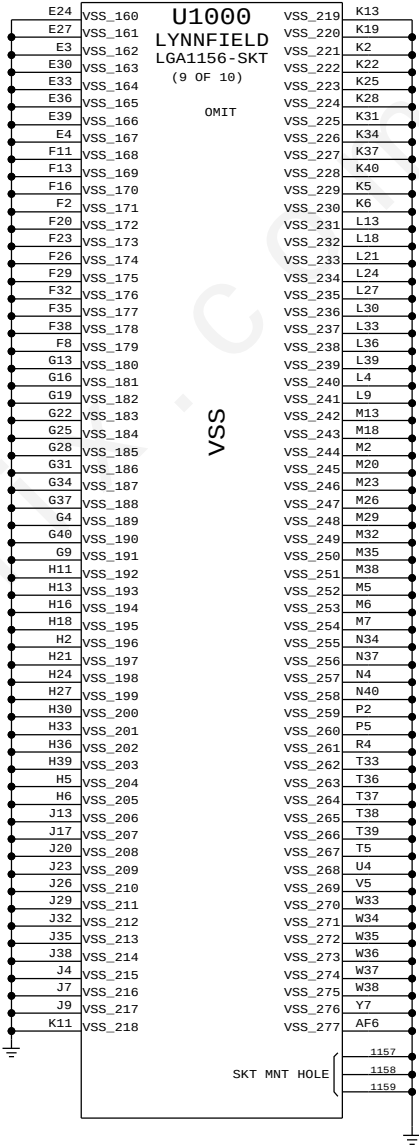
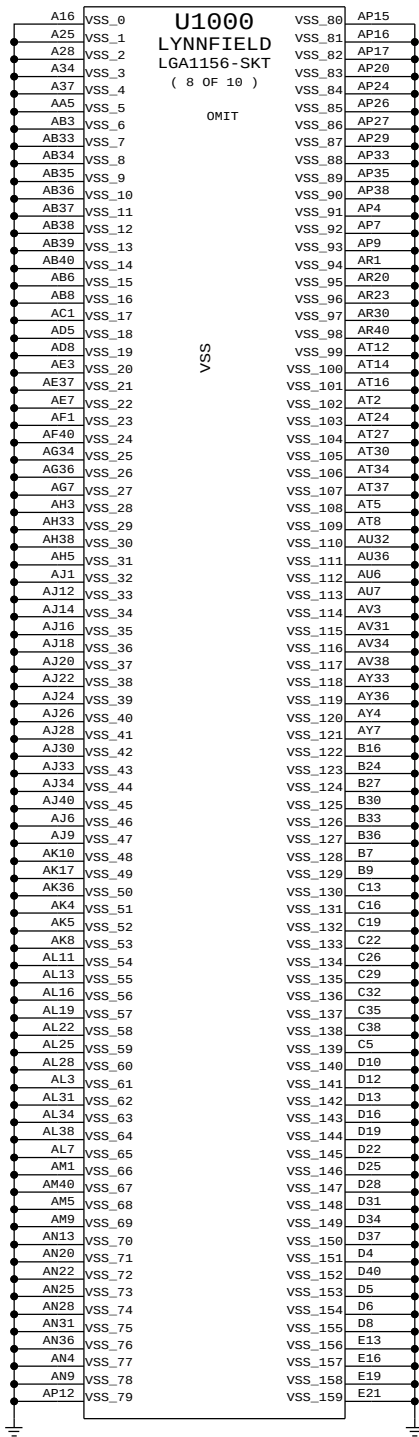
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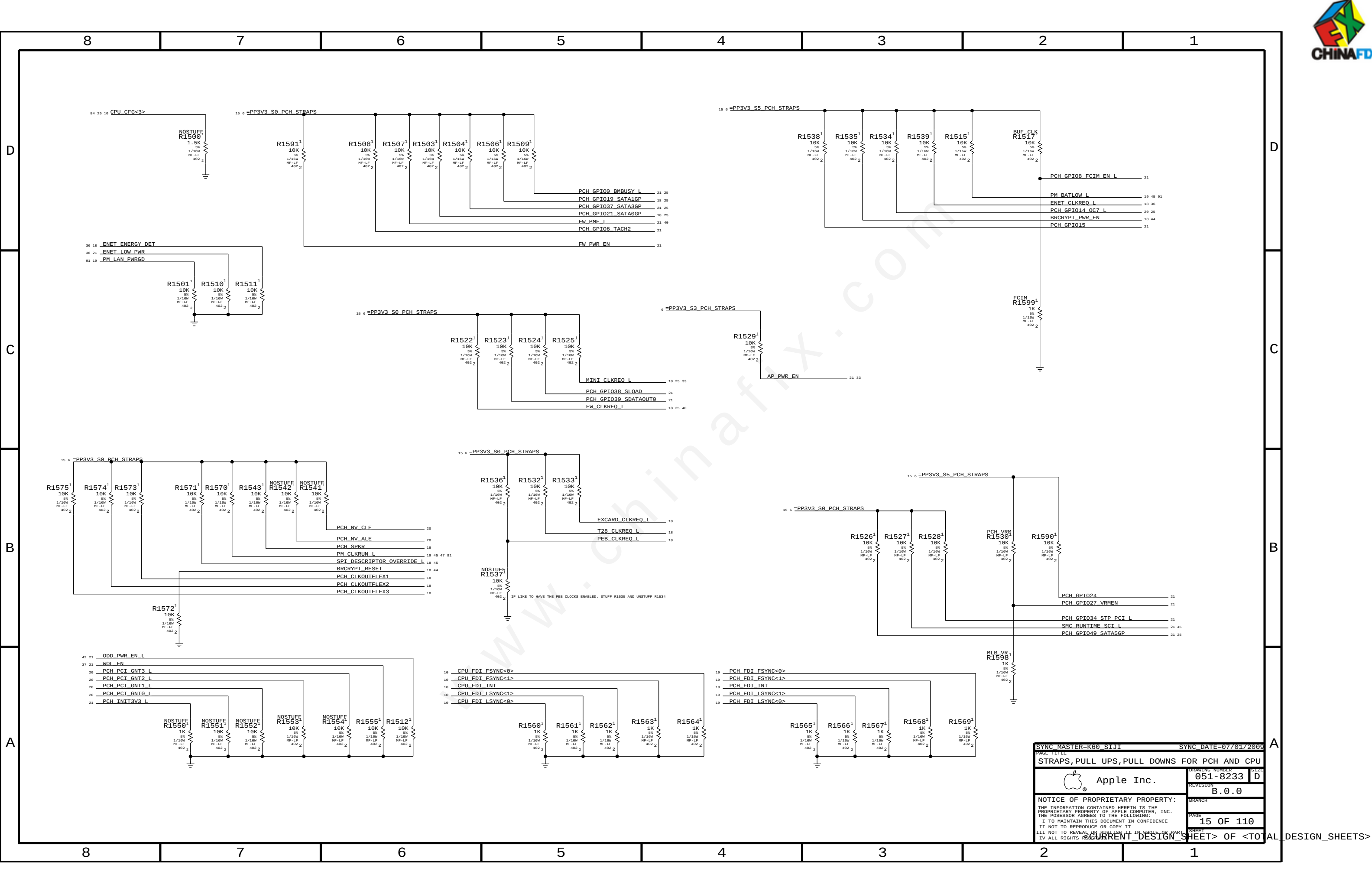


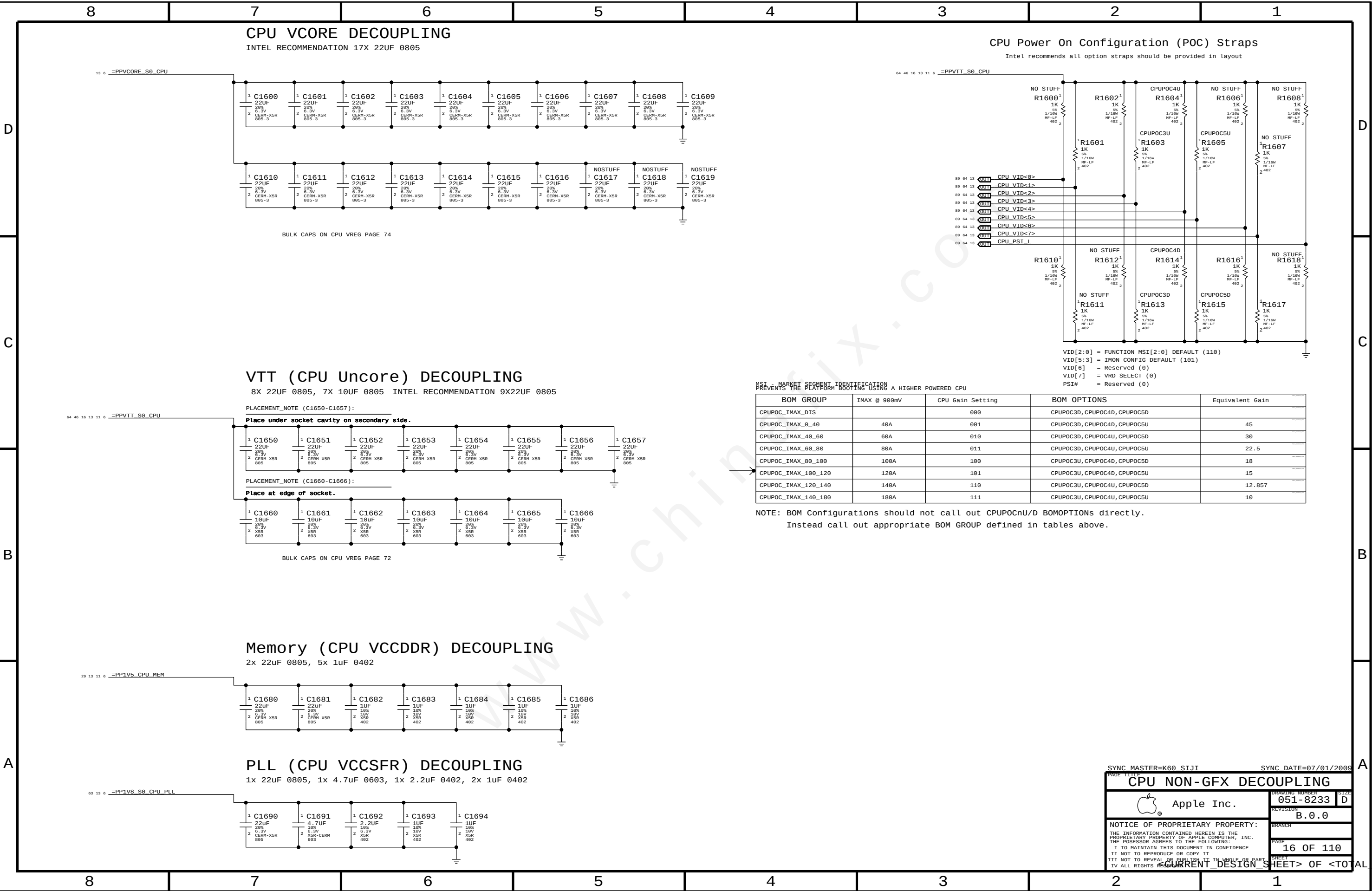


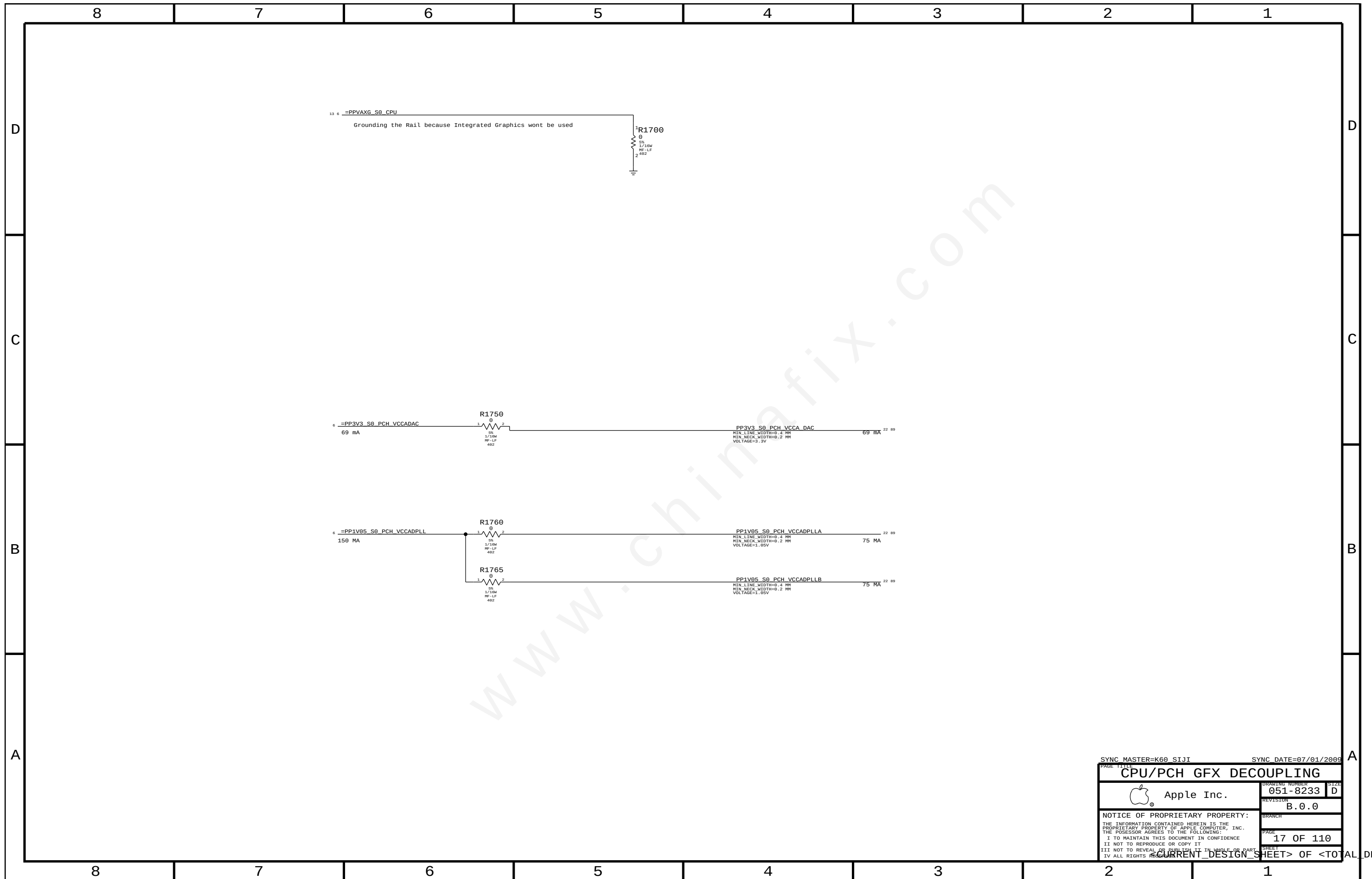


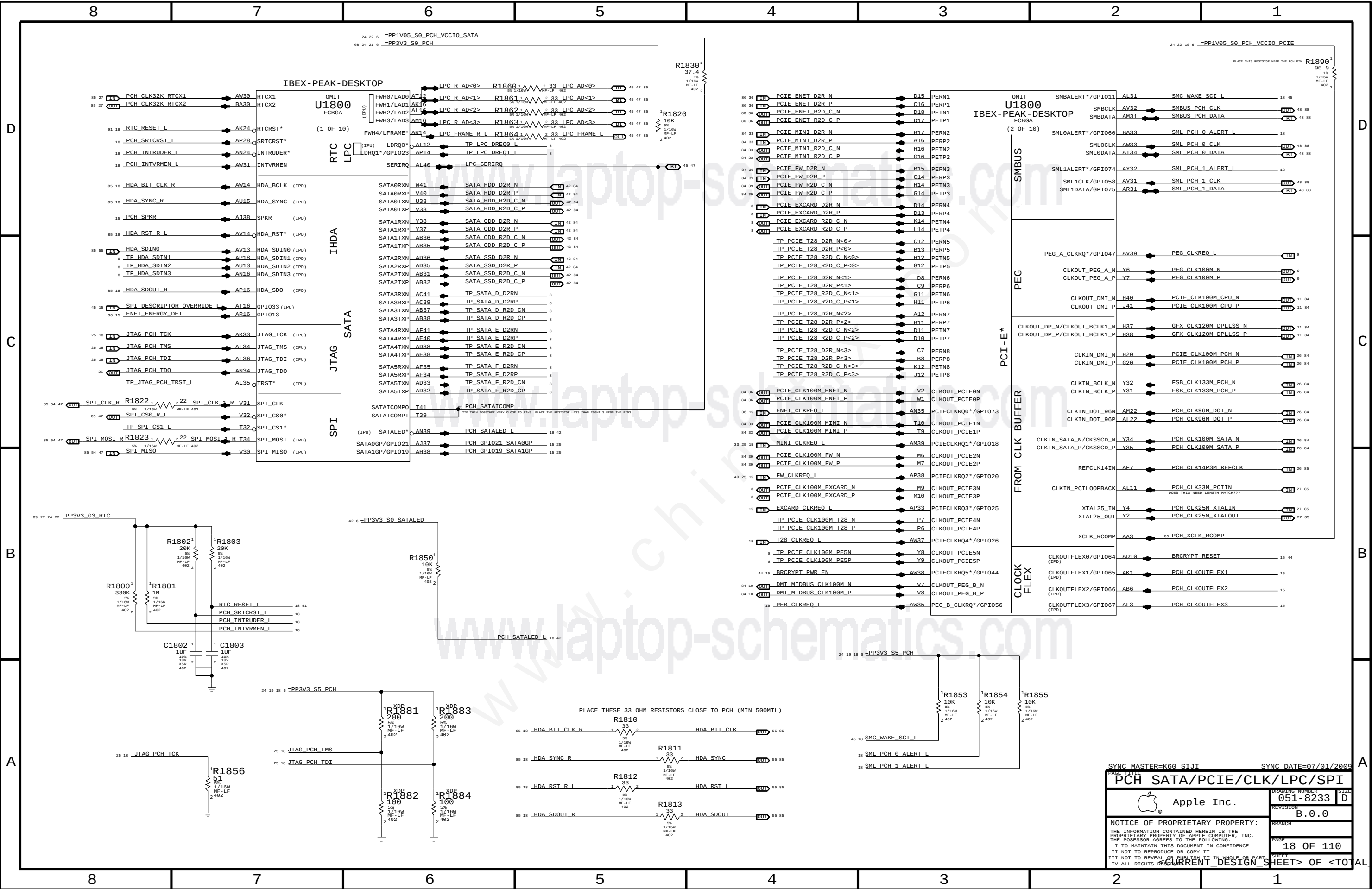


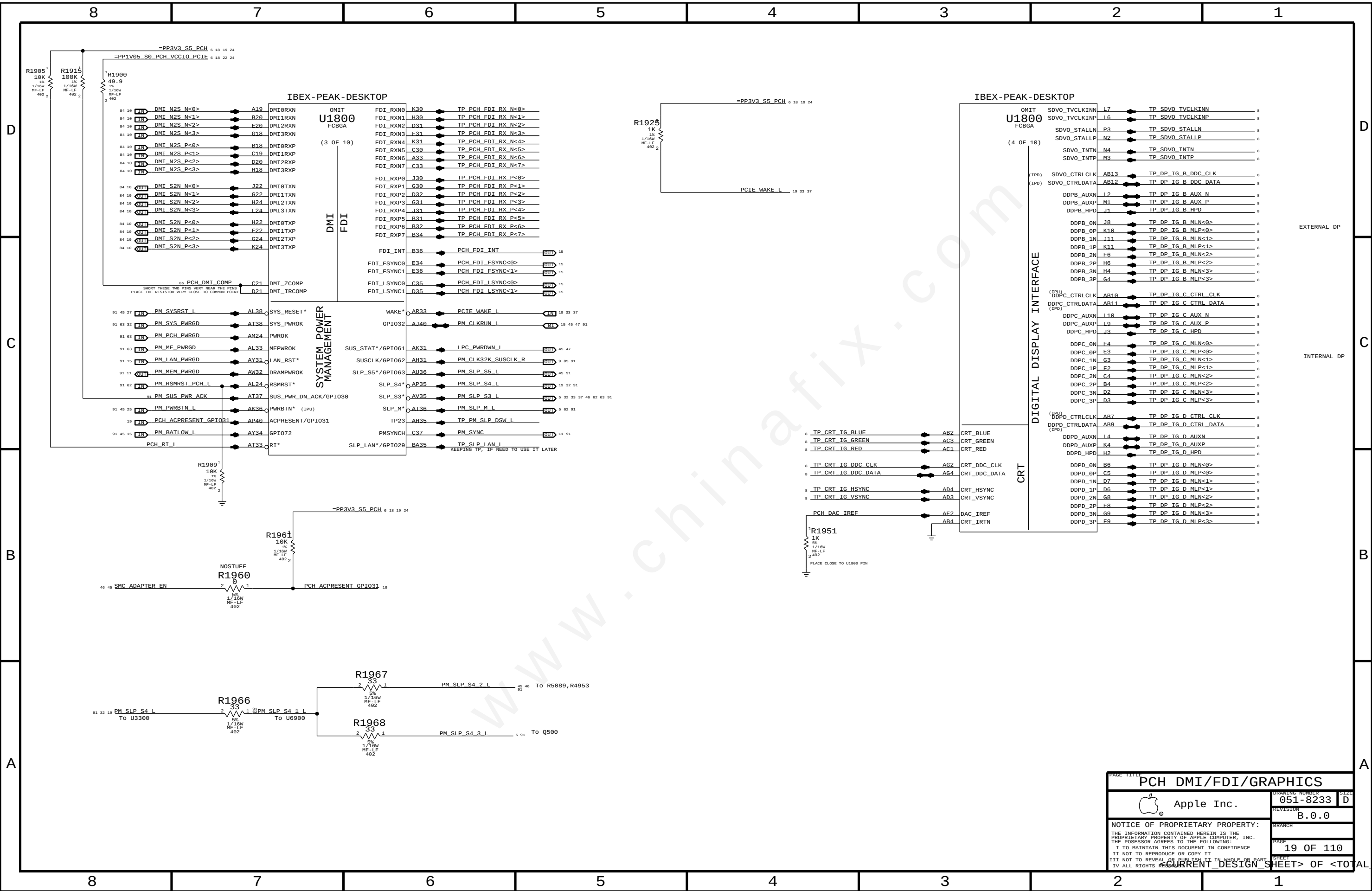


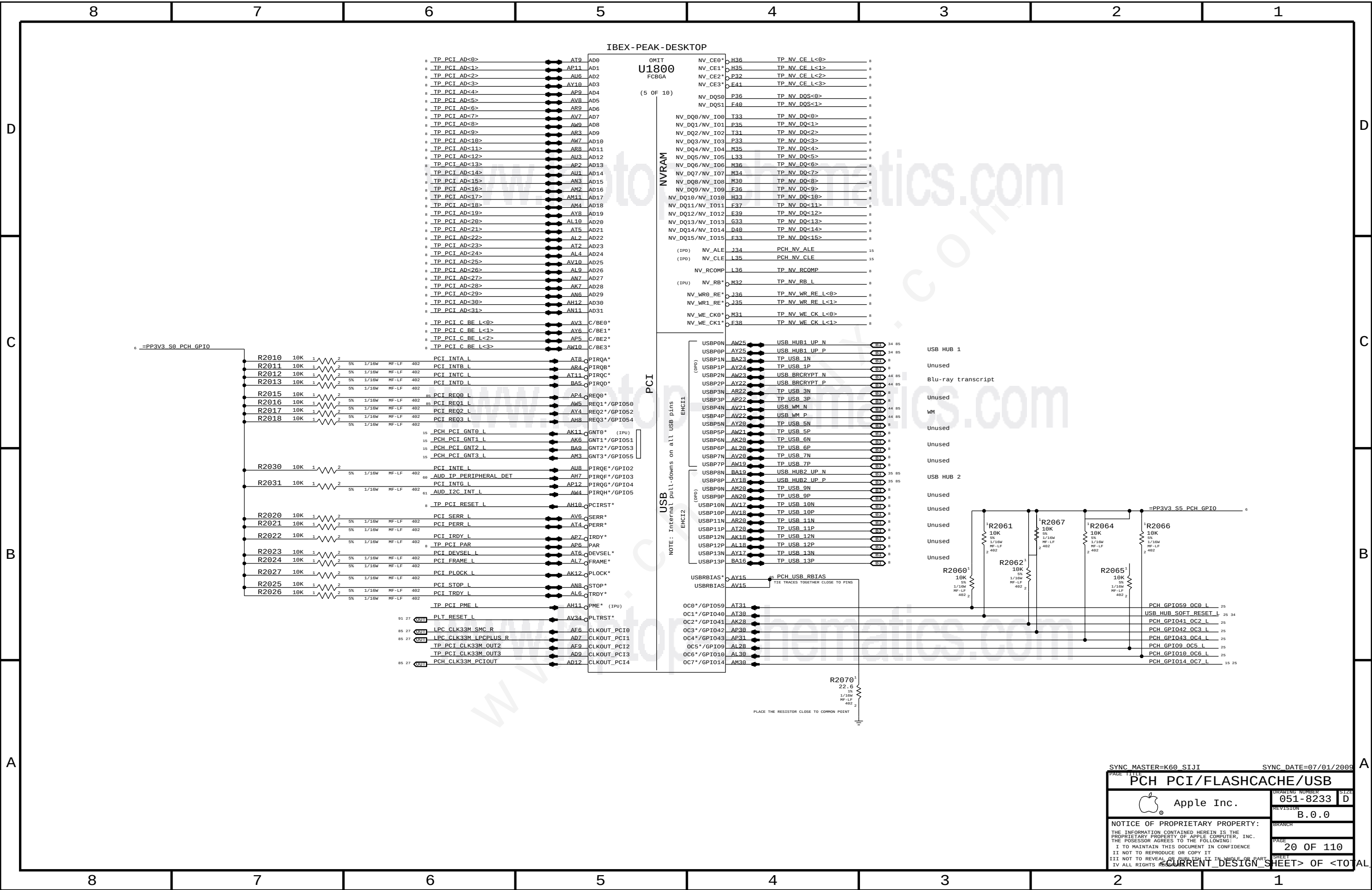


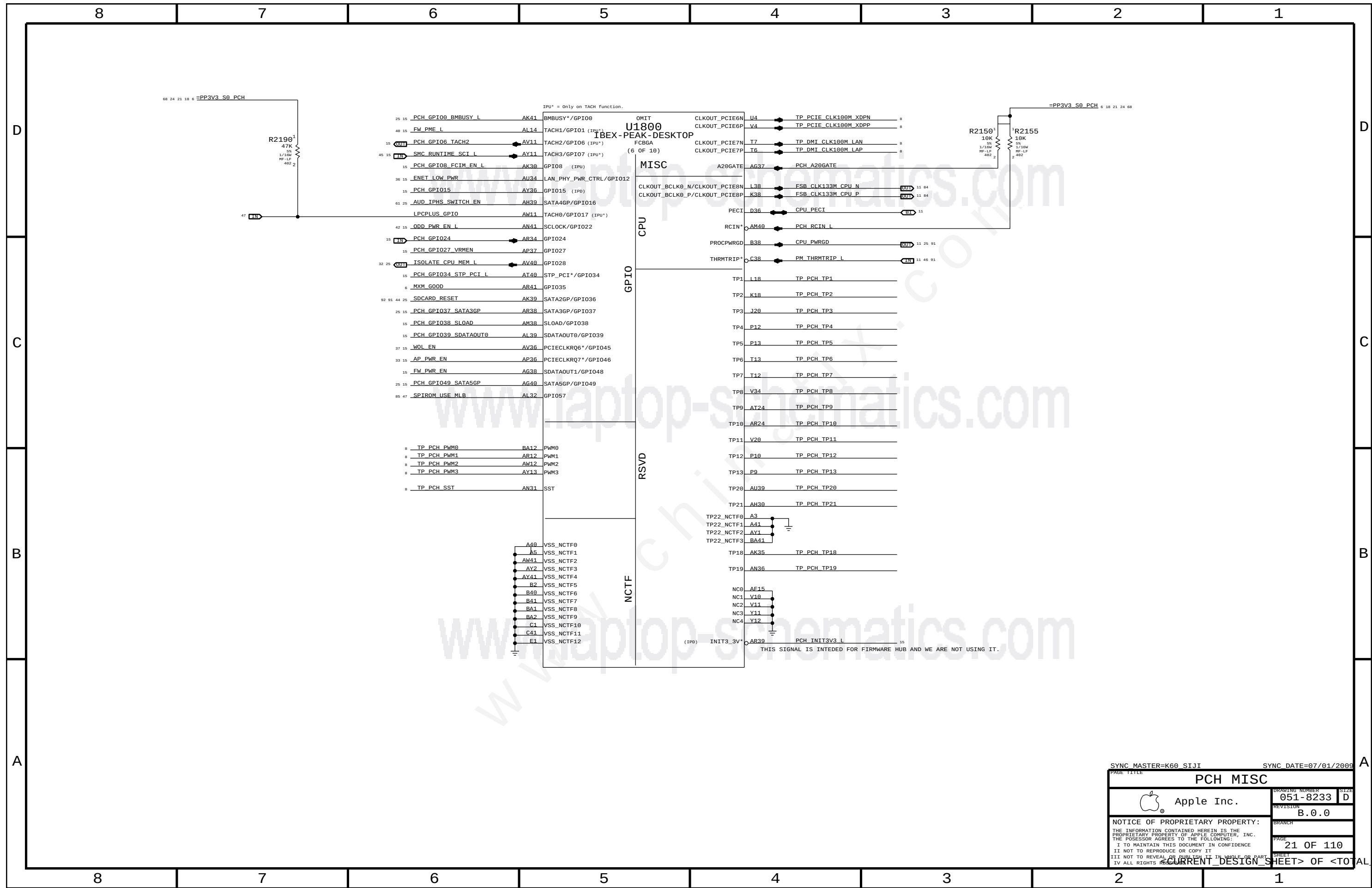


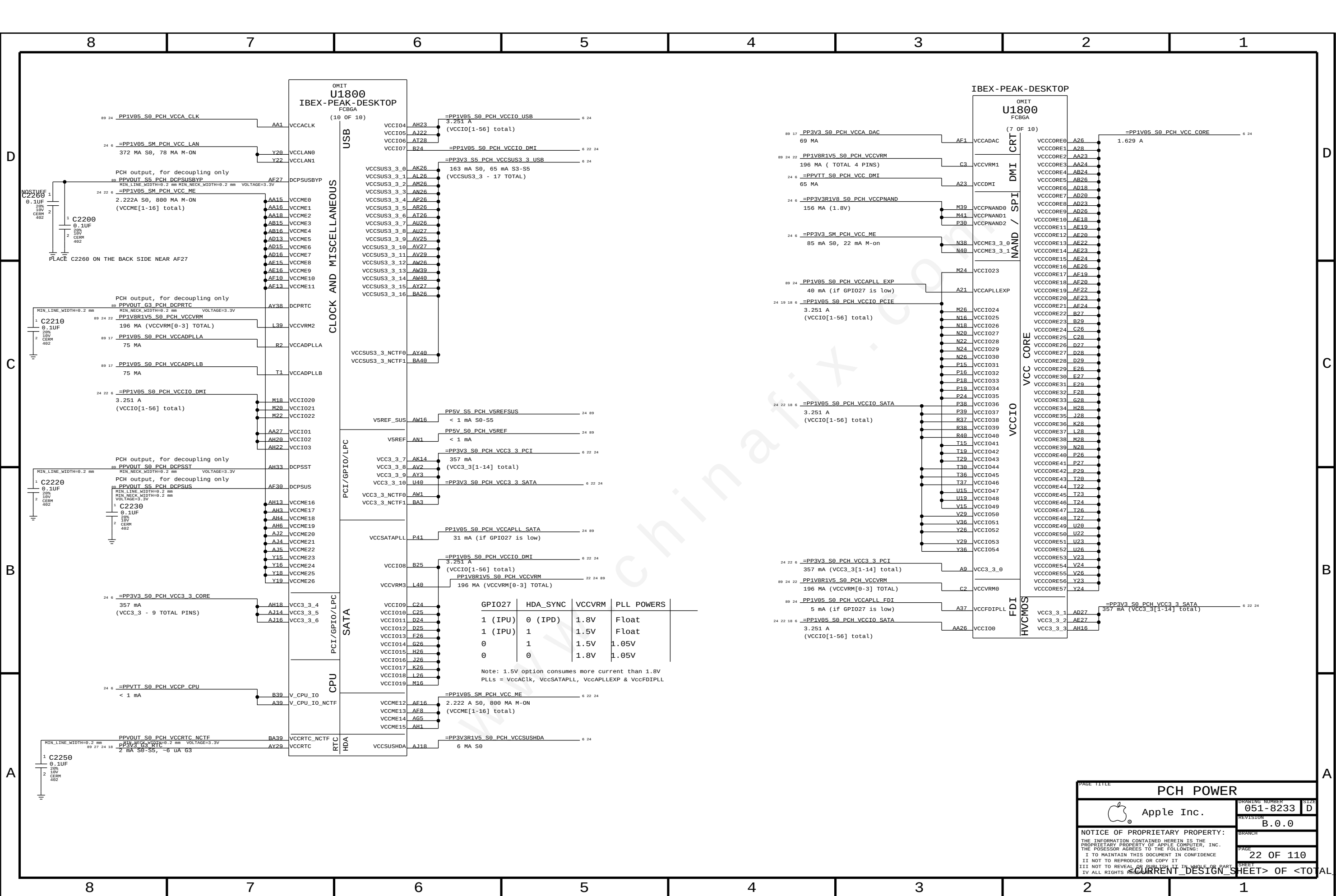


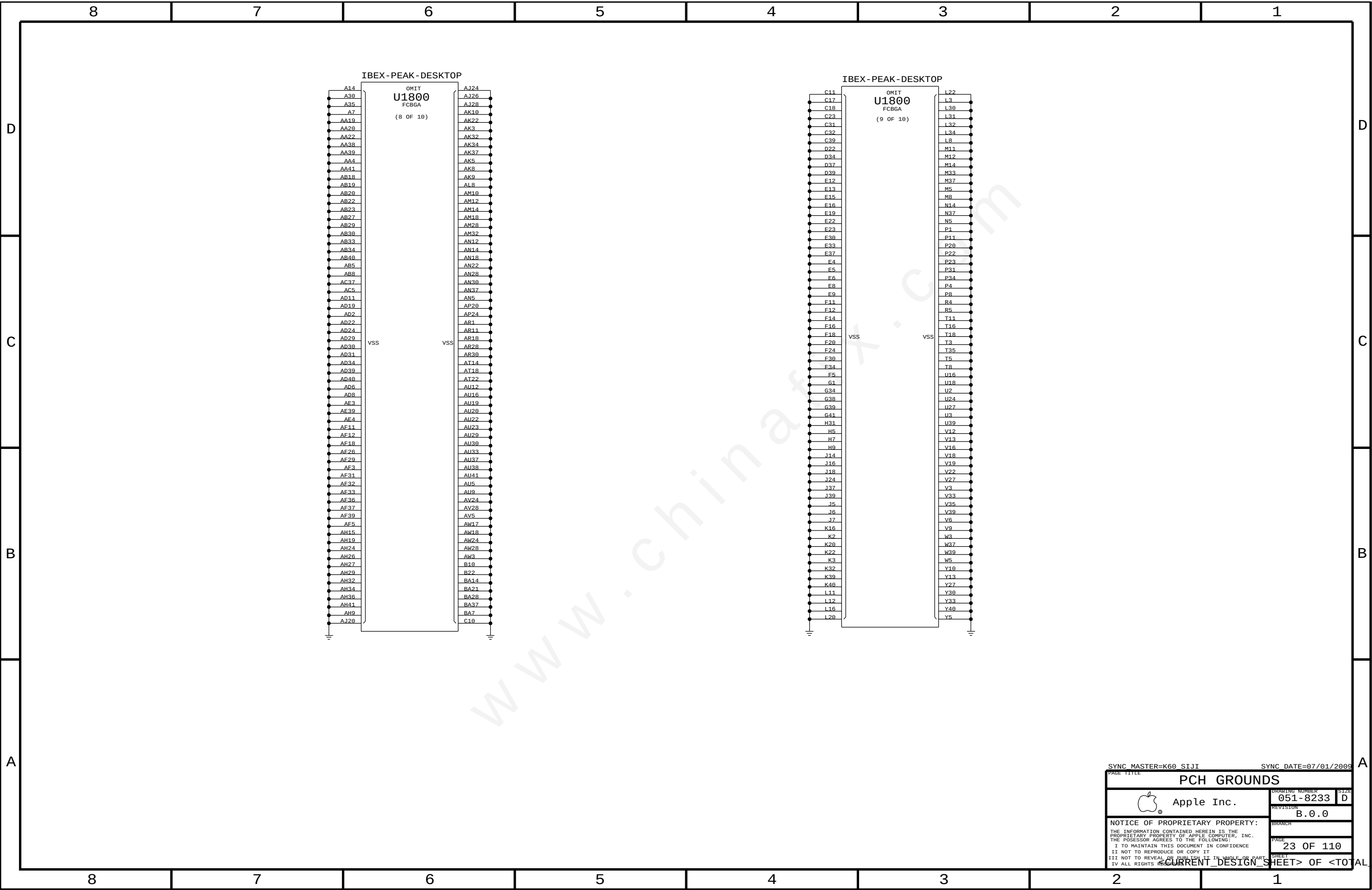








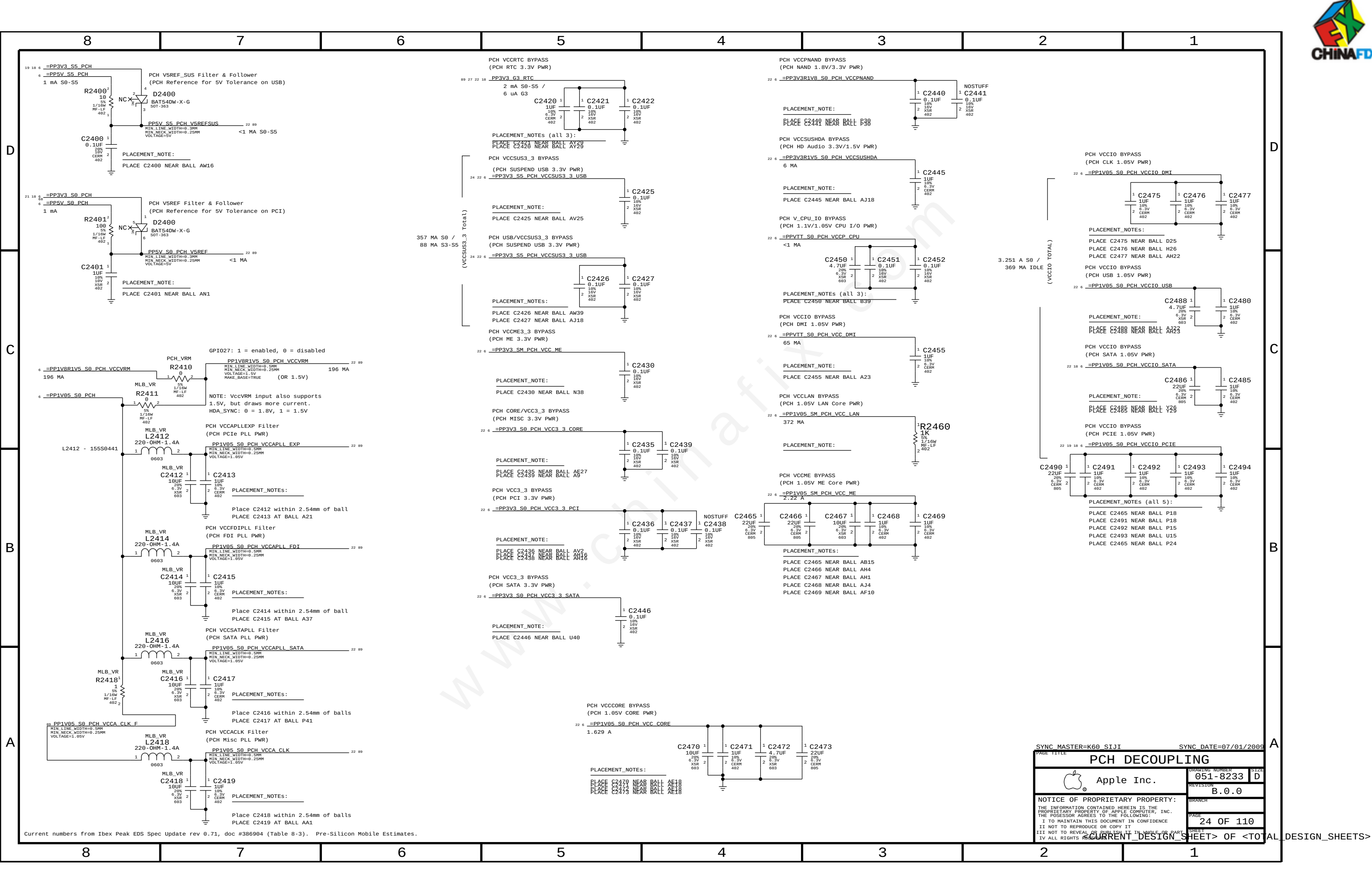


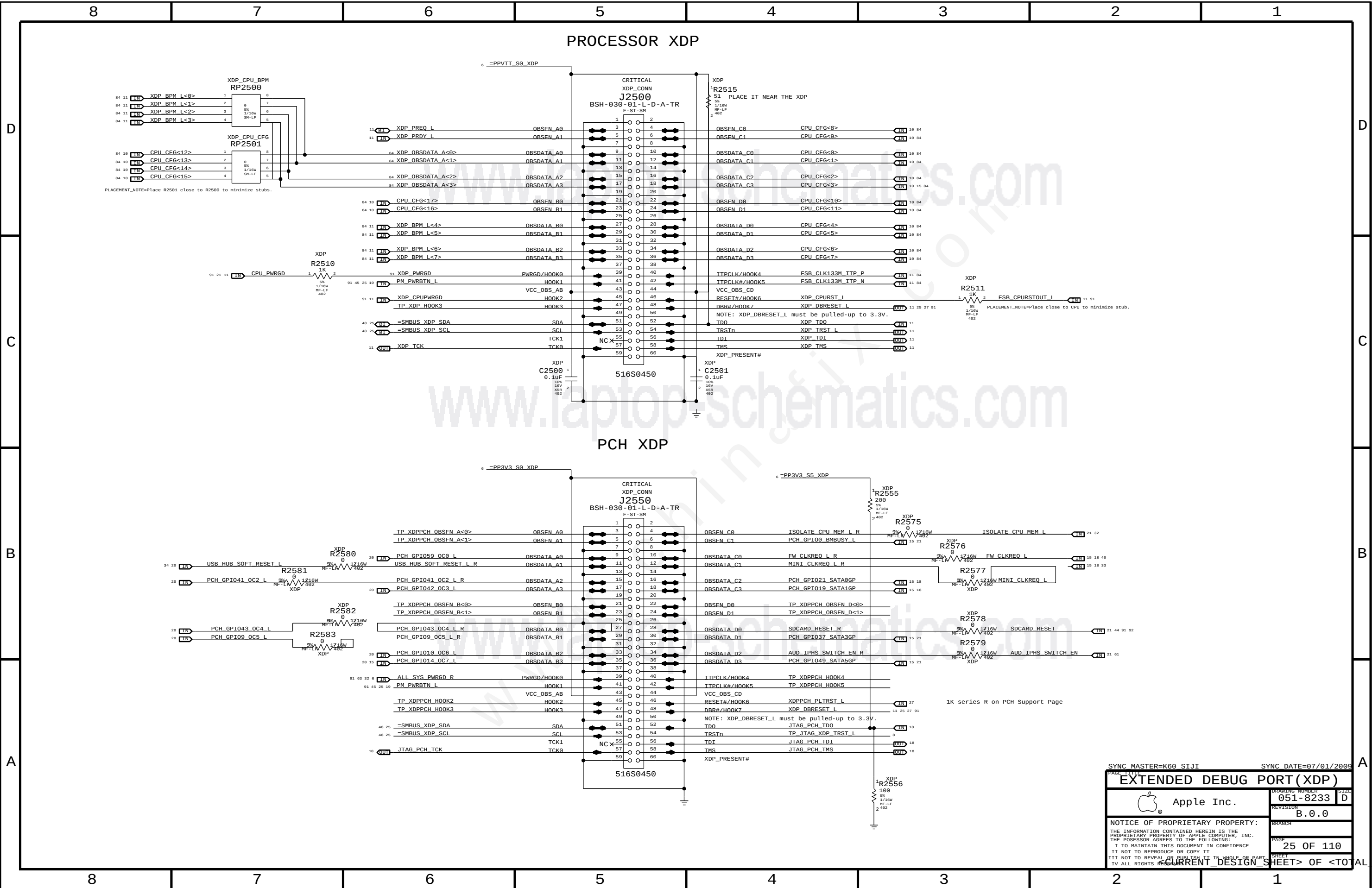


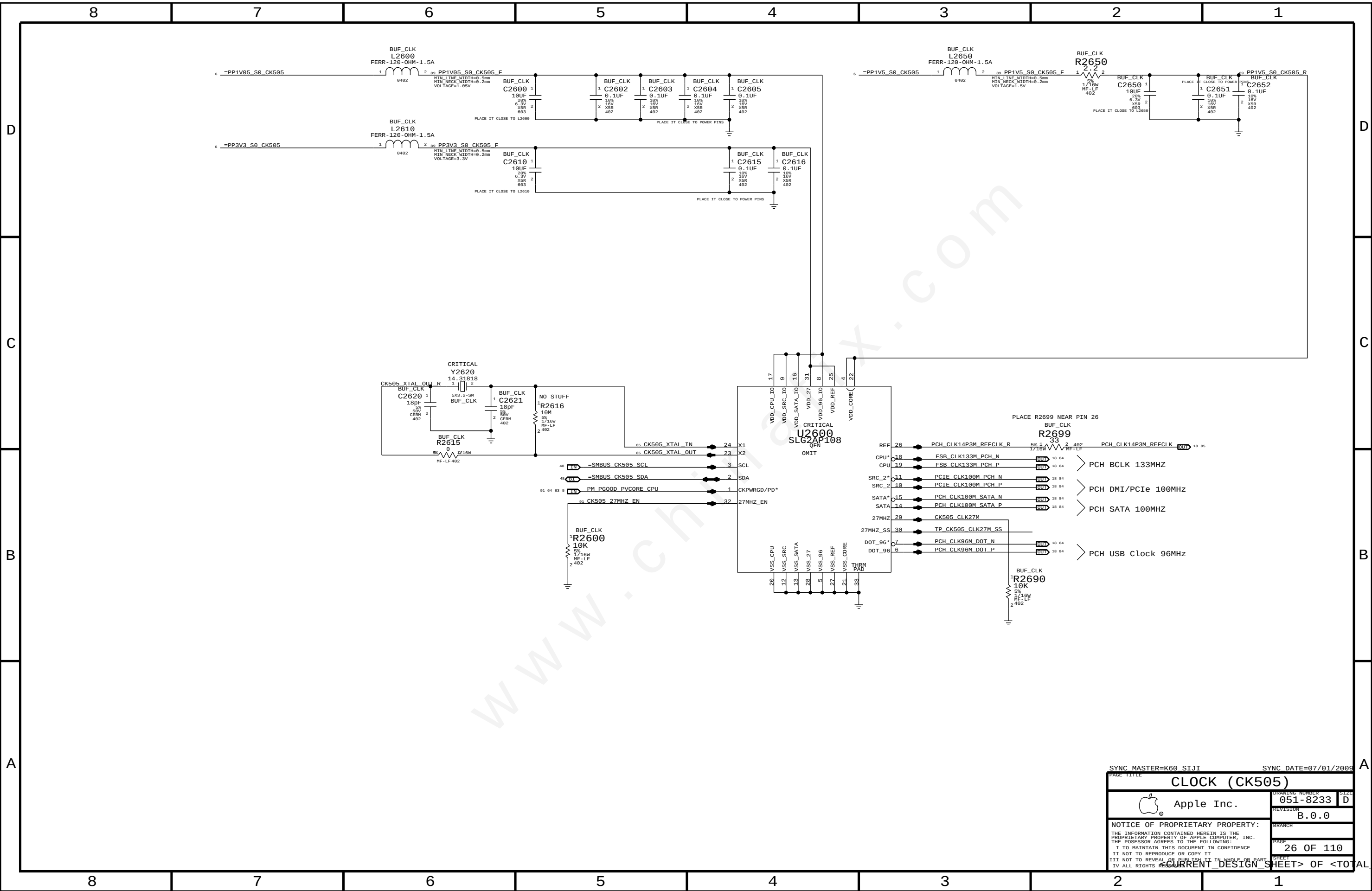
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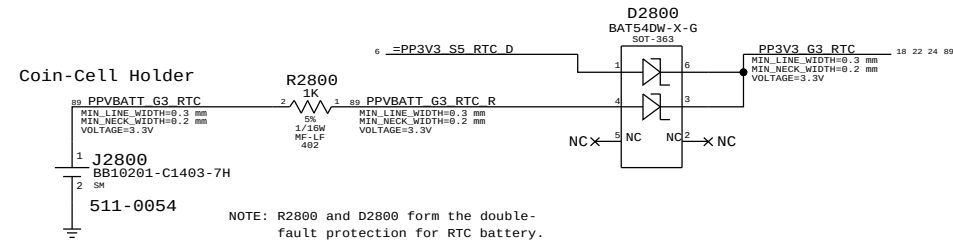
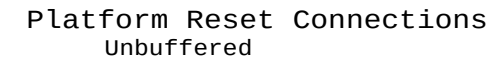
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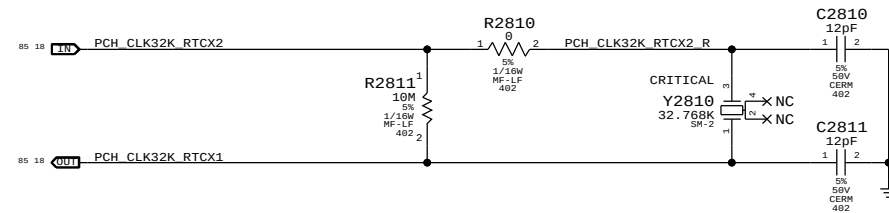




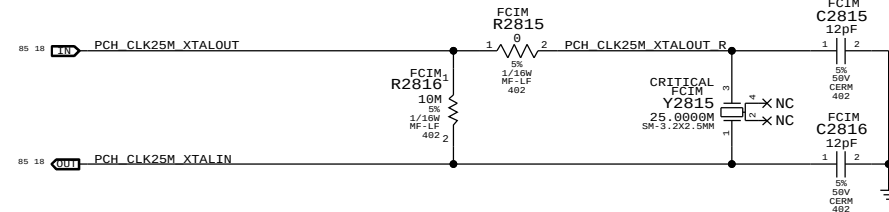




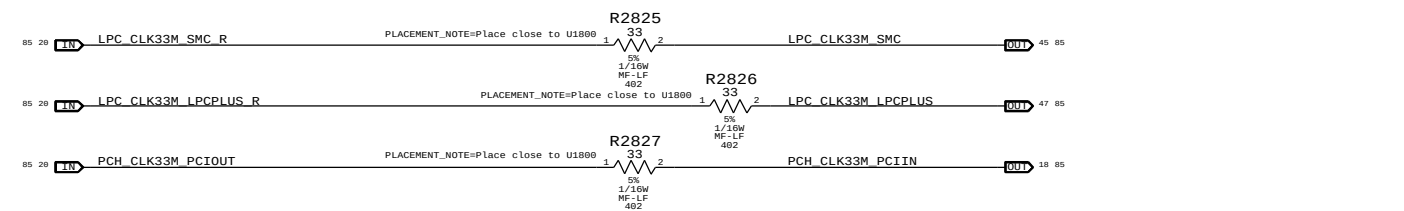
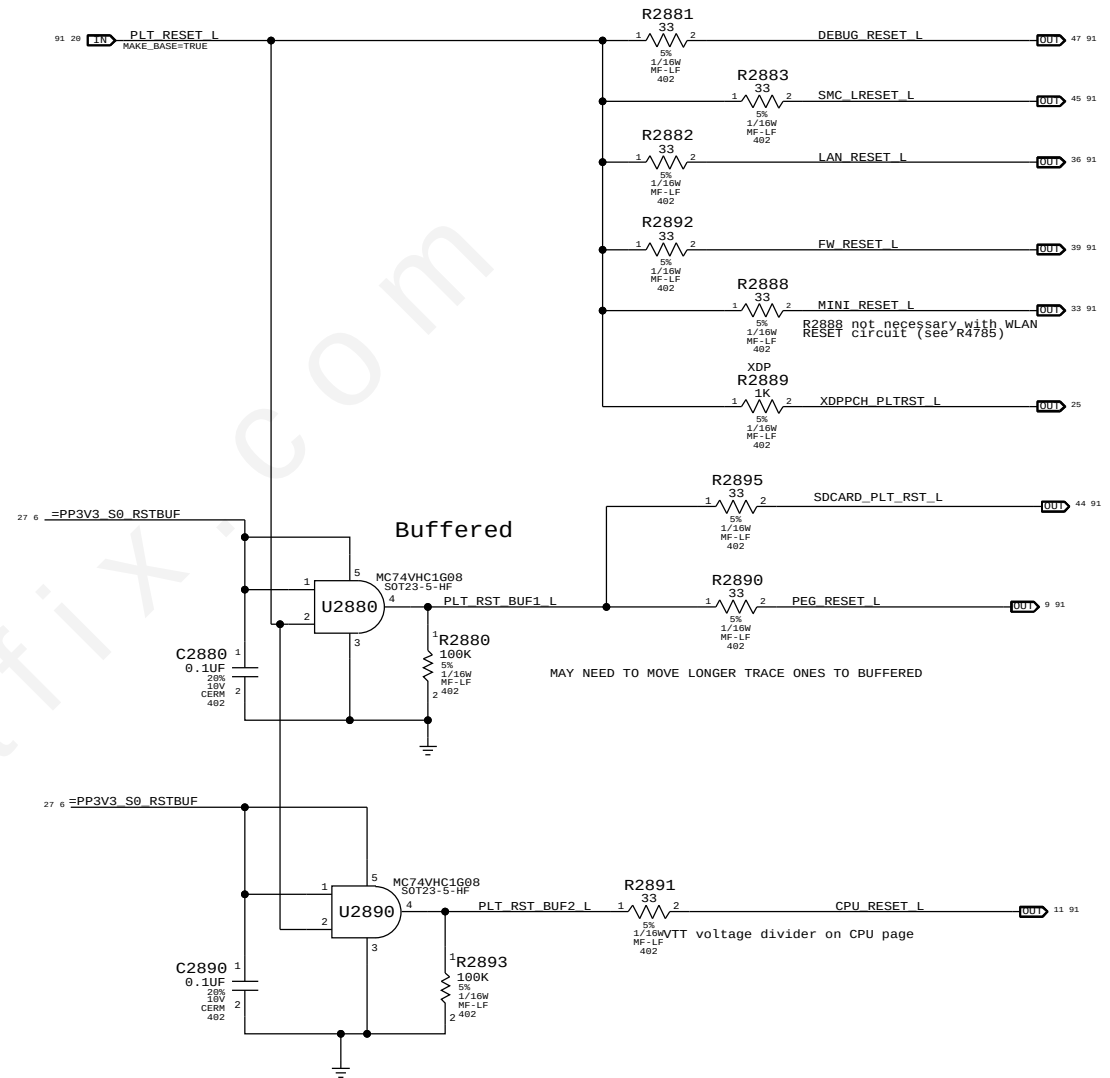
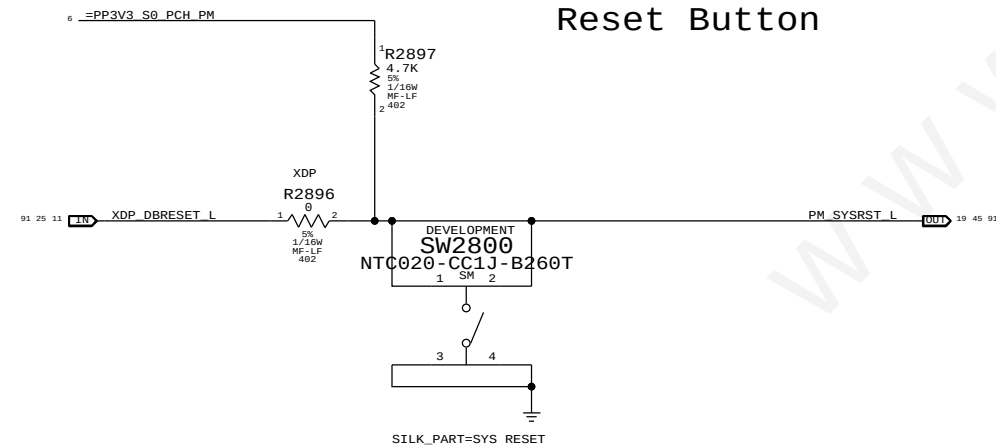
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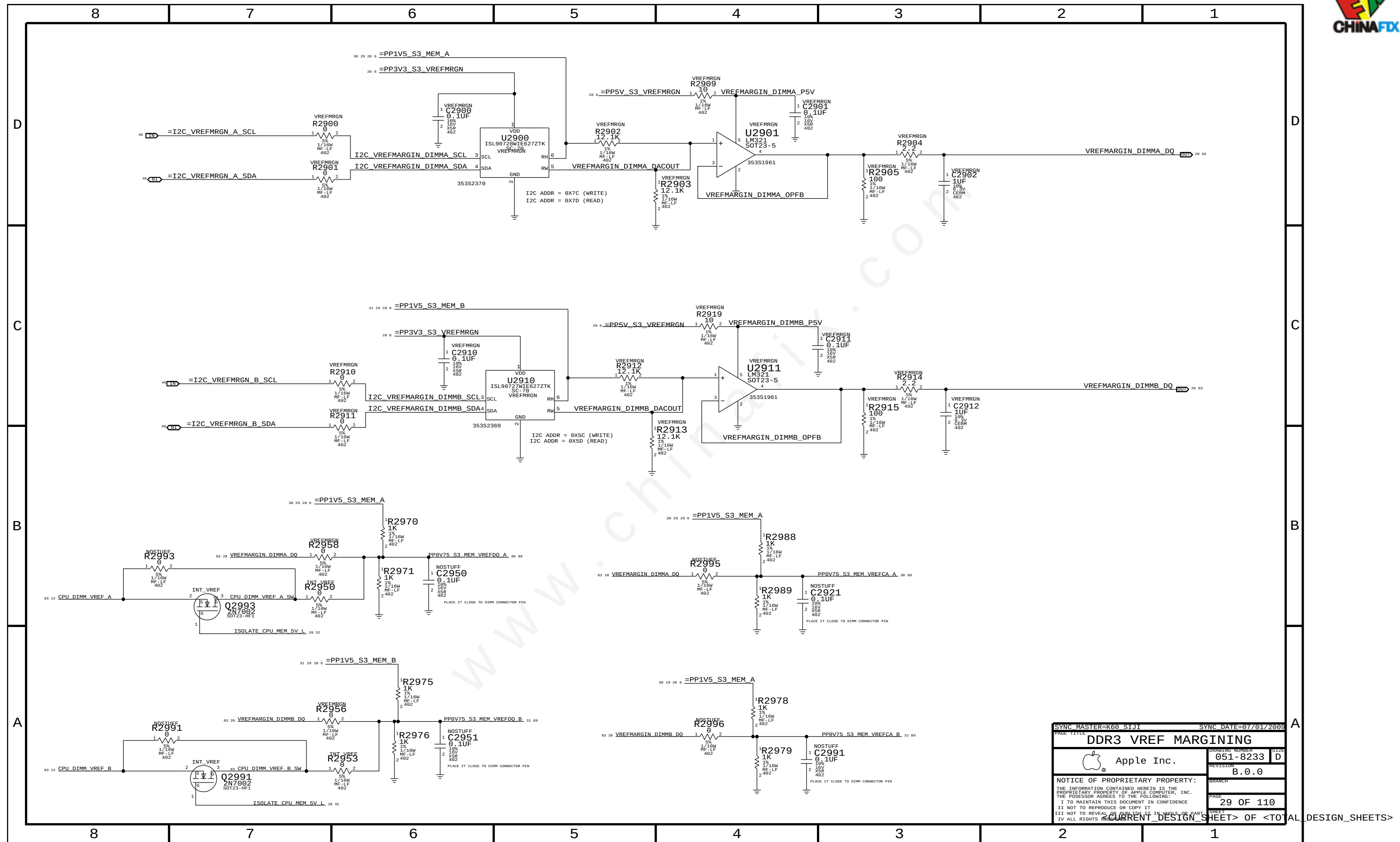


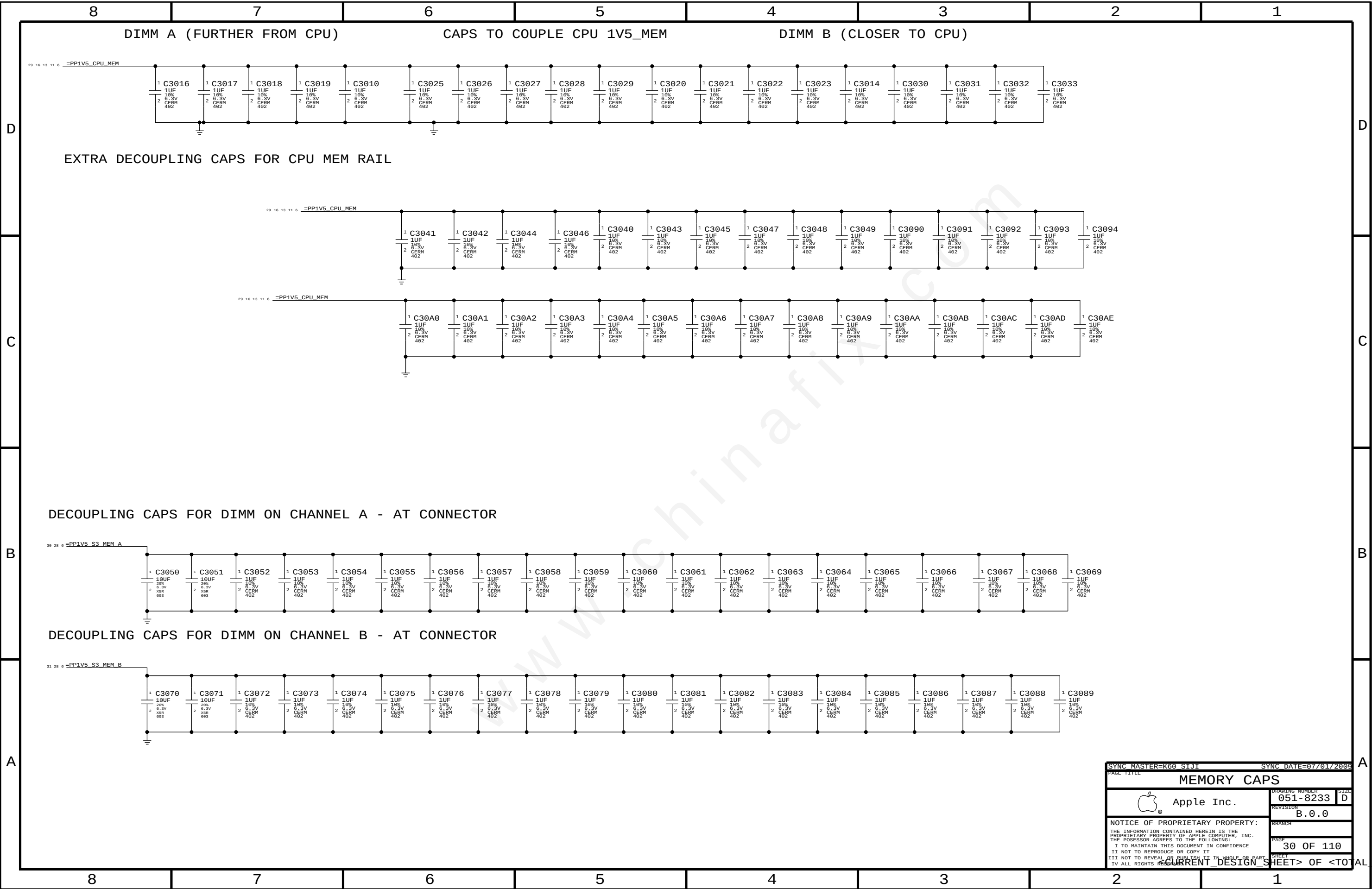
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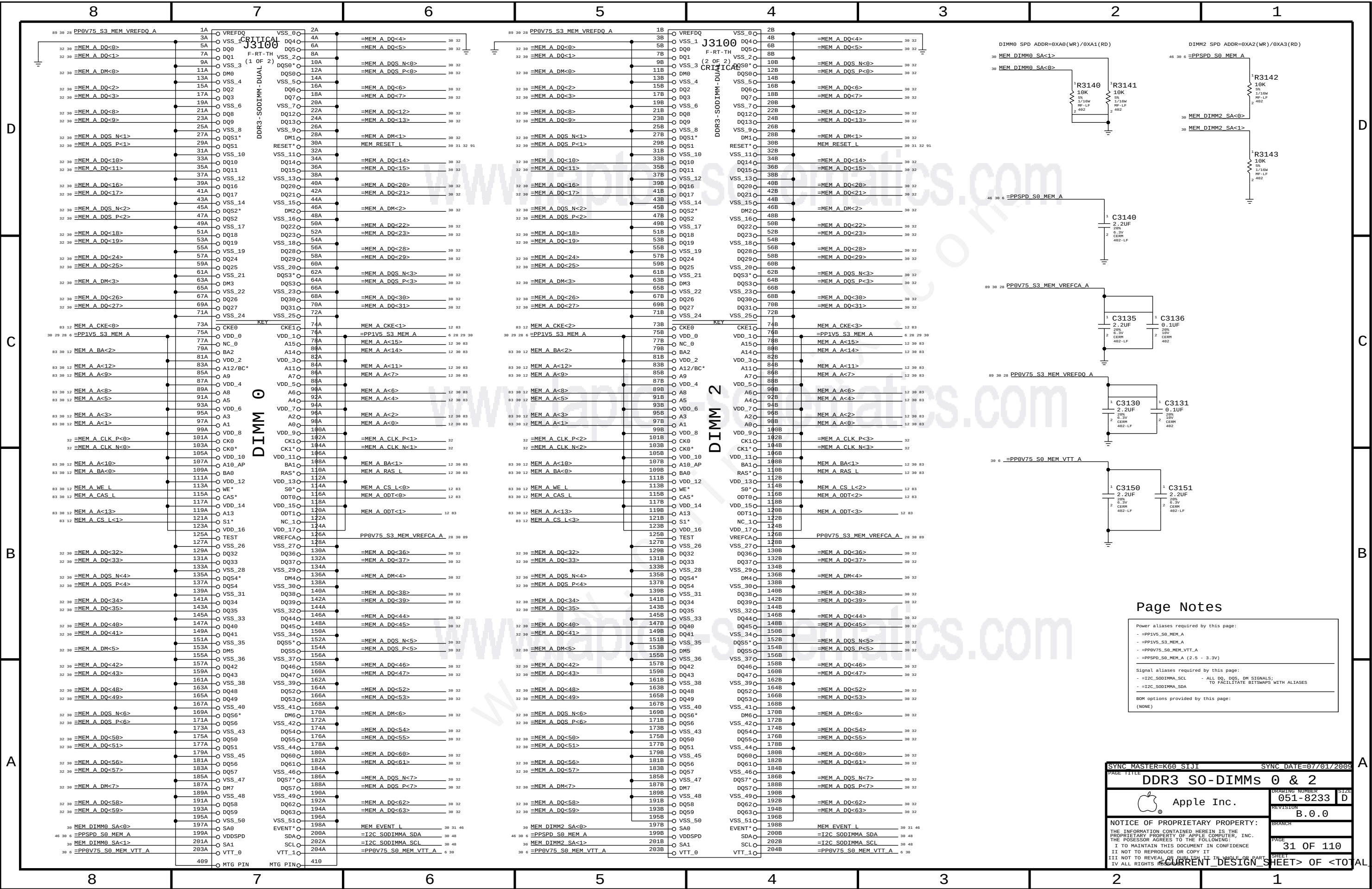
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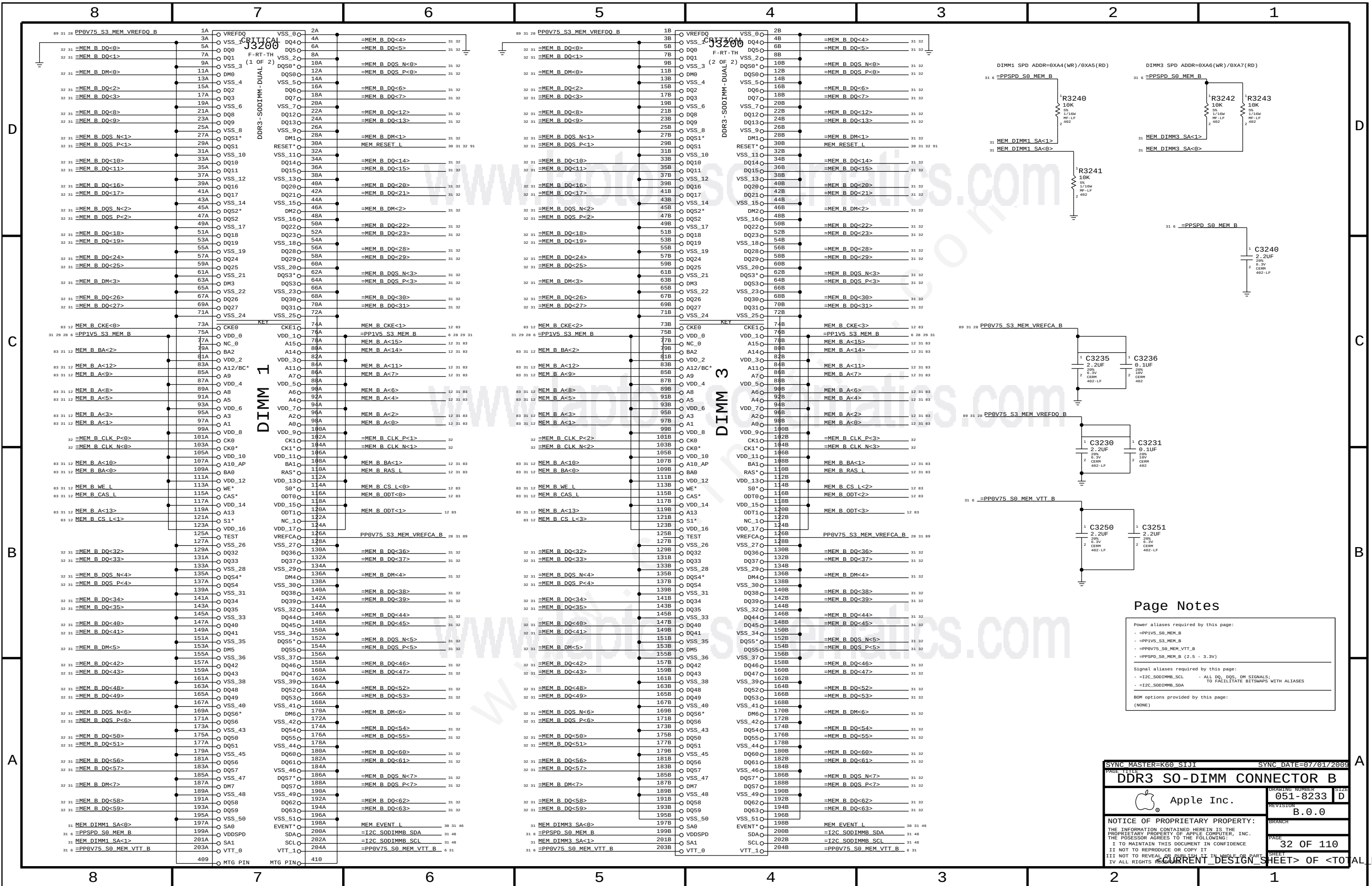






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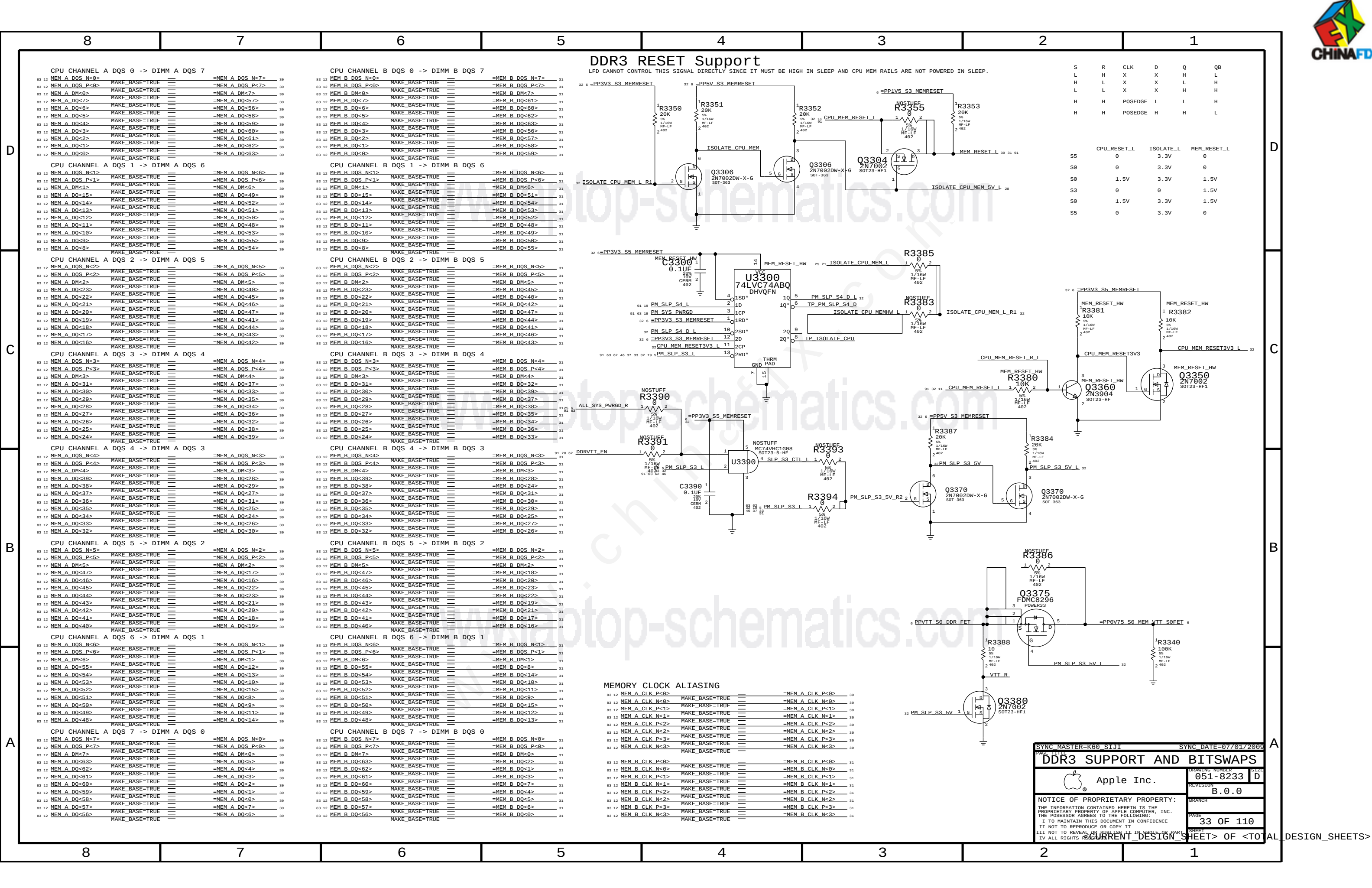




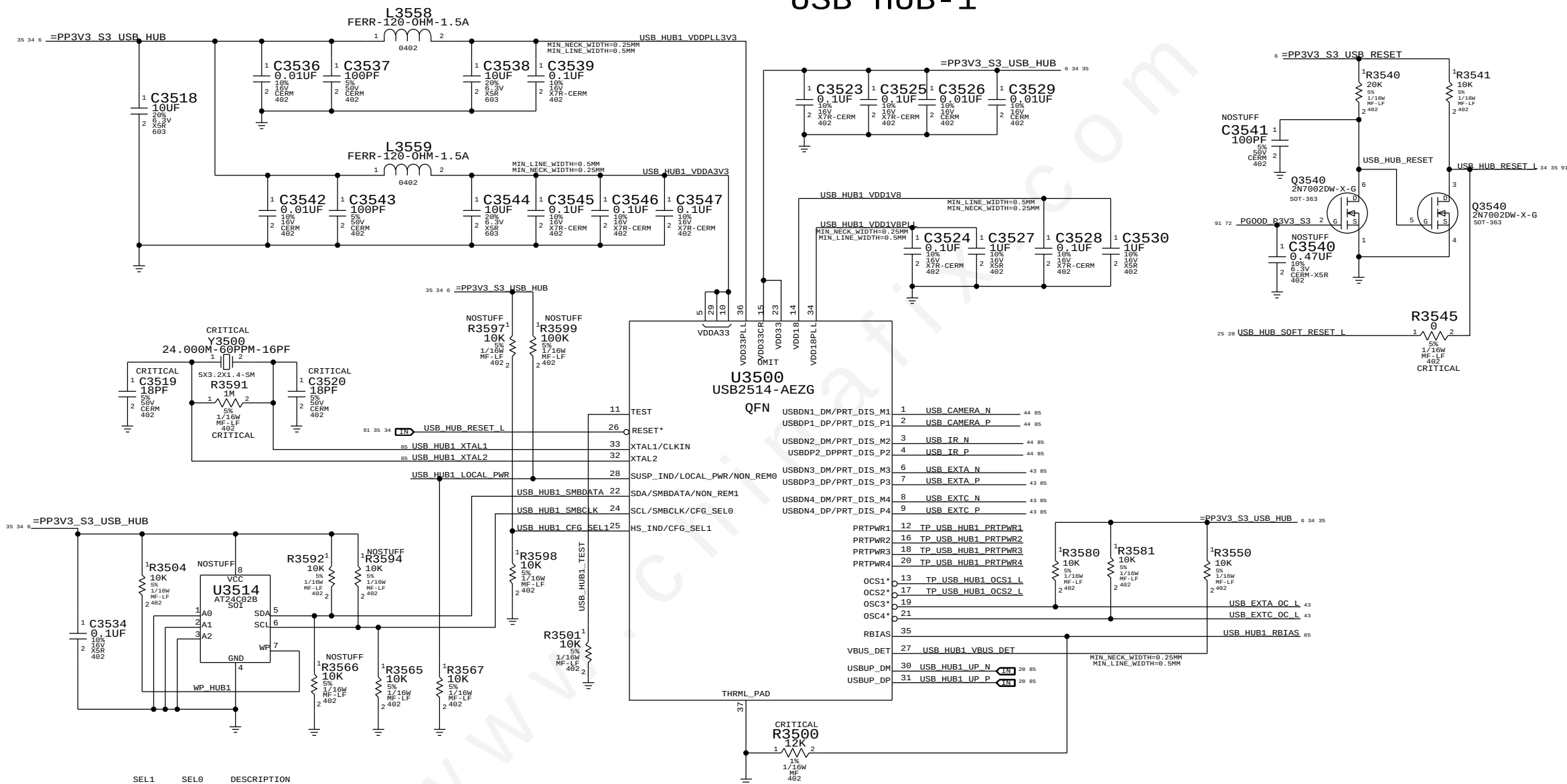
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 - PPSPD_S0_MEM_B (2.5 - 3.3V)
- Signal aliases required by this page:
- I2C_S0DIMMB_SCL - ALL DQ, DQS, DM SIGNALS TO FACILITATE BITSTREAMS WITH ALIASES
 - I2C_S0DIMMB_SDA
- BOM options provided by this page:
- (NONE)

SYNC MASTER=K60 SIJI		SYNC DATE=07/01/2009	
PAGE TITLE		DRAWING NUMBER	
DDR3 SO-DIMM CONNECTOR B		051-8233	
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USB HUB-1

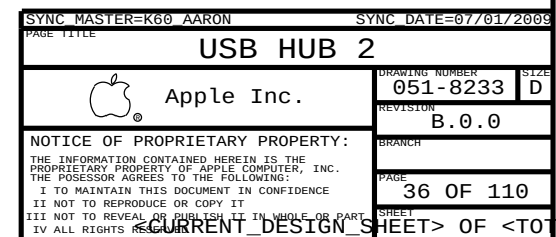


BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	SMSC USB2514	U3500, U3600	CRITICAL	

SYNC MASTER=K60 AARON		SYNC DATE=07/01/2009	
PAGE TITLE			
USB HUB 1			
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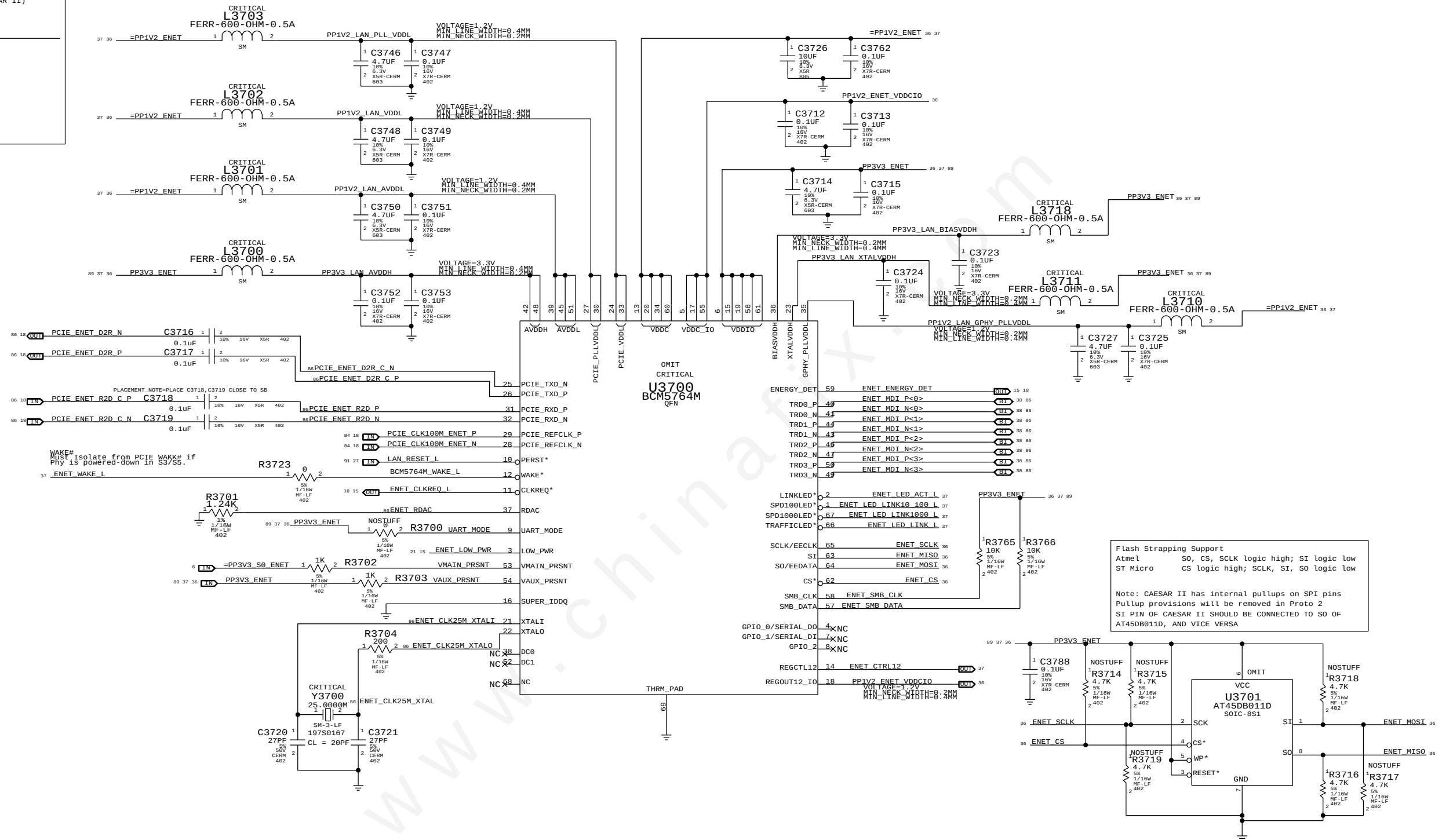




Page Notes

Power aliases required by this page:

- PP3V3_ENET (CAESAR II)
- =PP1V2_ENET



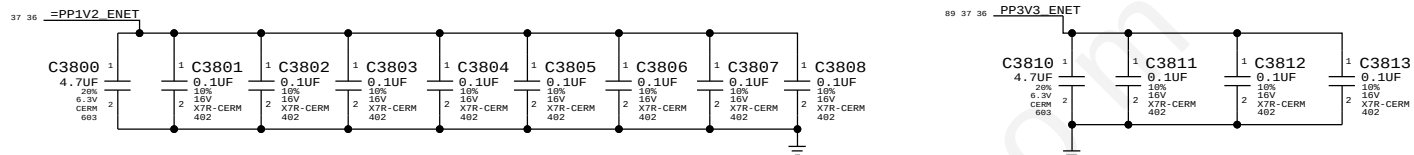
Flash Strapping Support	
Atmel	SO, CS, SCLK logic high; SI logic low
ST Micro	CS logic high; SCLK, SI, SO logic low
Note: CAESAR II has internal pullups on SPI pins Pullup provisions will be removed in Proto 2 SI PIN of CAESAR II SHOULD BE CONNECTED TO SO OF AT45DB011D, AND VICE VERSA	

SYNCH MASTER=K60 AARON		SYNCH DATE=07/01/2009	
PAGE TITLE			
ETHERNET (CAESAR II)			
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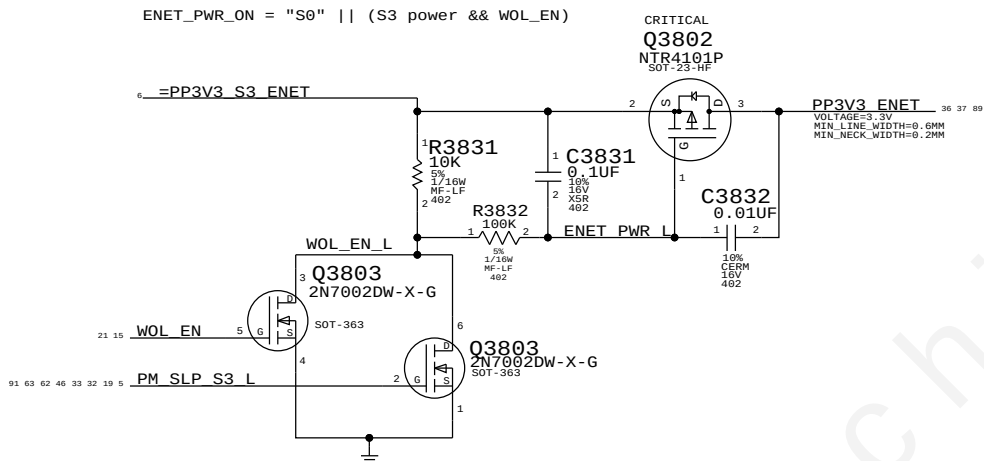
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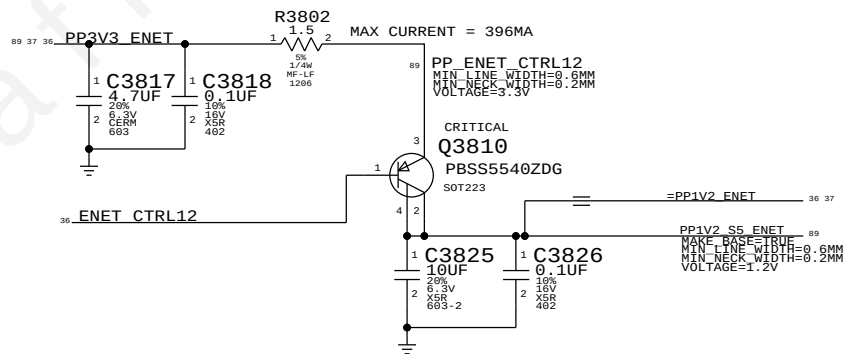
CAESAR II DECOUPLING



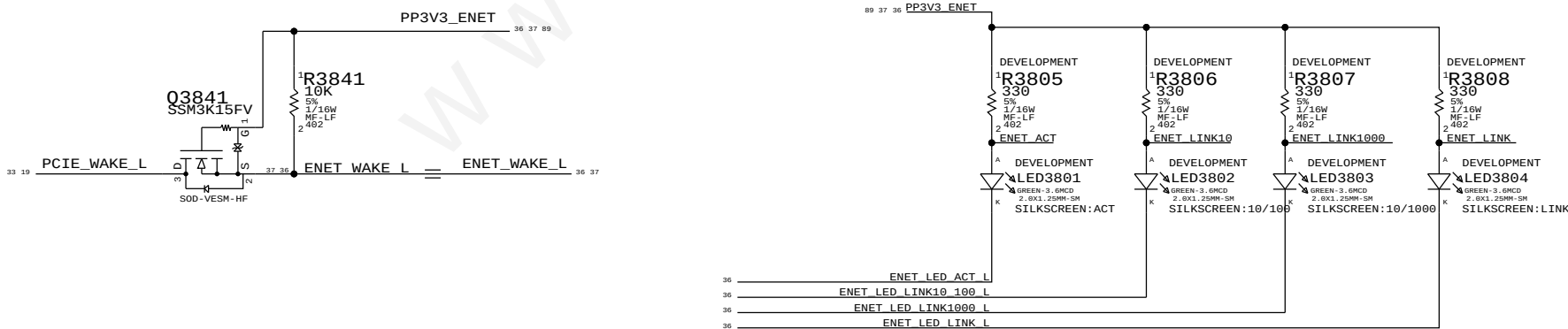
ENET POWER ENABLE CIRCUIT



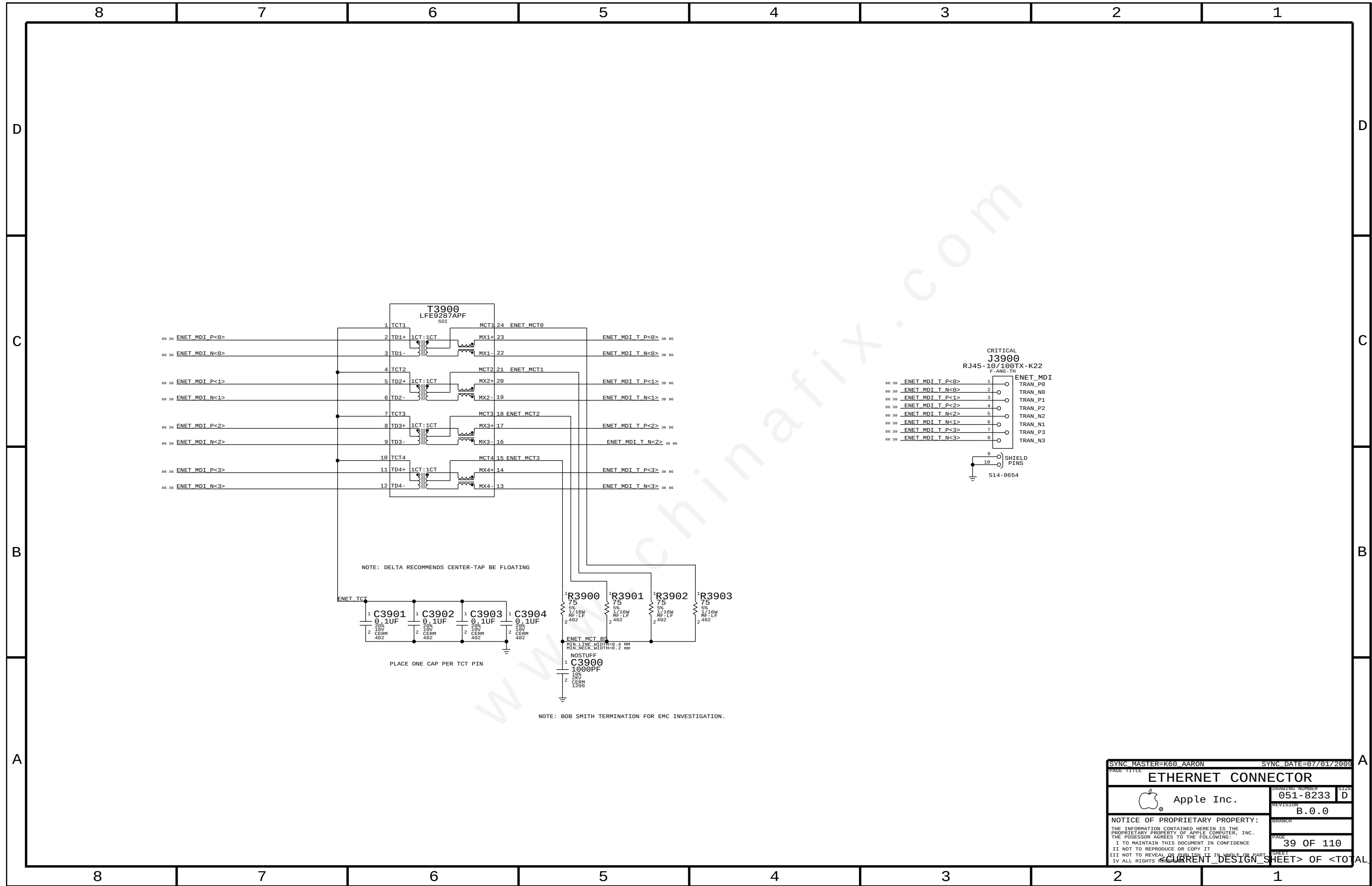
CAESAR II 1V2 RAIL SUPPLY



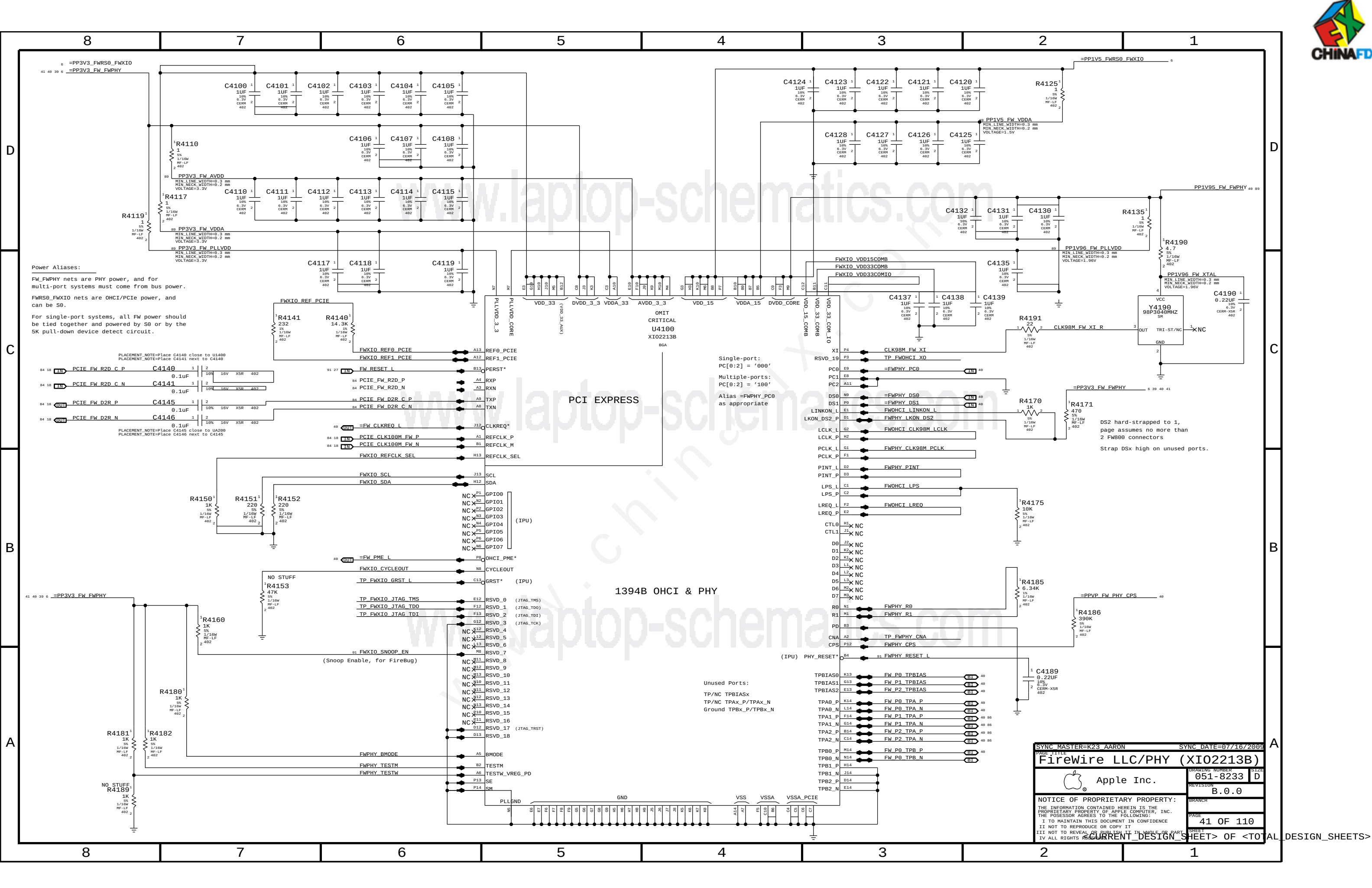
CAESAR II LED SUPPORT



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CAESAR II SUPPORT			
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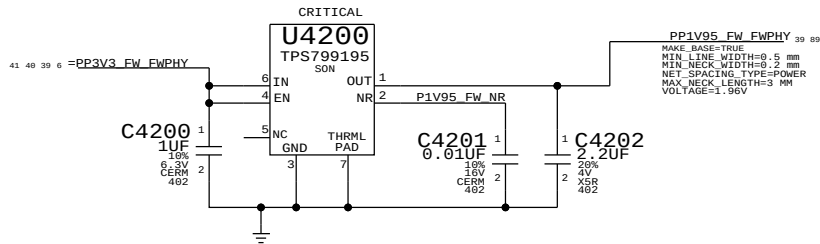
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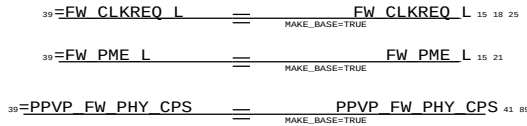
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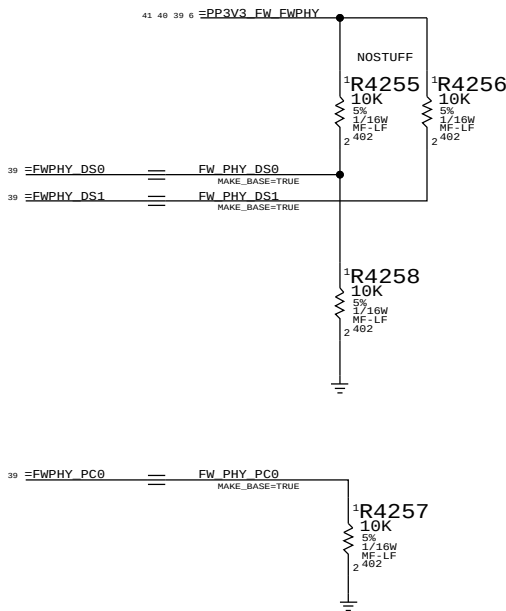
1394 PHY 1.95V SUPPLY



FireWire Aliases For Connectivity

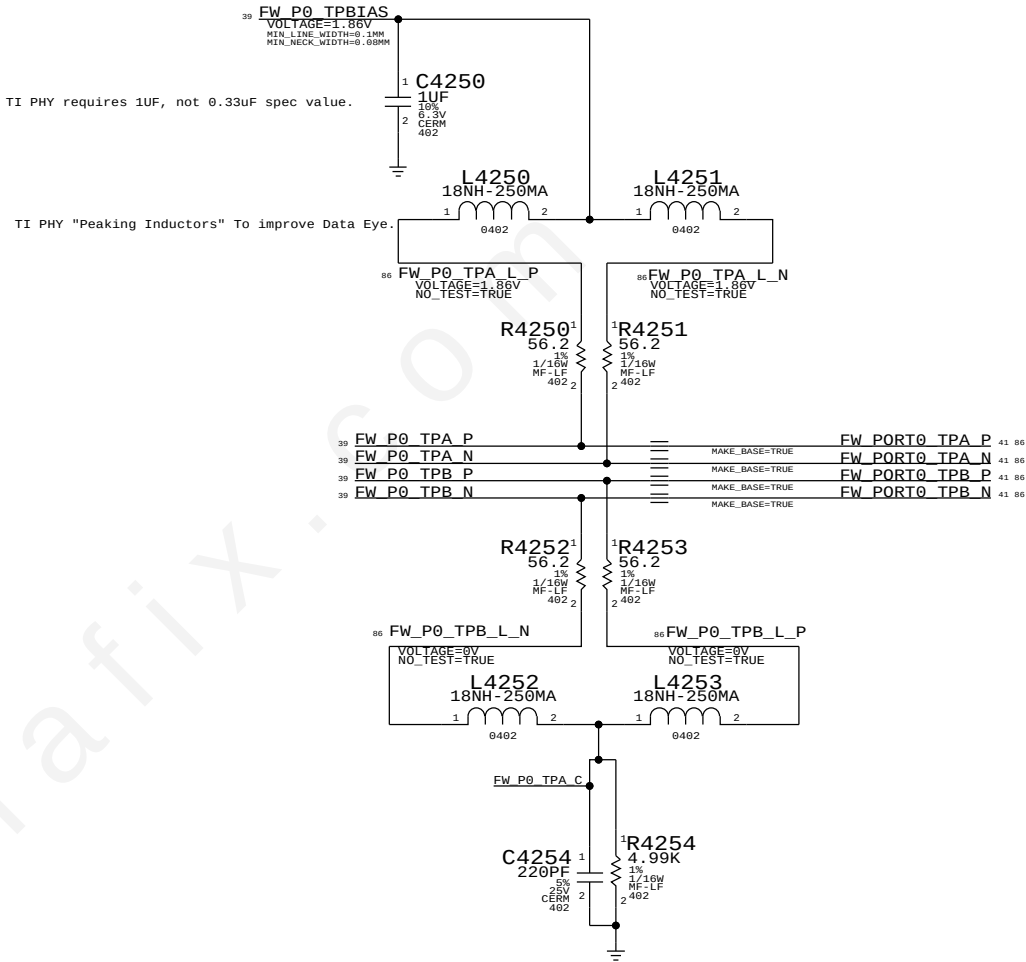


1394 PHY STRAPPING OPTIONS

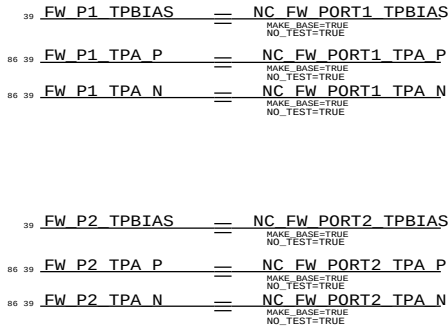


THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

iMacs are now one port only and have Power Code "000"

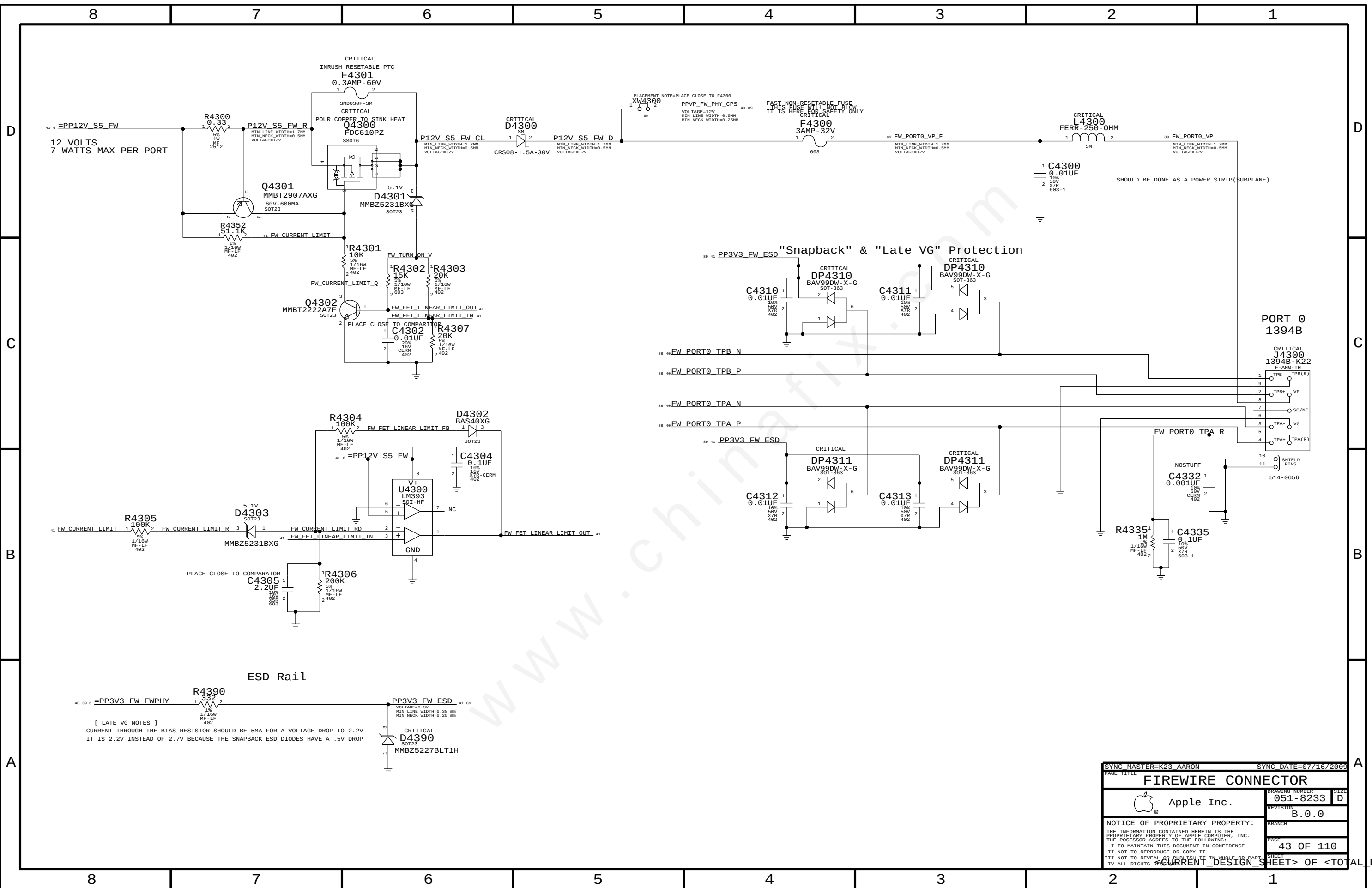
Termination
Place close to FireWire PHY

2ND & 3RD TPA/TPB PAIR UNUSED



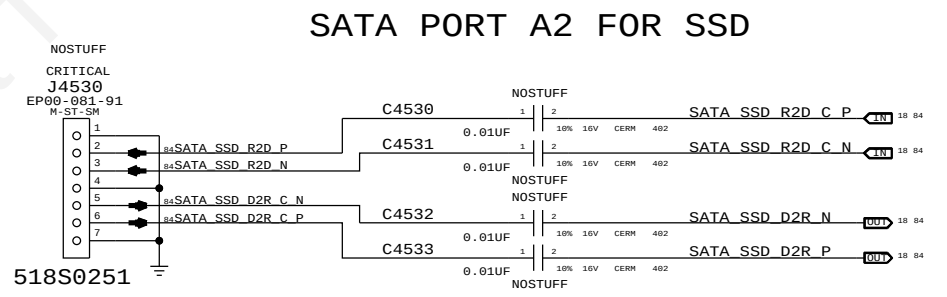
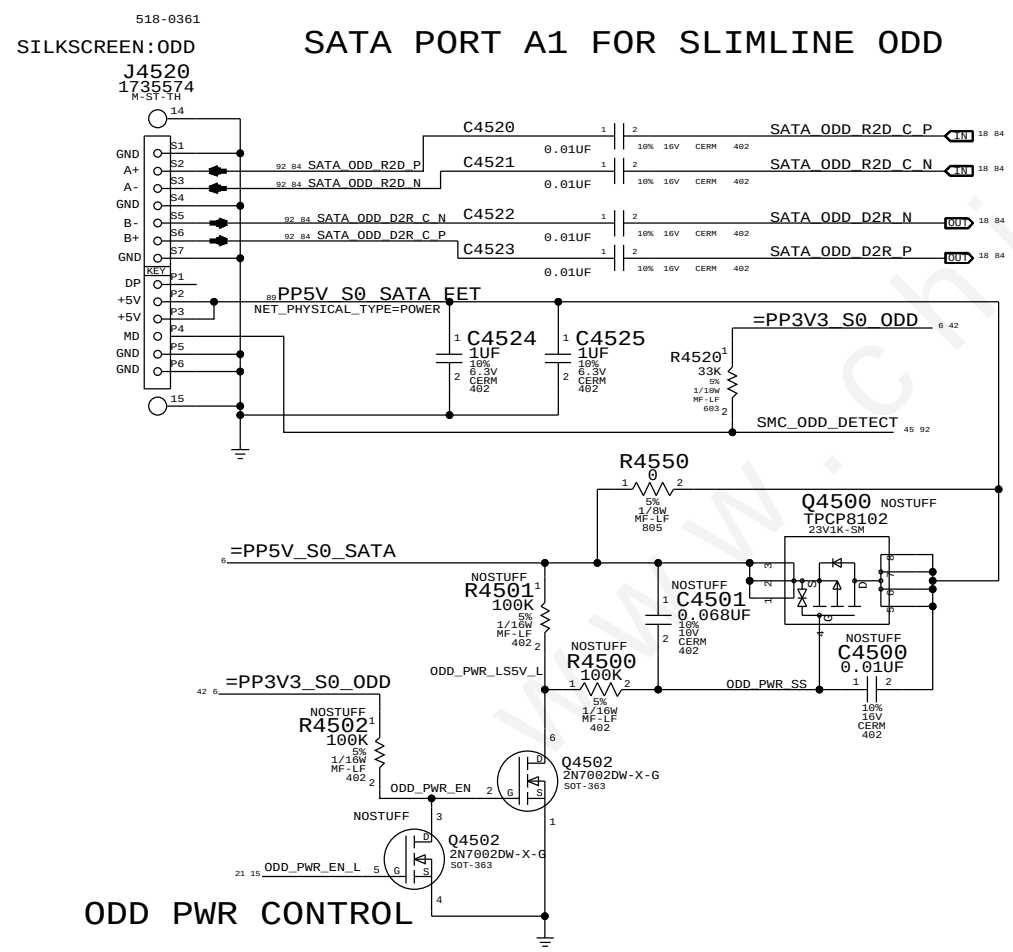
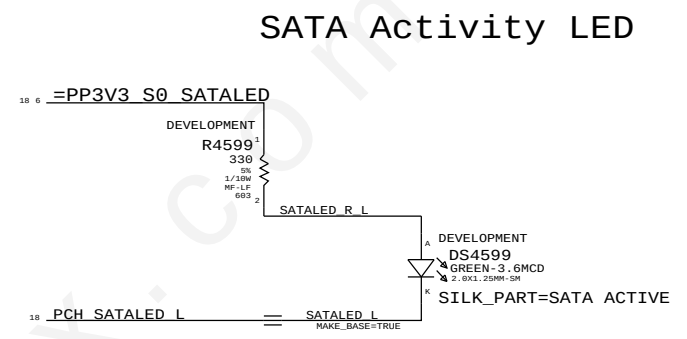
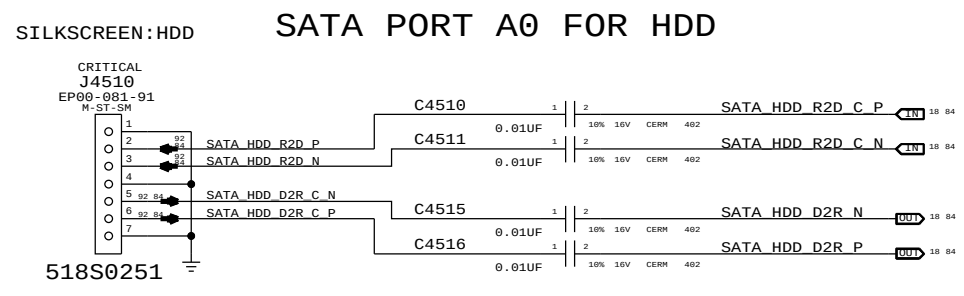
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FIREWIRE CONNECTOR			
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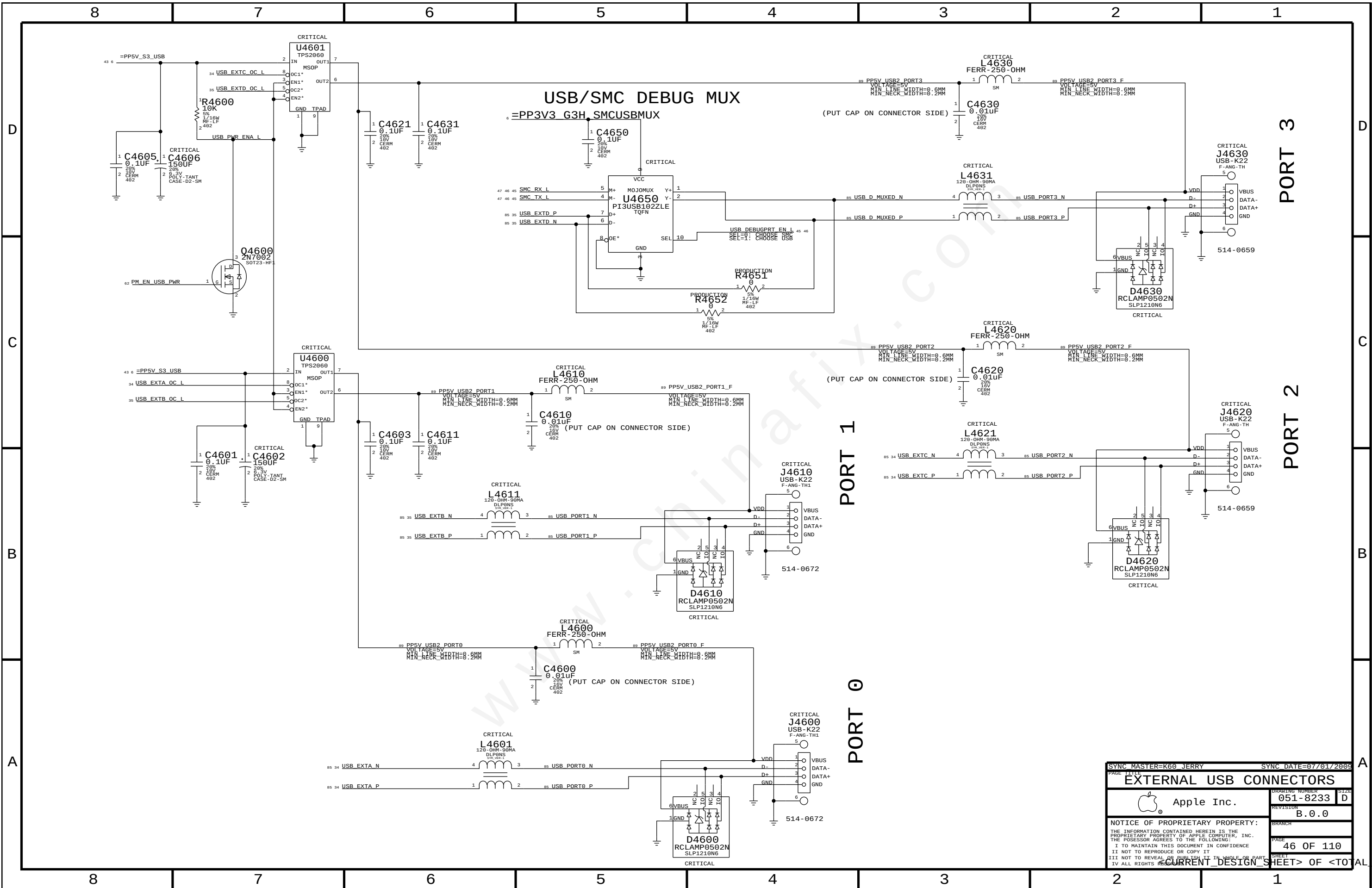
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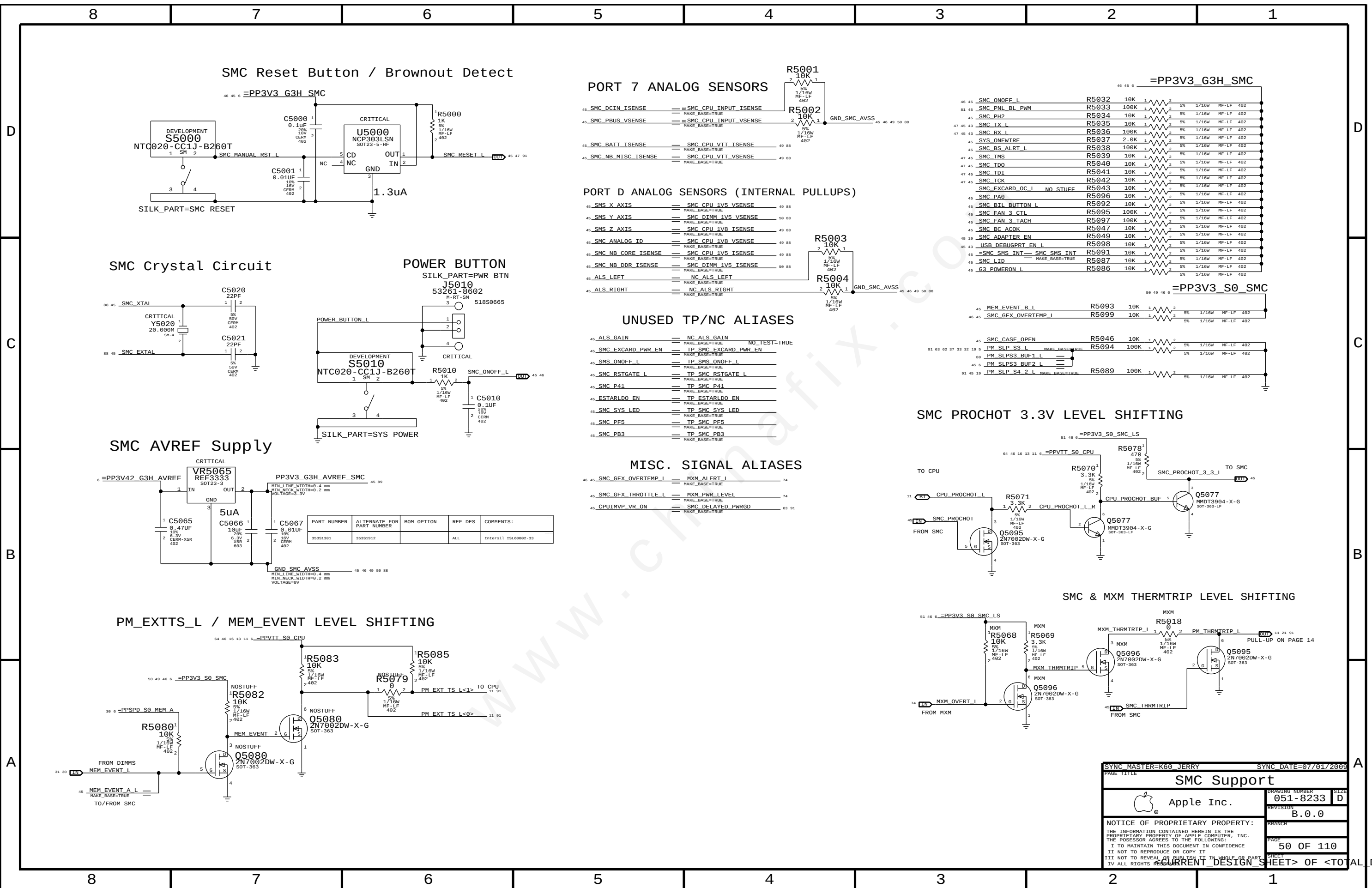


ODD PWR CONTROL

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SATA Connectors			
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SMC Support

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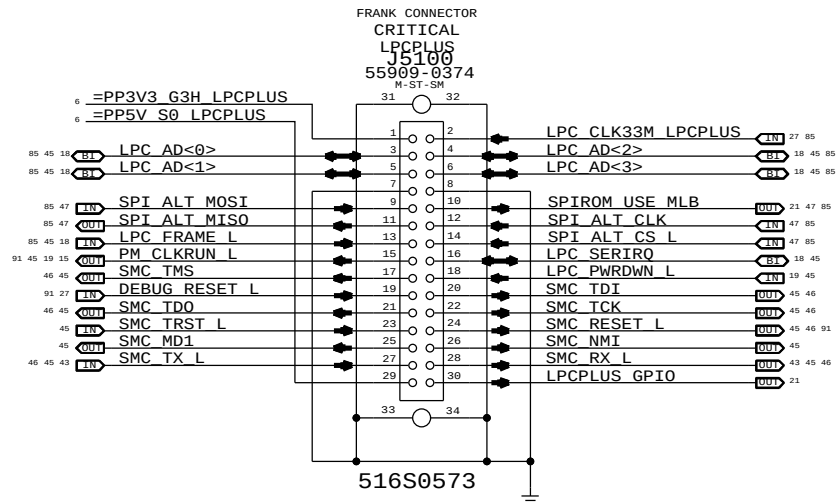
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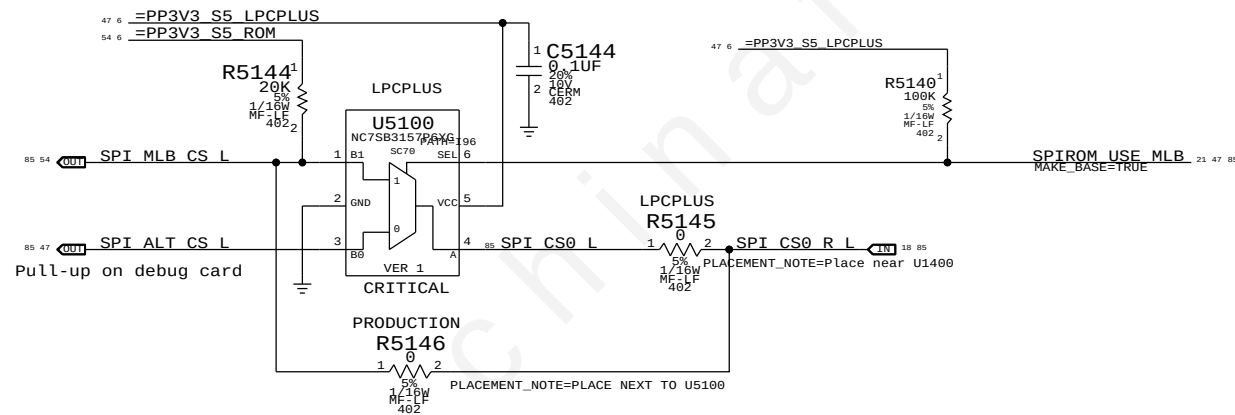
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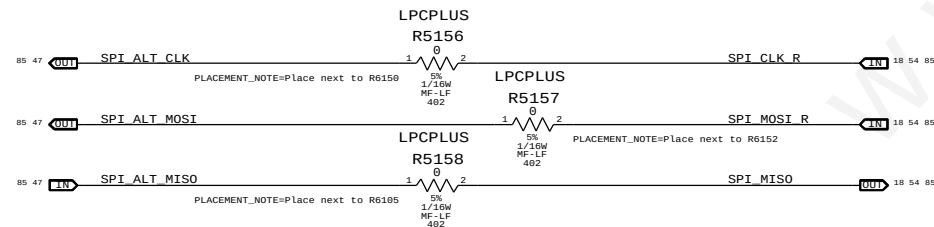
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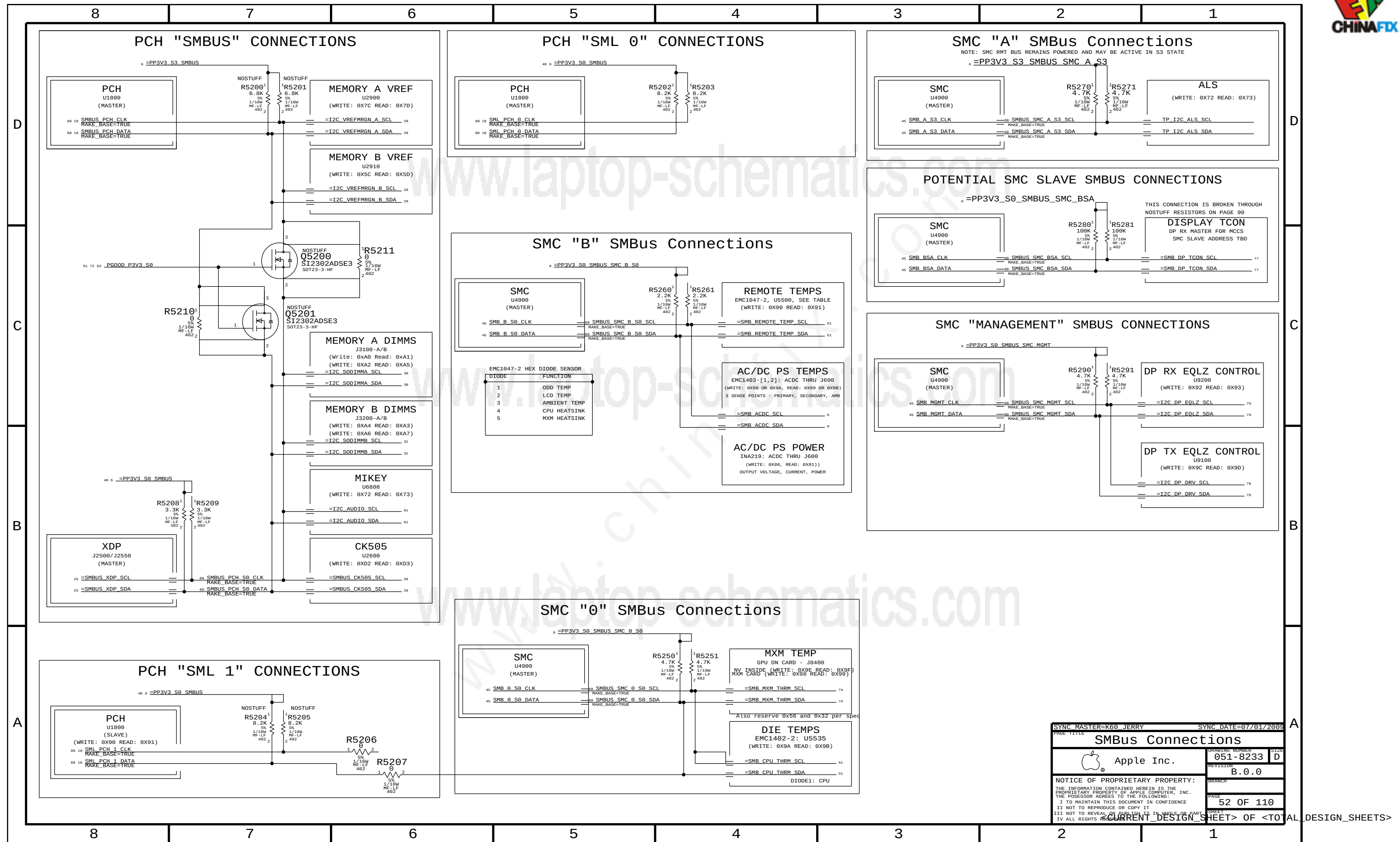


Alternate SPI ROM Support



SPI Bus Series Resistance Option





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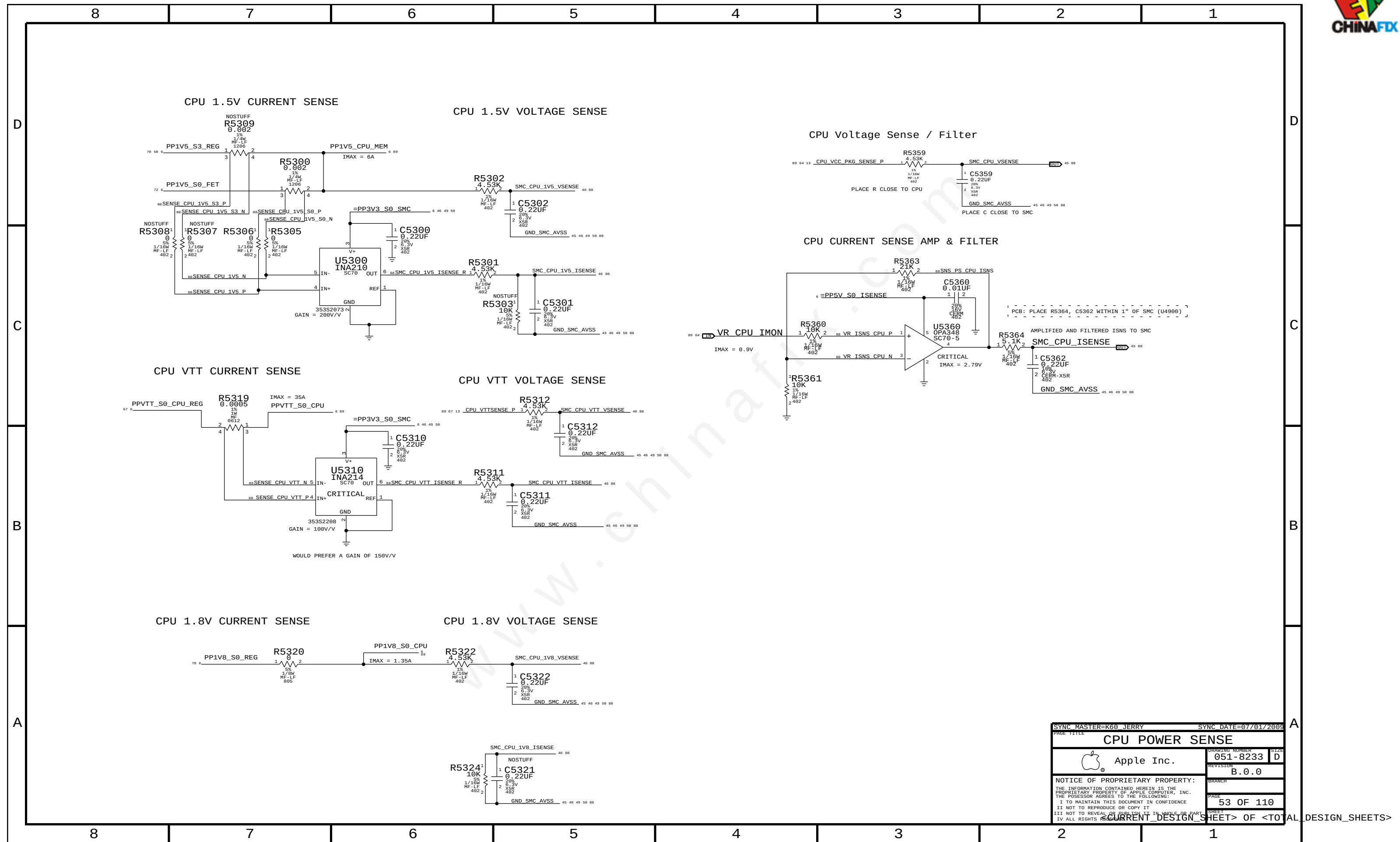
SMBus Connections

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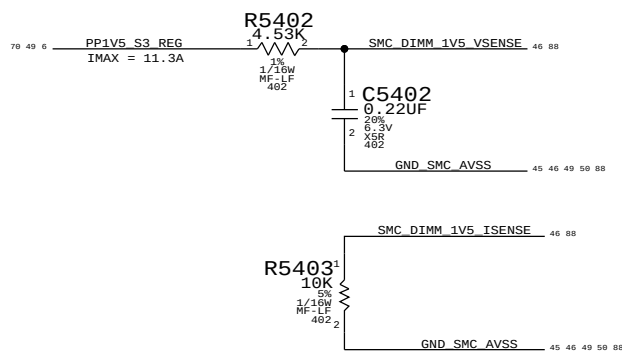
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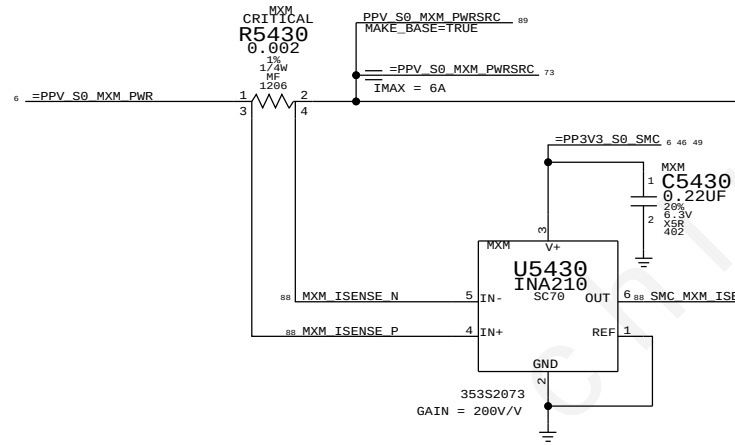
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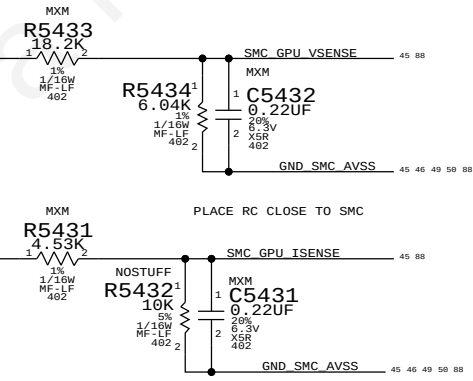
1.5V S3 VOLTAGE SENSE



MXM PWRSRC CURRENT SENSE



MXM PWRSRC VOLTAGE SENSE



SYNC MASTER=K60 JERRY

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GRAPHICS / DIMM POWER SENSE

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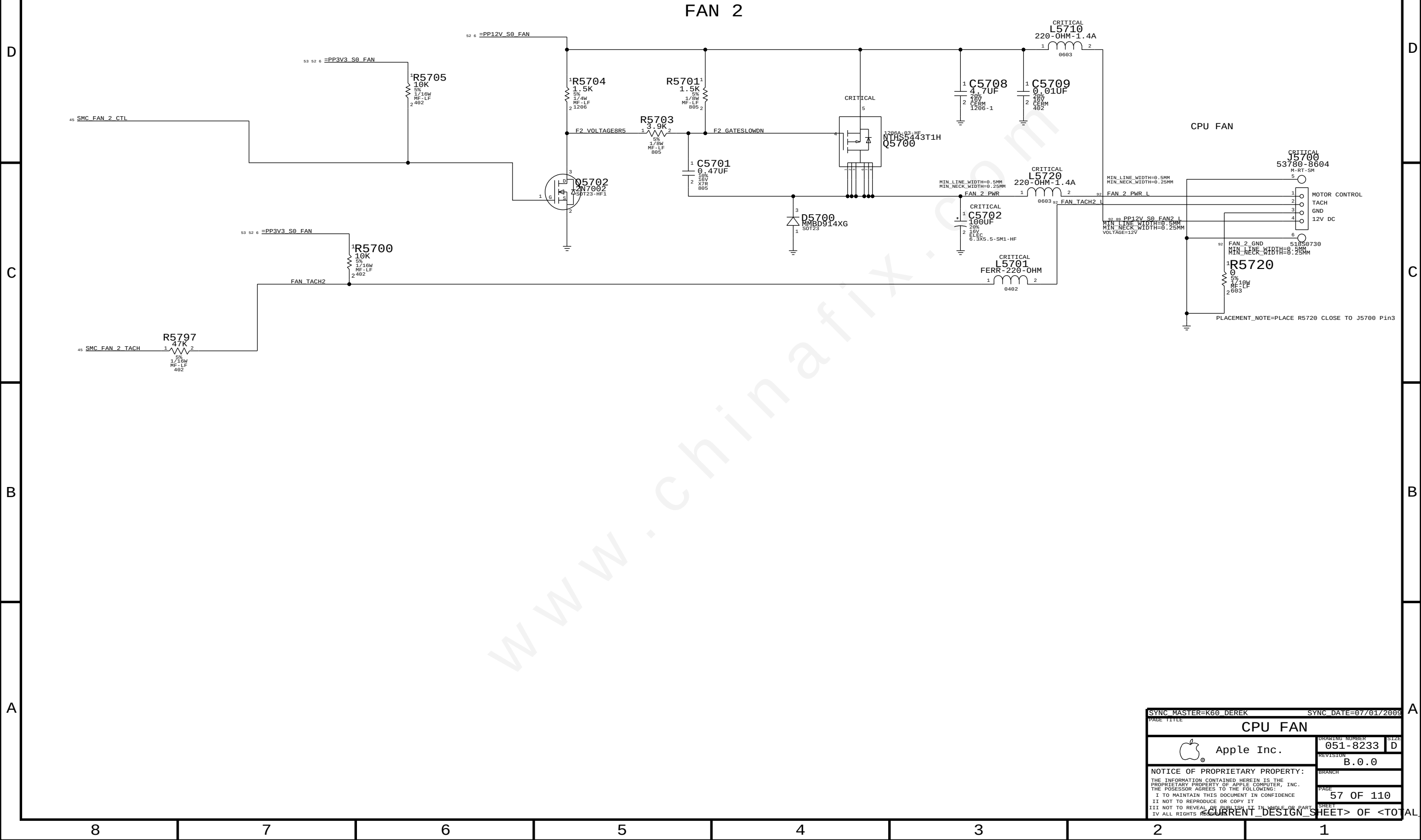
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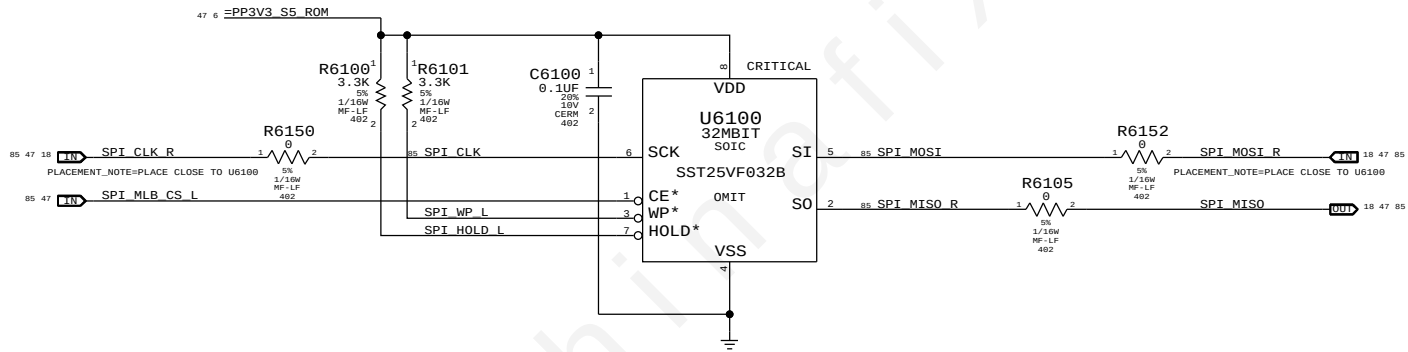
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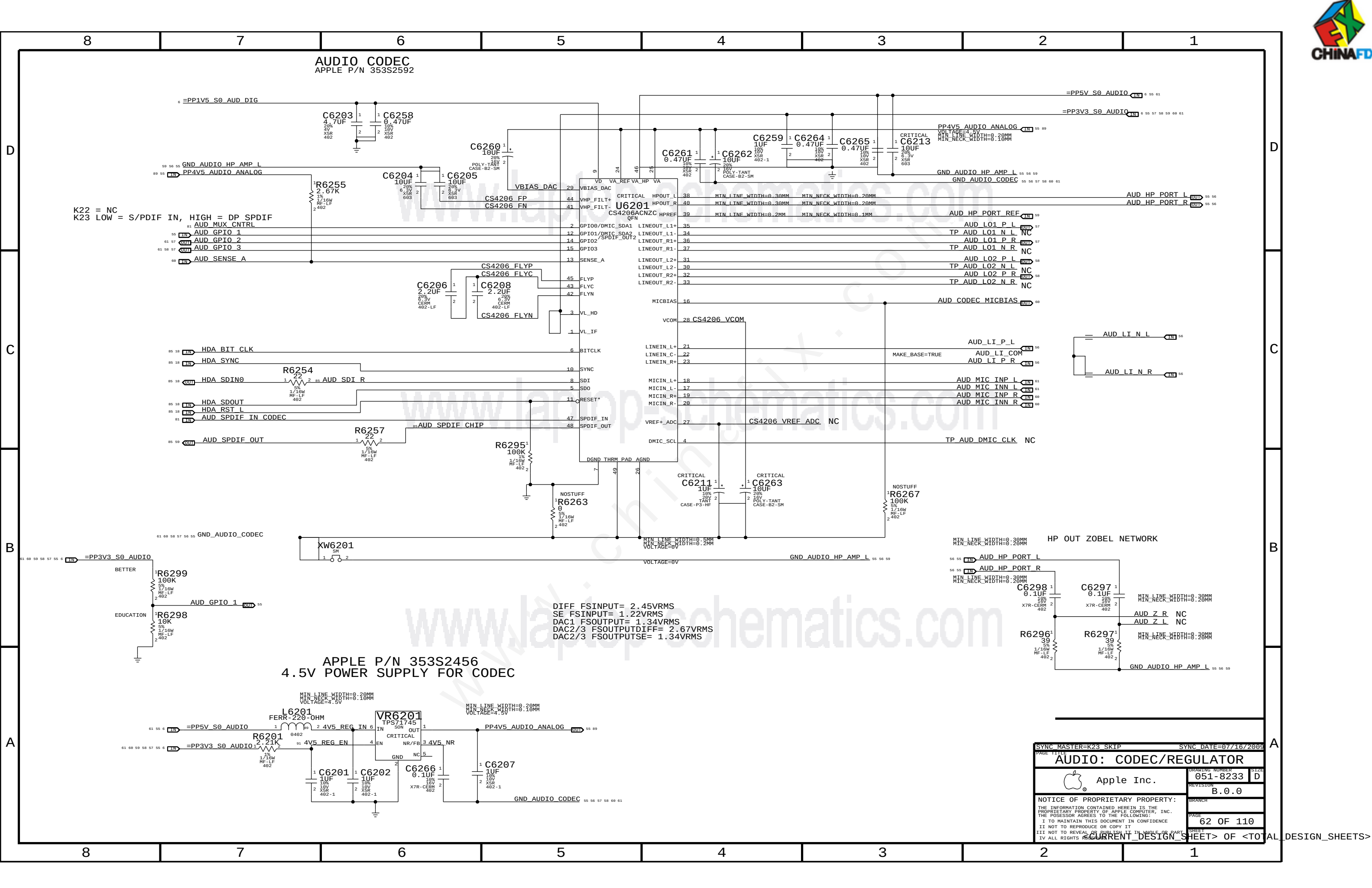
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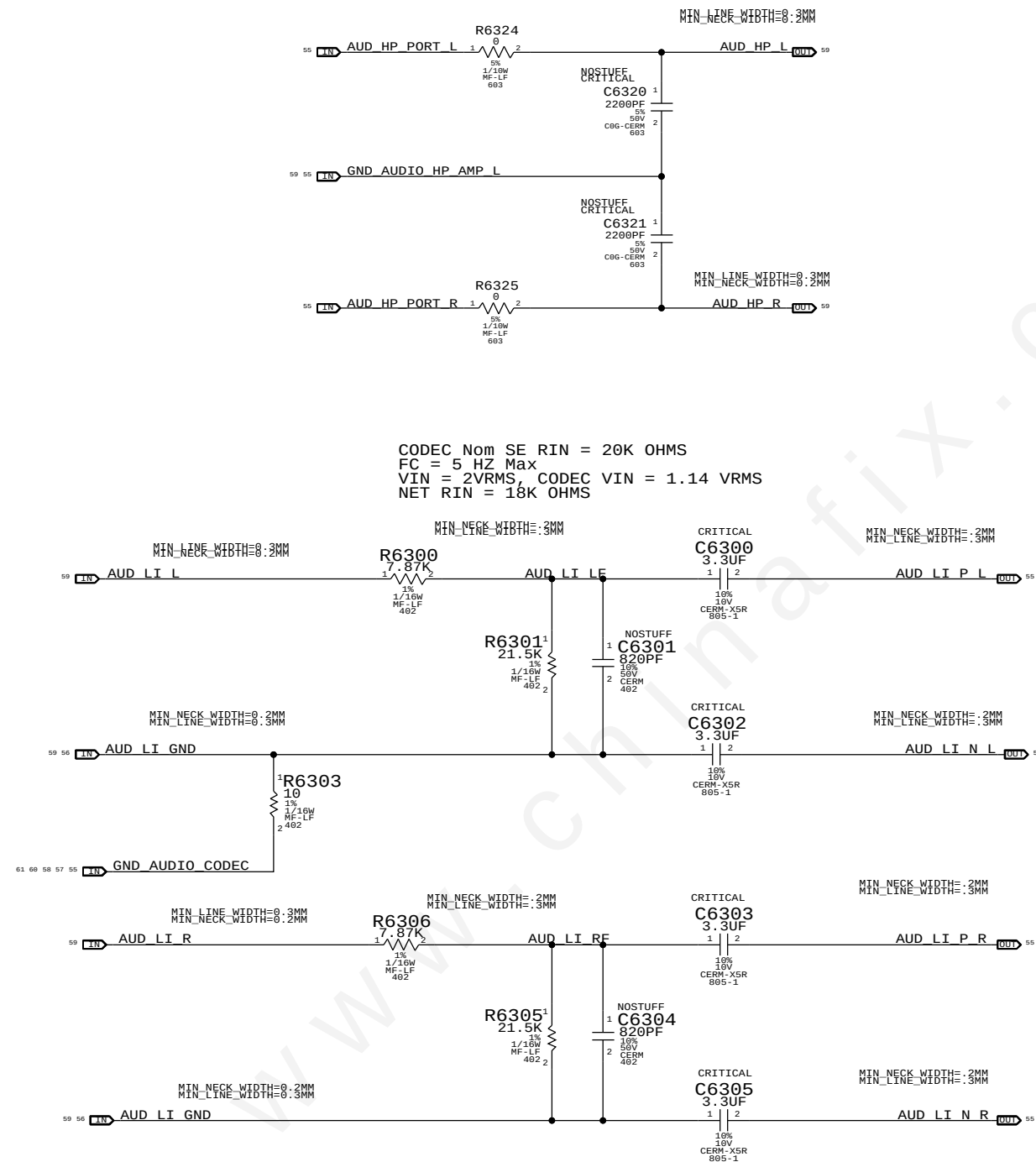


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Apple Inc.		REVISION	
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1ST ORDER DAC FILTER PLACEHOLDER

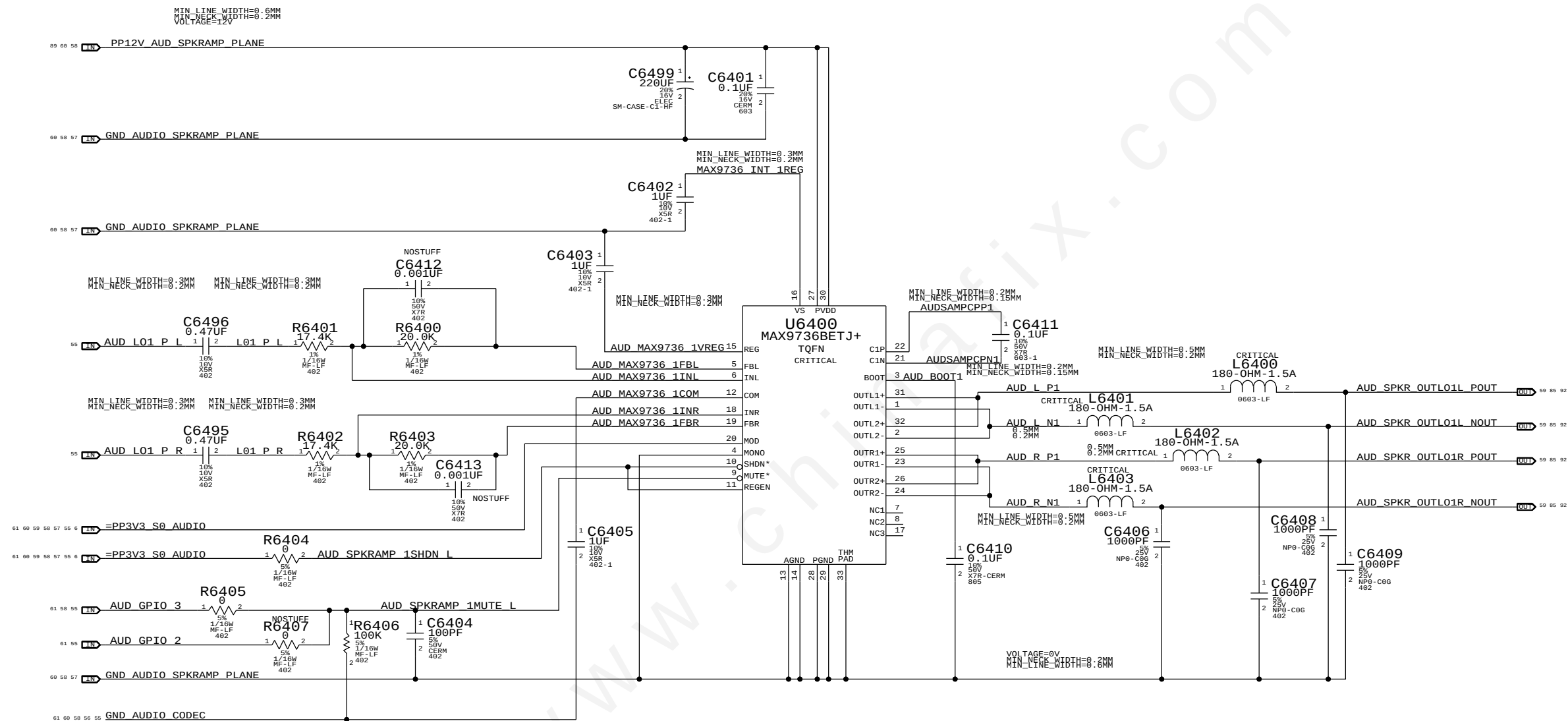


SYNCH MASTER=K23 SKIP		SYNCH DATE=07/16/2005	
PAGE TITLE			
AUDIO: FILTER/BUFFER			
 Apple Inc.	DRAWING NUMBER	SIZE	
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BRANCH		PAGE	
		63 OF 110	
SHEET		SHEET	
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TWEETER SPEAKER AMPLIFIER

MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
 CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
 AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
 FC = 19.5 HZ
 POUT = 6.76 W INTO 8 OHMS @ 1% THD+N

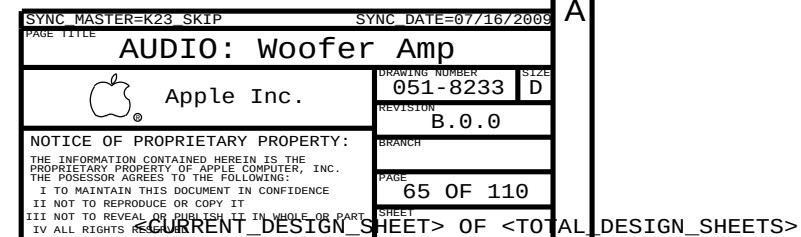


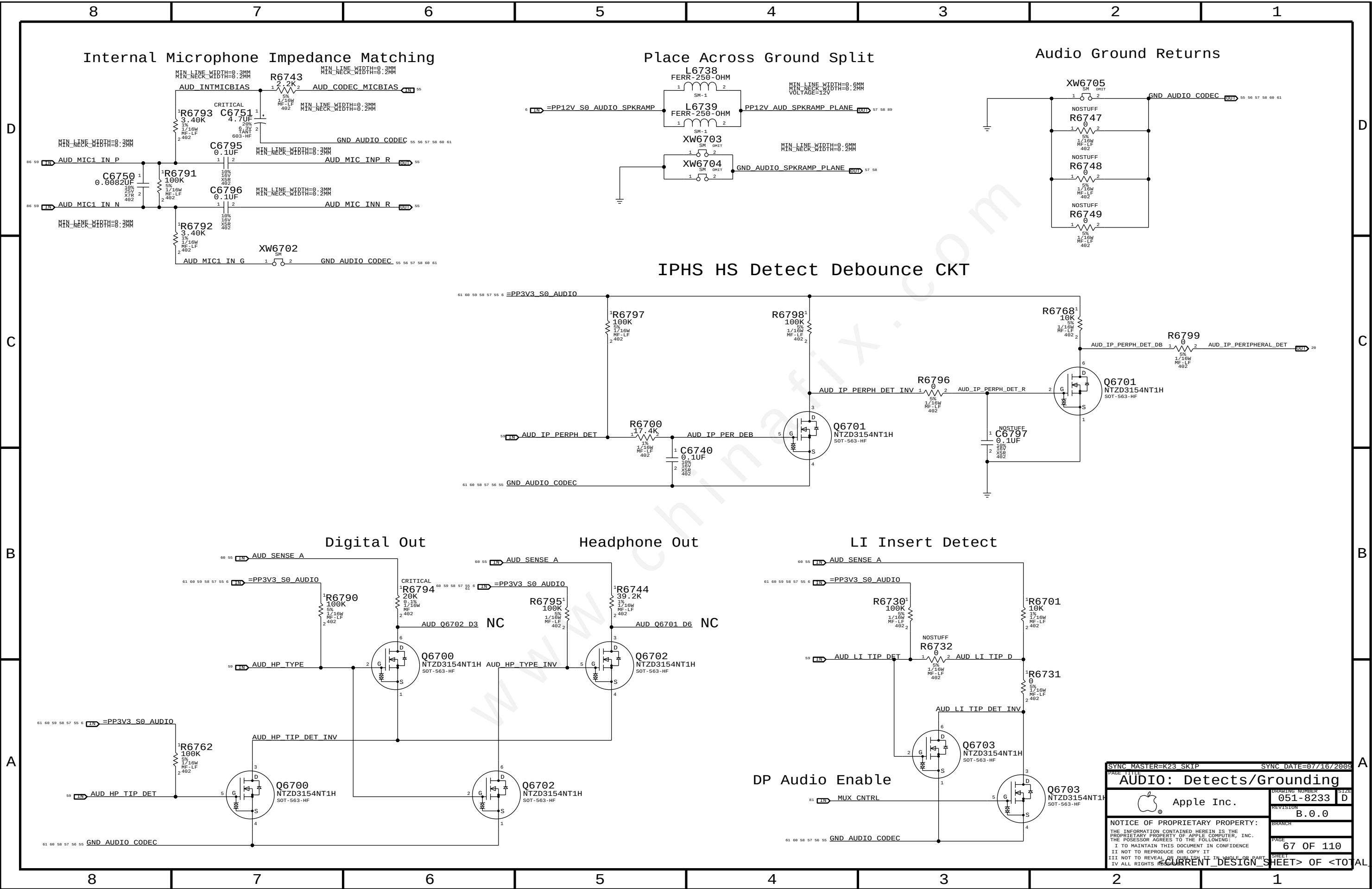
SYNC MASTER=K23 SKIP		SYNC DATE=07/16/2009	
PAGE TITLE			
AUDIO: Tweeter Amp 1			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
		B.0.0	
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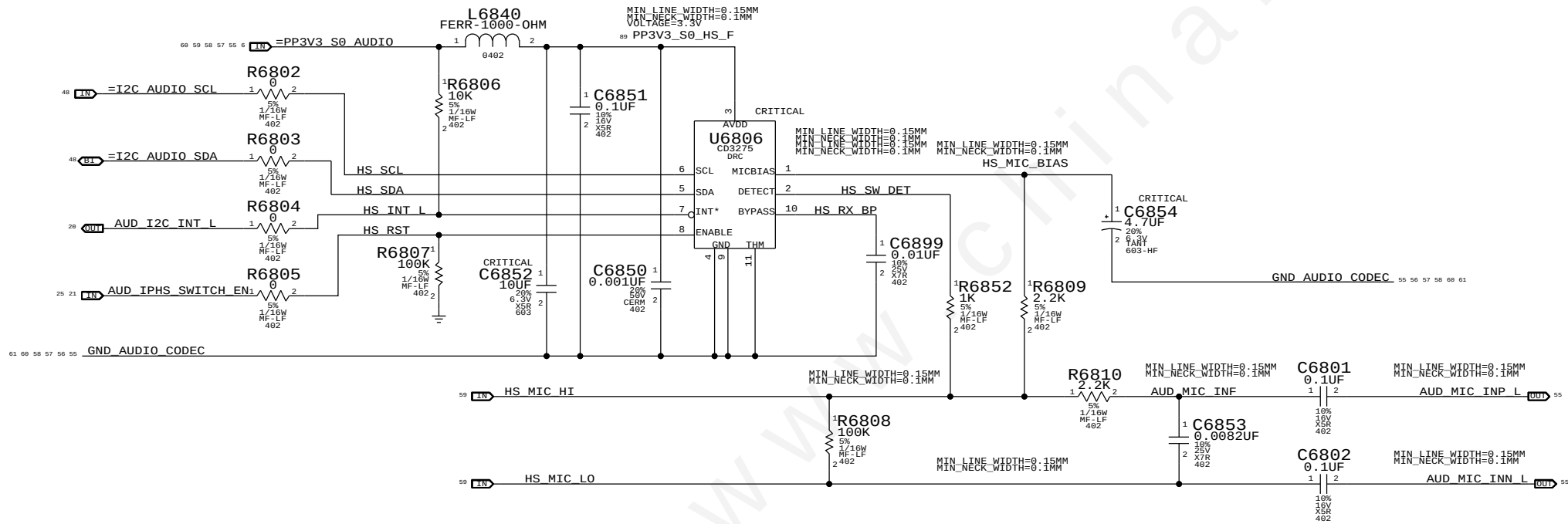


SYNC MASTER=K23 SKIP		SYNC DATE=07/16/2009	
PAGE TITLE		AUDIO: Detects/Grounding	
Apple Inc.		DRAWING NUMBER	051-8233 D
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FUNCTION	PIN	CONVERTER	VOLUME	ENABLE/ CNTRL TYPE	DETECT/INTERRUPT
PRIMARY	0X0B	0X04	0X04	GPIO 3	N/A
SECONDARY	0X0A	0X03	0X03	GPIO 3	N/A
HEADPHONES	0X09	0X02	0X02	N/A	0X09 (A)
LINE INPUT	0X0C	0X05	0X05	N/A	LINE IN
BUILT-IN MICROPHONE	0X0D(13,B,RIGHT)	0X06	0X06	MICBIAS 80%	N/A
HEADSET MICROPHONE	0X0D (13,V22,B,LEFT)	0X06	0X06	MIKEY	MIKEY
SPDIF OUT	0X10	0X08	N/A	N/A	0X0C (B)
SPDIF IN	0X0F	0X07	N/A	N/A	N/A
MIKEY	N/A	N/A	N/A	MCP GPIO_38	MCP GPIO_5

MIKEY RECEIVER CKT

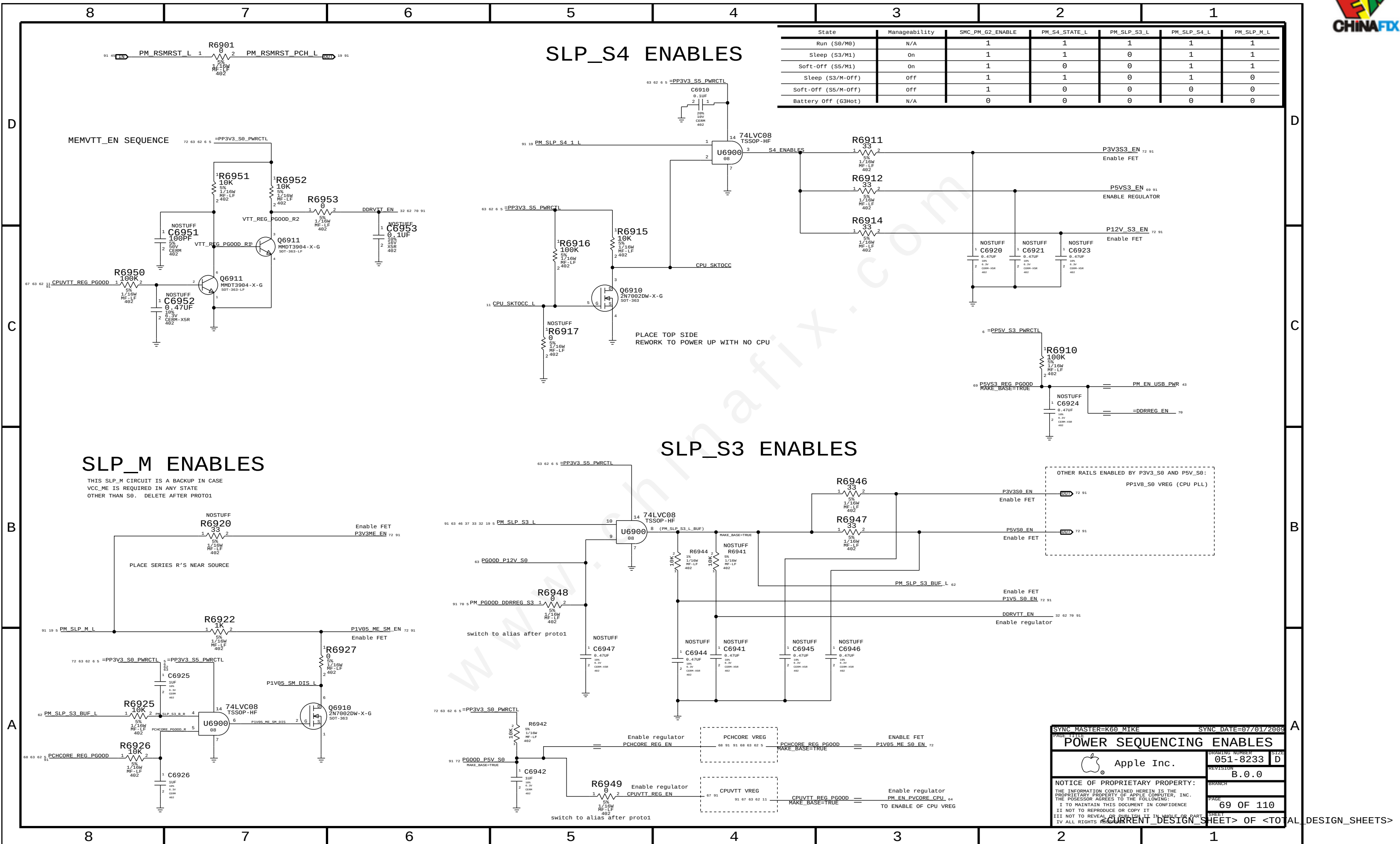
WRITE: 0X72 READ: 0X73 APN 353S2256

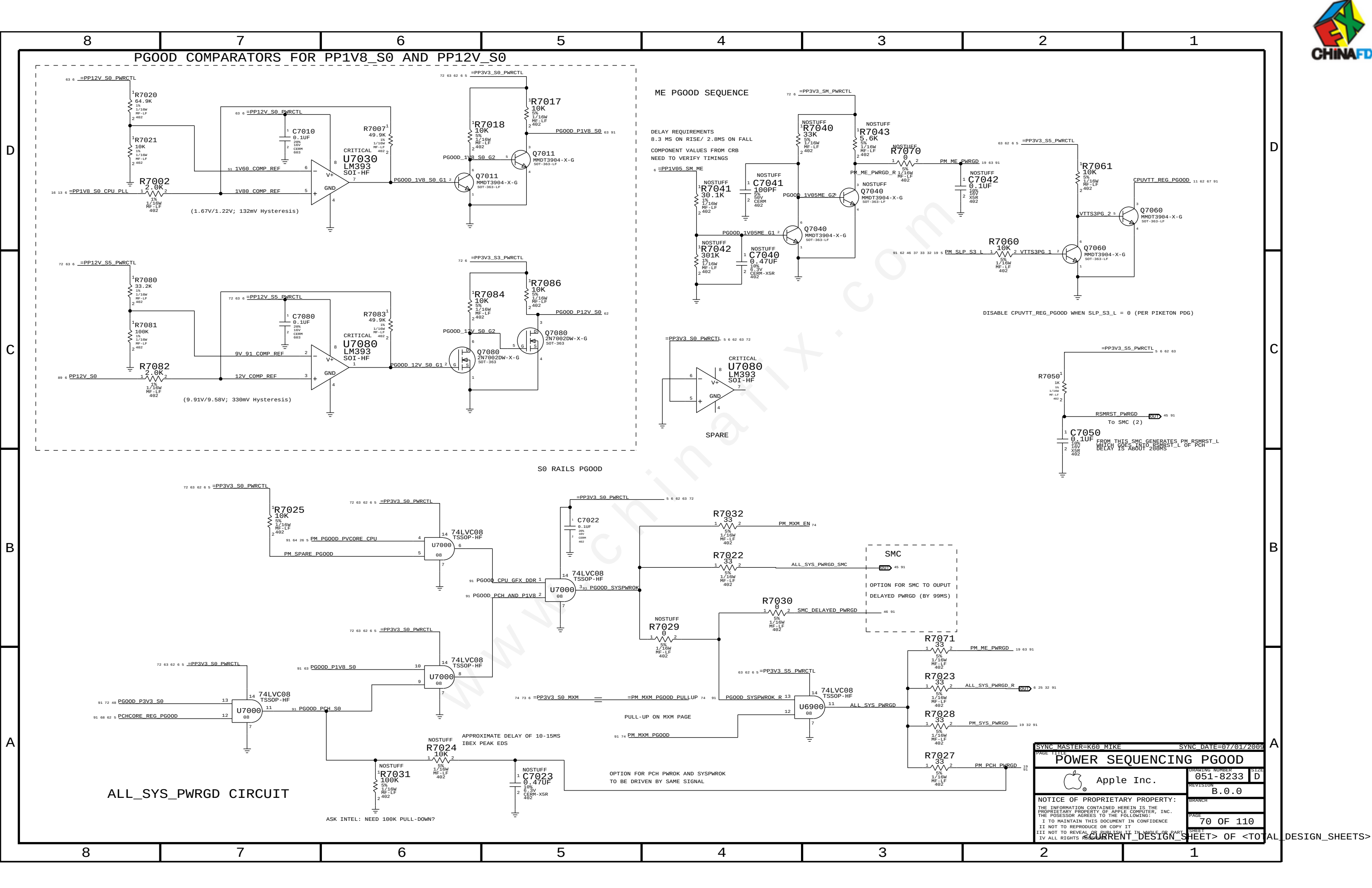


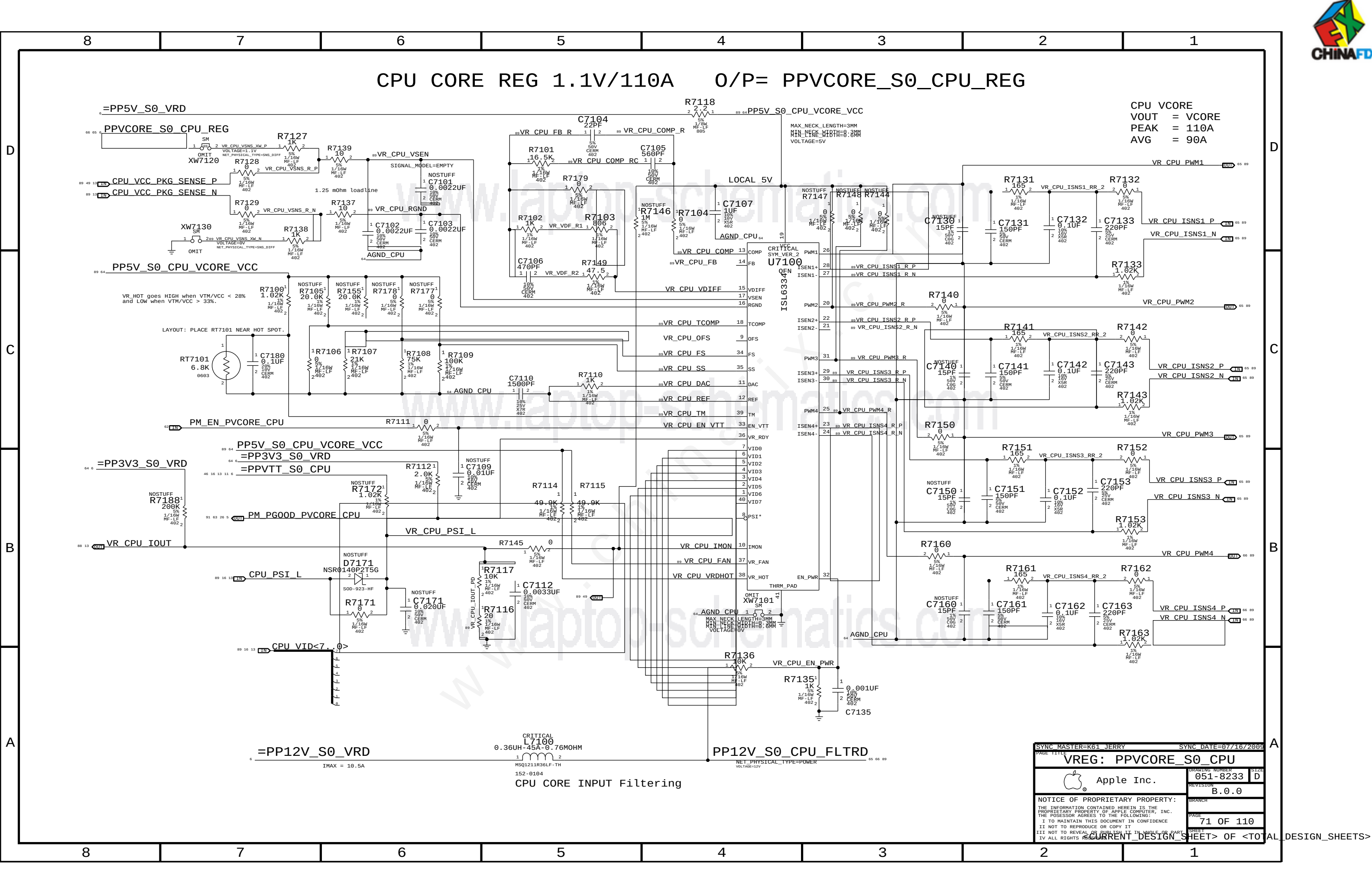
FLP = 8.82 KHZ
FHP = 80 HZ

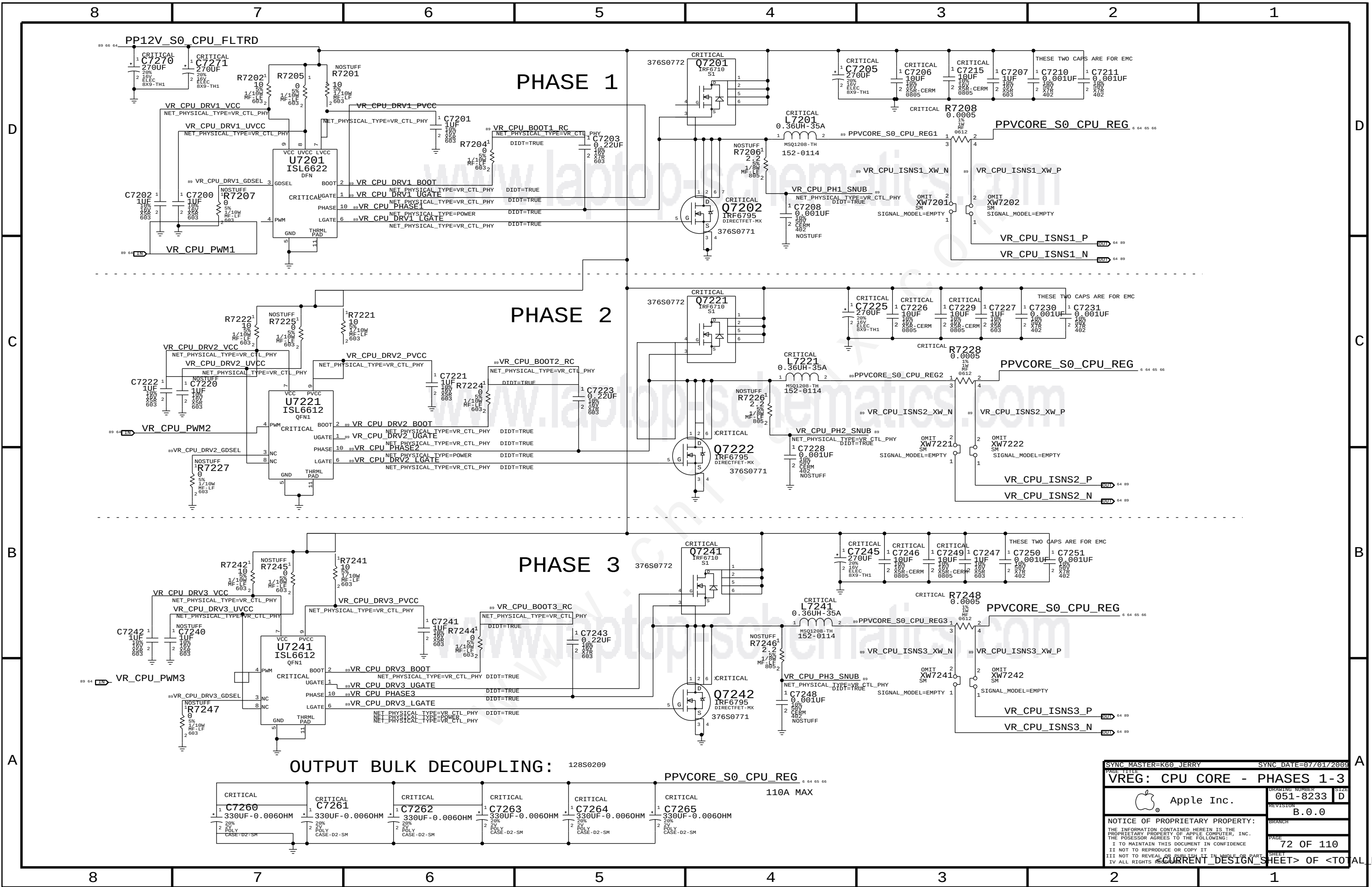
SYNC MASTER=K23 SKIP		SYNC DATE=07/16/2009	
PAGE TITLE			
AUDIO: Mikey			
DRAWING NUMBER		SIZE	
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REVISION		B.0.0	
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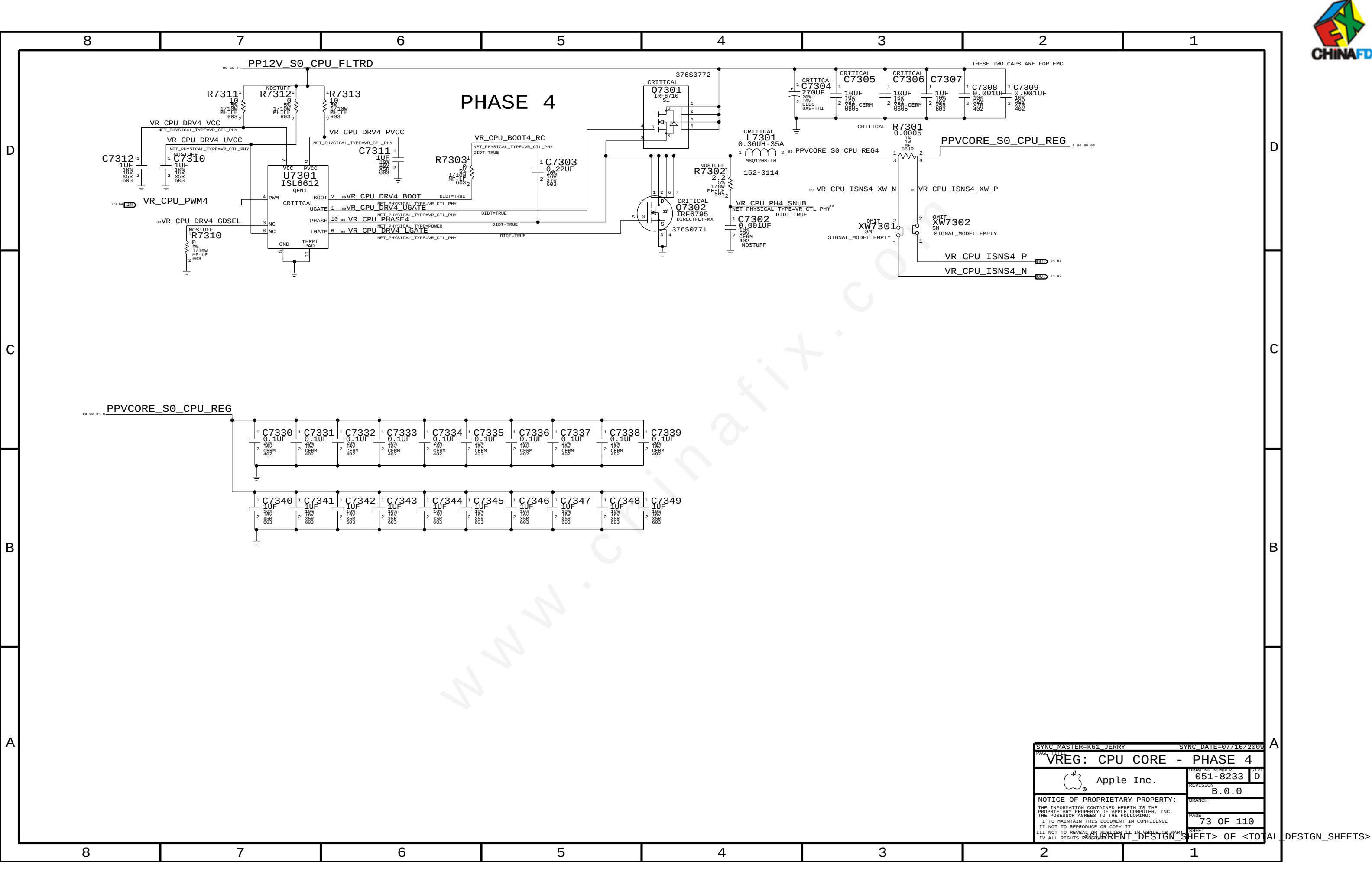






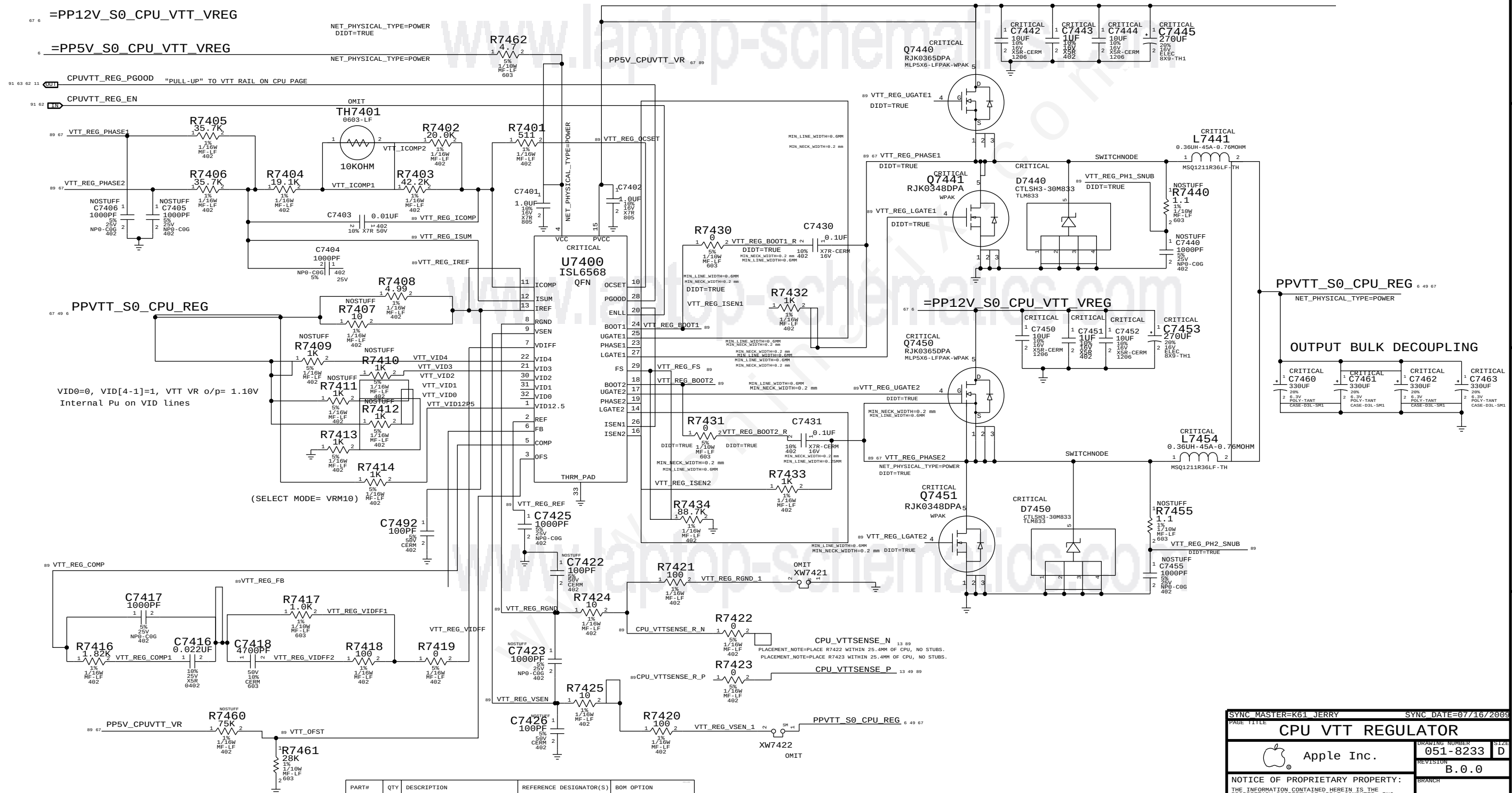
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SYNC MASTER=K60 JERRY		SYNC DATE=07/01/2009	
PAGE TITLE			
VREG: CPU CORE -		PHASES 1-3	
 Apple Inc.		DRAWING NUMBER	051-8233
		SIZE	D
		REVISION	B.0.0
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REGULATORY DESIGN SHEET		SHEET	
> OF		< TOTAL	



O/P= PPVTT_S0_CPU_REG

VOUT = 1.1V OR 1.05V
PEAK = 35A
AVG = 30A



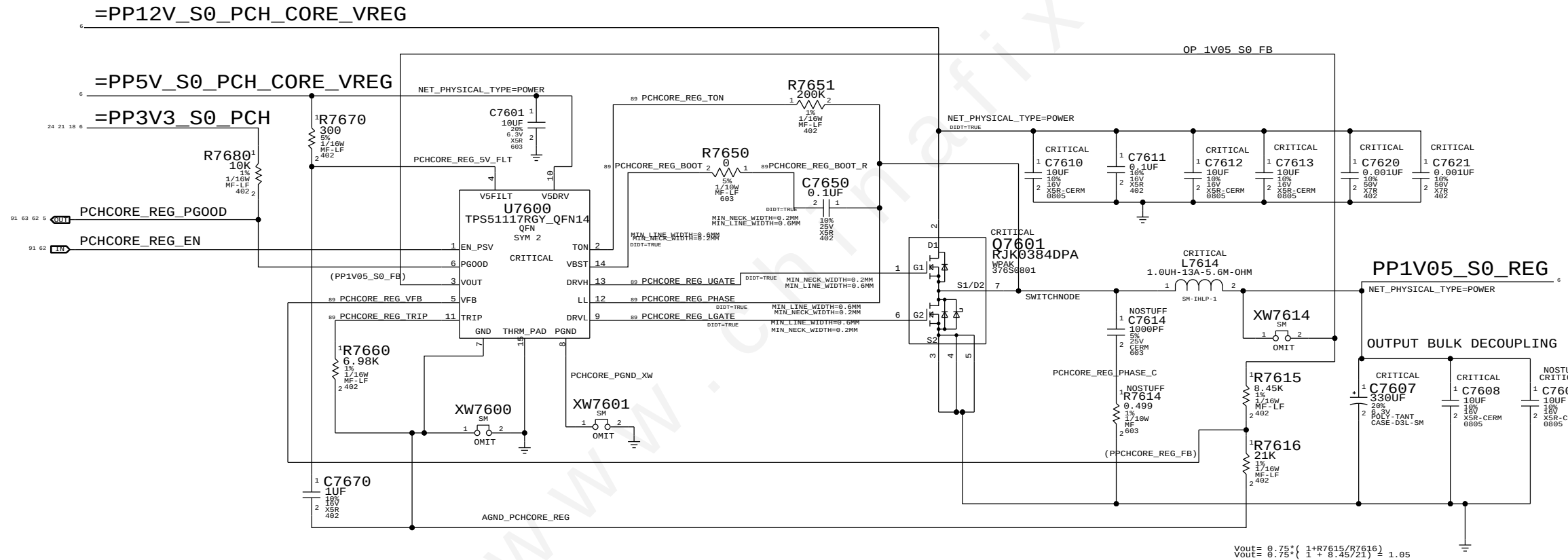
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
113S0127	1	RES, 68k, 0603, 5%	TH7401	

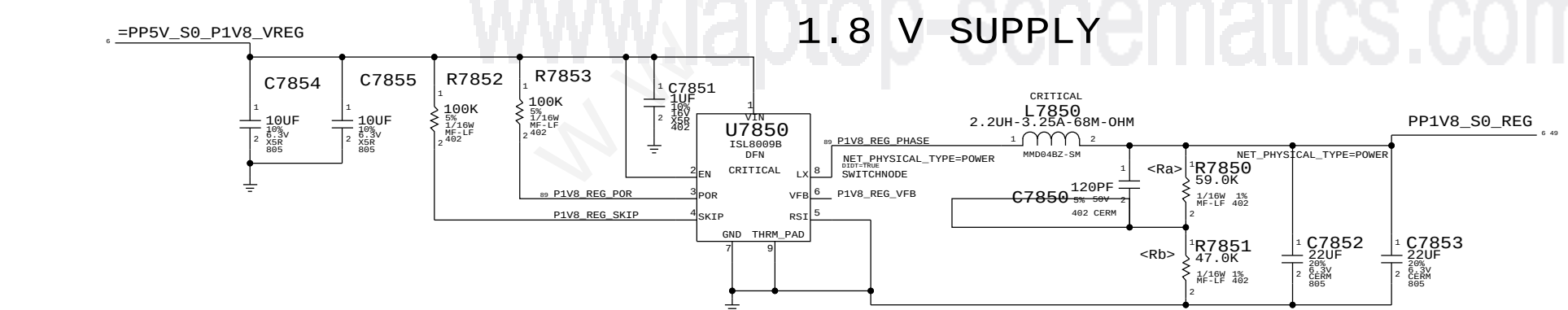
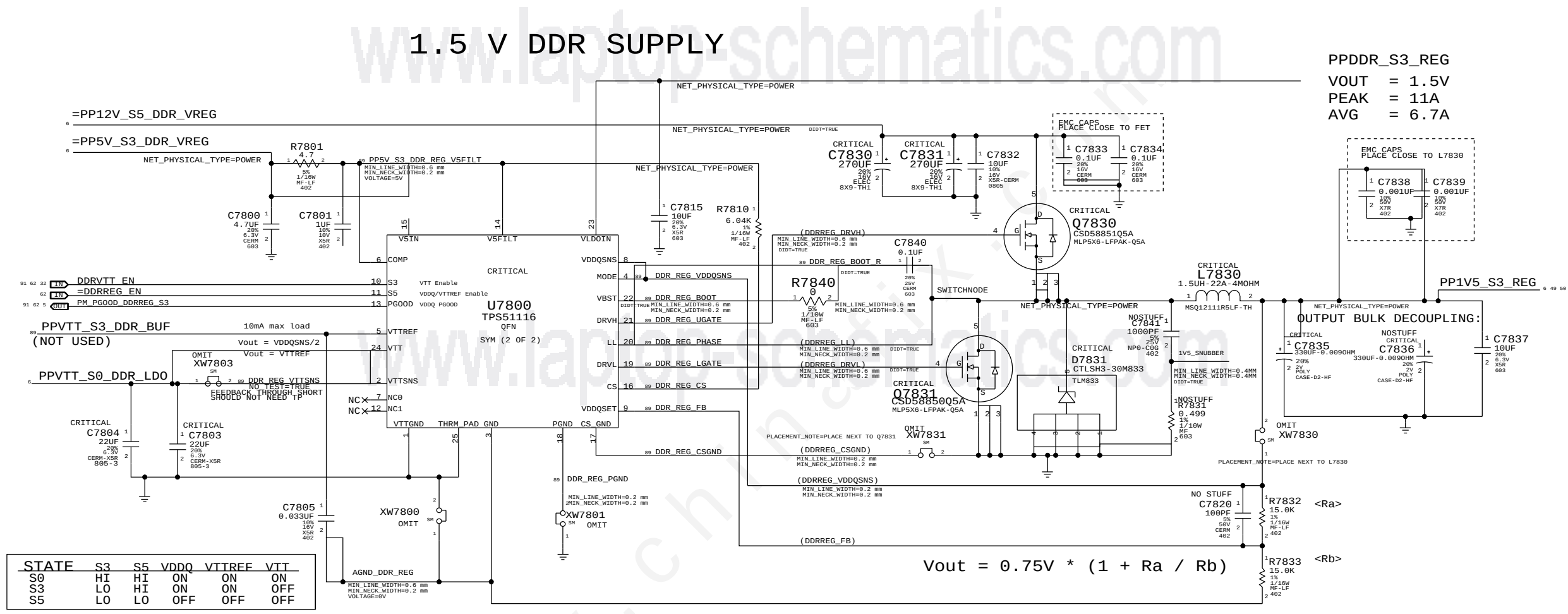
SYNC MASTER=K61 JERRY		SYNC DATE=07/16/2009	
PAGE TITLE			
CPU VTT REGULATOR			
 Apple Inc.		DRAWING NUMBER 051-8233	
		SIZE D	
		REVISION B.0.0	
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IBEX PEAK CORE REG 1.05V OUTPUT = PP1V05_S0_REG

PP1V05_S0_REG
VOUT = 1.05V
PEAK = 7.5A
AVG = 3A





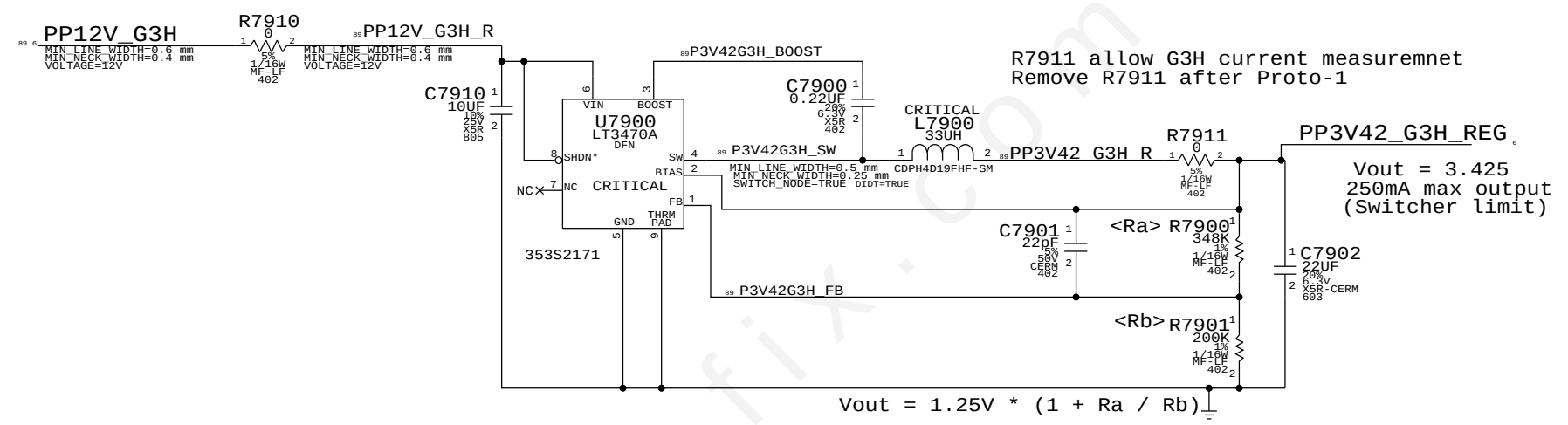
$$V_o = 0.8 * (1 + R_a/R_b)$$
$$V_o = 0.8 * (1 + 59/47) = 1.804V$$

SYNC MASTER=K60 JERRY		SYNC DATE=07/01/2009	
PAGE TITLE			
1.5V / 1.8V VREGS			
 Apple Inc.	DRAWING NUMBER		SIZE
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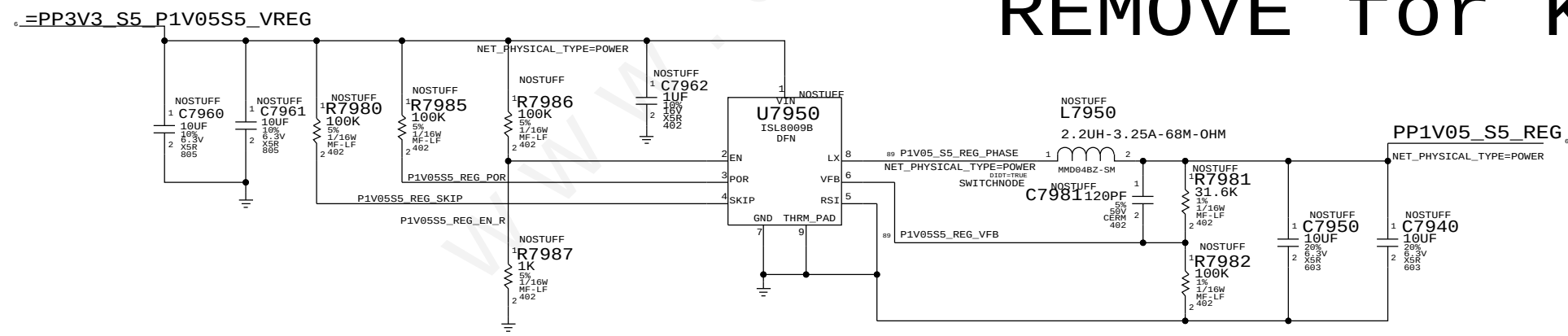
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

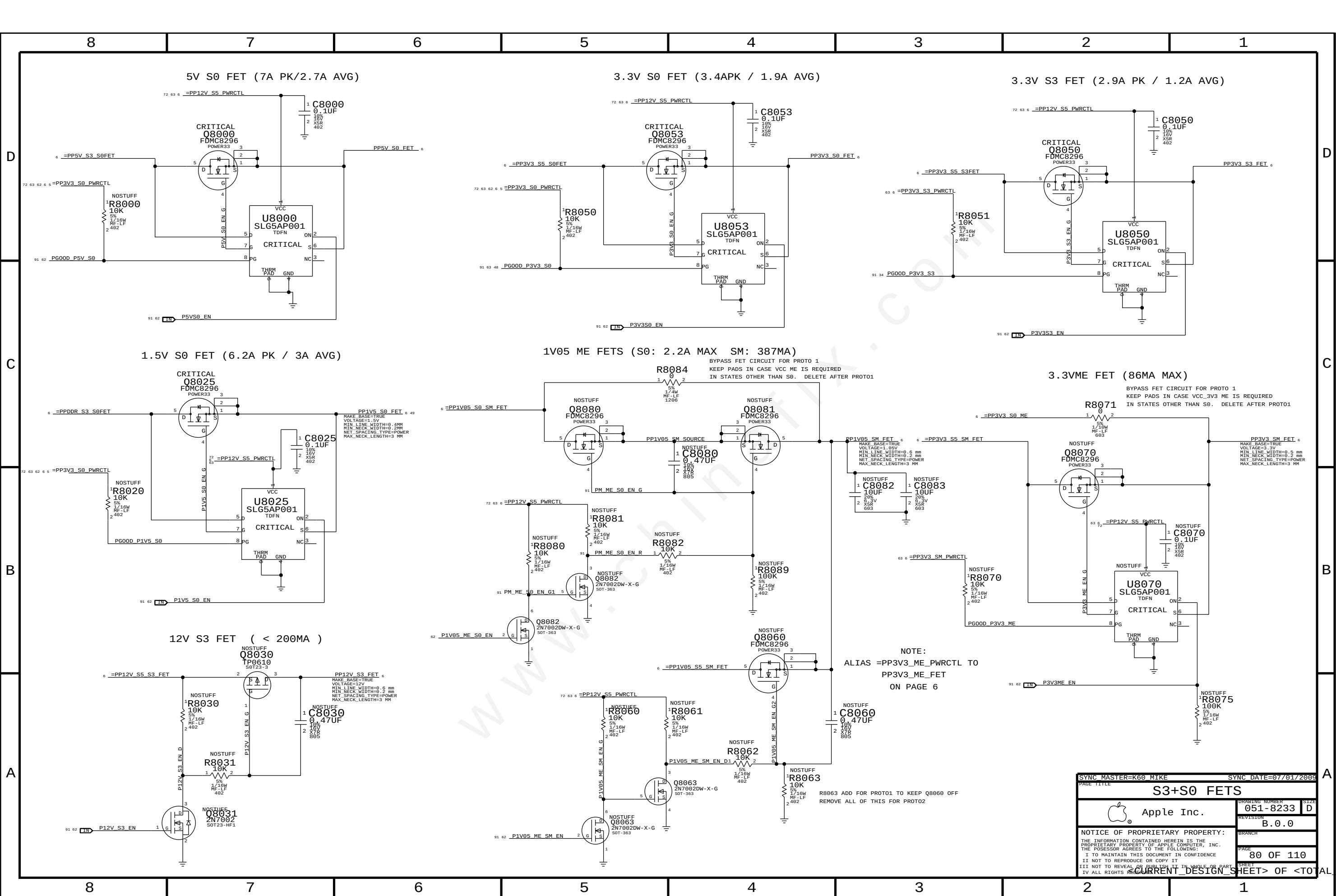


1.05V S5 SUPPLY

REMOVE for K60/K61



SYNC MASTER=K60 JERRY		SYNC DATE=07/01/2009	
PAGE TITLE			
1.05 S5 SUPPLY			
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		1	OF 1



Page Notes

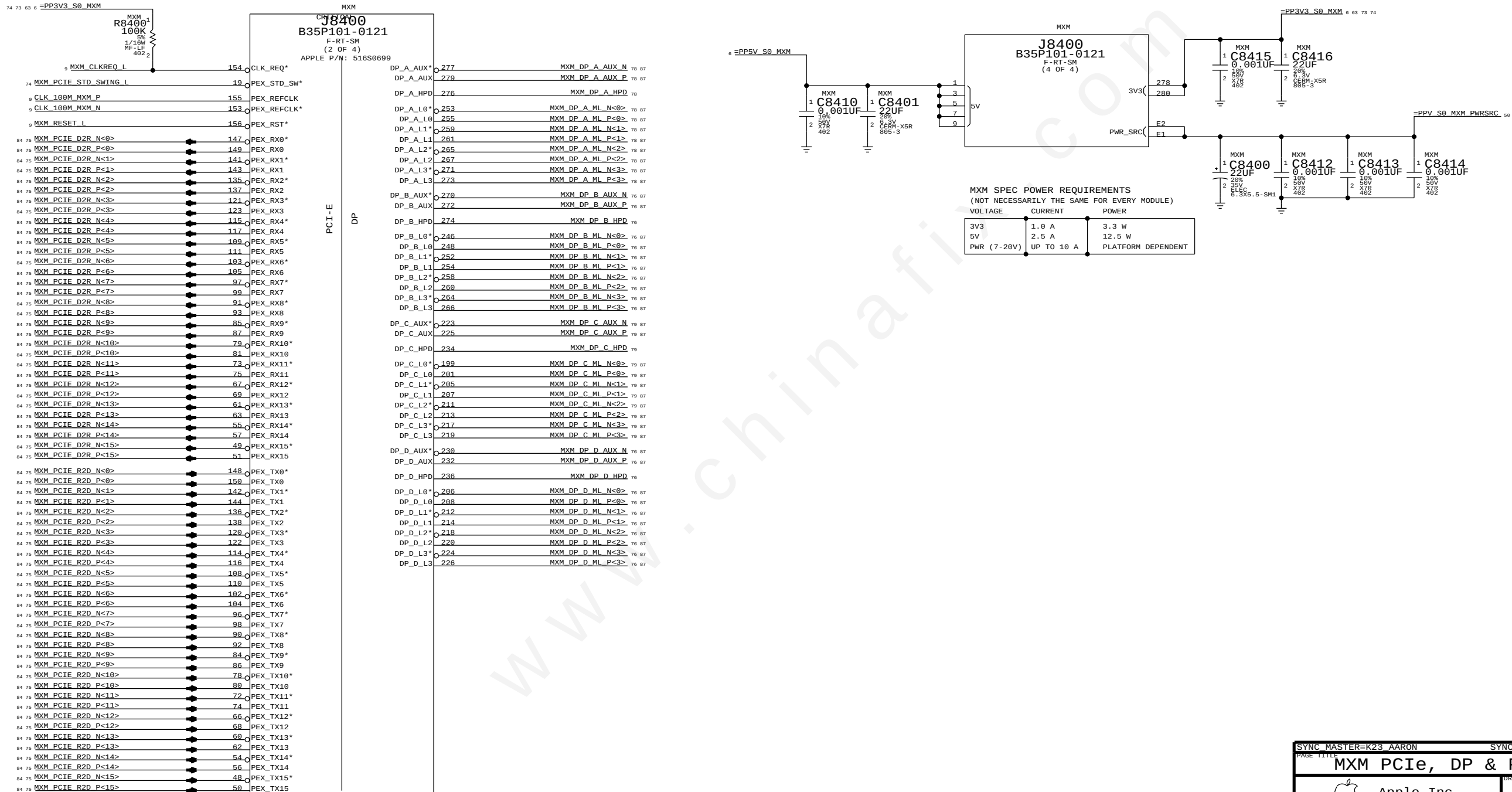
Power aliases required by this page:

- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

- MXM



PAGE TITLE		SYNC DATE=07/16/2009	
MXM PCIe, DP & Power			
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Page Notes

Power aliases required by this page:

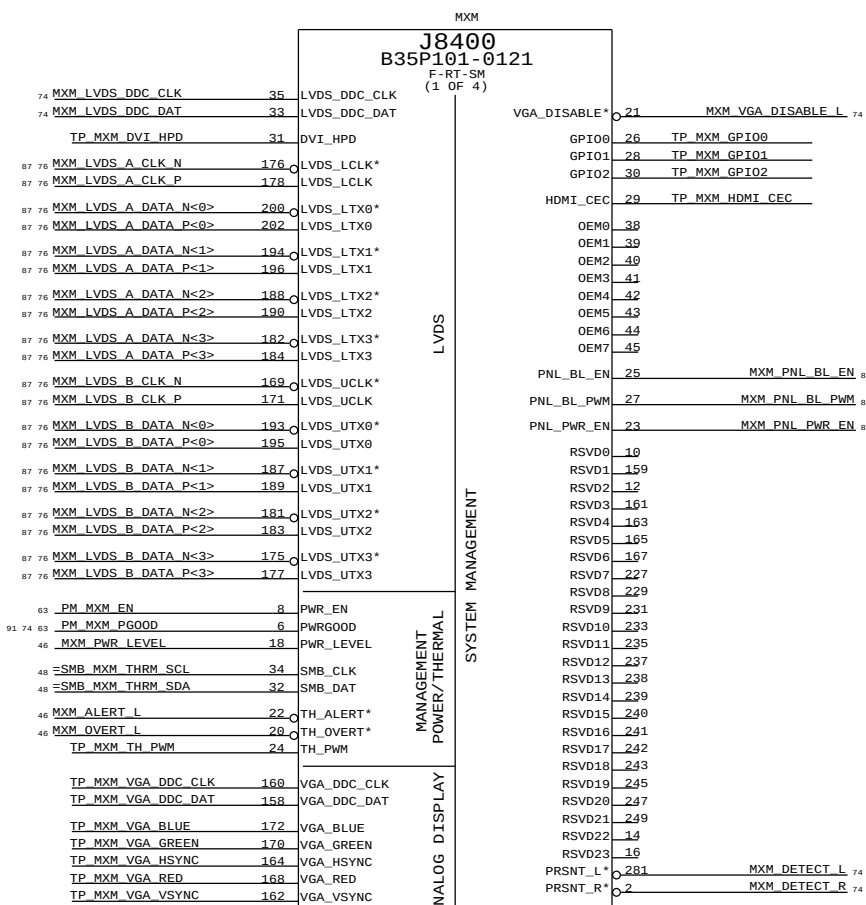
- =PP3V3_S0_MXM

Signal aliases required by this page:

- =SMB_MXM_THRM_DATA - =PM_MXM_PG00D_PULLUP
- =SMB_MXM_THRM_CLK

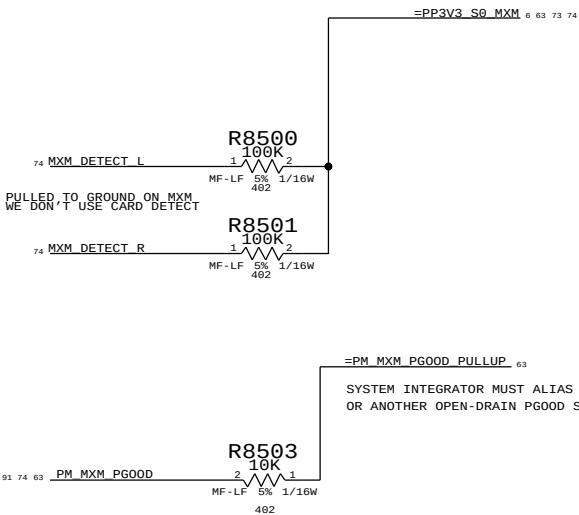
BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



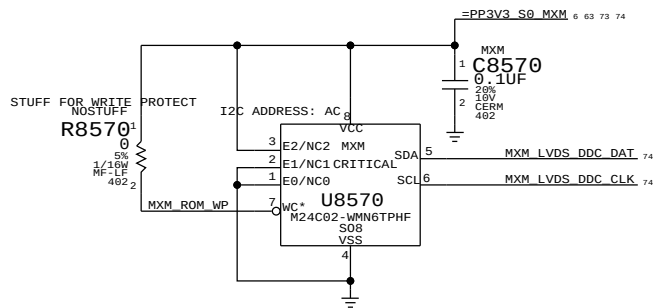
FLOAT = NORMAL VGA MODE
GND = SECONDARY DISPLAY CARD

FLOAT = LOW SWING
GND = HIGH SWING



MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J7800



SYNC MASTER=K23 AARON SYNC DATE=07/16/2009

PAGE TITLE

MXM I/O

Apple Inc.

051-8233 D

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MXM TX CAPS

84 9	PEG_R2D_C_P<0>	MXM C8600 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<15>	73 84
84 9	PEG_R2D_C_N<0>	MXM C8601 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_N<15>	73 84
84 9	PEG_R2D_C_N<1>	MXM C8602 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<14>	73 84
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84 9	PEG_R2D_C_N<2>	MXM C8604 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<13>	73 84
84 9	PEG_R2D_C_P<2>	MXM C8605 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_N<13>	73 84
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84 9	PEG_R2D_C_N<6>	MXM C8613 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_N<9>	73 84
84 9	PEG_R2D_C_N<7>	MXM C8614 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<8>	73 84
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MXM RX CAPS

84 73	MXM_PCIE_D2R_P<15>	MXM C8632 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<0>	9 84
84 73	MXM_PCIE_D2R_N<15>	MXM C8633 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_P<0>	9 84
84 73	MXM_PCIE_D2R_P<14>	MXM C8634 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<1>	9 84
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84 73	MXM_PCIE_D2R_P<4>	MXM C8654 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<11>	9 84
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84 73	MXM_PCIE_D2R_N<3>	MXM C8657 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_P<12>	9 84
84 73	MXM_PCIE_D2R_P<2>	MXM C8658 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<13>	9 84
84 73	MXM_PCIE_D2R_N<2>	MXM C8659 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_P<13>	9 84
84 73	MXM_PCIE_D2R_P<1>	MXM C8662 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<14>	9 84
84 73	MXM_PCIE_D2R_N<1>	MXM C8663 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_P<14>	9 84
84 73	MXM_PCIE_D2R_P<0>	MXM C8660 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<15>	9 84
84 73	MXM_PCIE_D2R_N<0>	MXM C8661 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_P<15>	9 84

SYNC MASTER=K60 AARON		SYNC DATE=07/01/2009	
PAGE TITLE			
MXM PCIE CAPS			
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REVISION		B.0.0	
BRANCH			
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CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			



Page Notes

Power aliases required by this page:
- =PP3V3_S0_DP

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Unused MXM Interfaces

87 74	MXM LVDS A CLK N	==	NC MXM LVDS A CLK N	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A CLK P	==	NC MXM LVDS A CLK P	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA N<0>	==	NC MXM LVDS A DATA N<0>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA P<0>	==	NC MXM LVDS A DATA P<0>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA N<1>	==	NC MXM LVDS A DATA N<1>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA P<1>	==	NC MXM LVDS A DATA P<1>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA N<2>	==	NC MXM LVDS A DATA N<2>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA P<2>	==	NC MXM LVDS A DATA P<2>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA N<3>	==	NC MXM LVDS A DATA N<3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS A DATA P<3>	==	NC MXM LVDS A DATA P<3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B CLK N	==	NC MXM LVDS B CLK N	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B CLK P	==	NC MXM LVDS B CLK P	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA N<0>	==	NC MXM LVDS B DATA N<0>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA P<0>	==	NC MXM LVDS B DATA P<0>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA N<1>	==	NC MXM LVDS B DATA N<1>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA P<1>	==	NC MXM LVDS B DATA P<1>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA N<2>	==	NC MXM LVDS B DATA N<2>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA P<2>	==	NC MXM LVDS B DATA P<2>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA N<3>	==	NC MXM LVDS B DATA N<3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM LVDS B DATA P<3>	==	NC MXM LVDS B DATA P<3>	==	MAKE_BASE=TRUE NO_TEST=TRUE

Unused MXM DP Interfaces

87 73	MXM DP B MI P<0..3>	==	NC MXM DP B MI P<0..3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP B MI N<0..3>	==	NC MXM DP B MI N<0..3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP B AUX P	==	NC MXM DP B AUX P	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP B AUX N	==	NC MXM DP B AUX N	==	MAKE_BASE=TRUE NO_TEST=TRUE
73	MXM DP B HPD	==	NC MXM DP B HPD	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP D MI P<0..3>	==	NC MXM DP D MI P<0..3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP D MI N<0..3>	==	NC MXM DP D MI N<0..3>	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP D AUX P	==	NC MXM DP D AUX P	==	MAKE_BASE=TRUE NO_TEST=TRUE
87 73	MXM DP D AUX N	==	NC MXM DP D AUX N	==	MAKE_BASE=TRUE NO_TEST=TRUE
73	MXM DP D HPD	==	NC MXM DP D HPD	==	MAKE_BASE=TRUE NO_TEST=TRUE

Display: Aliases

SYNC_MASTER=K61_AARON SYNC_DATE=07/01/2009

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D	051-7997	3.0.0
SCALE	SHT	OF
NONE	76	92



Page Notes

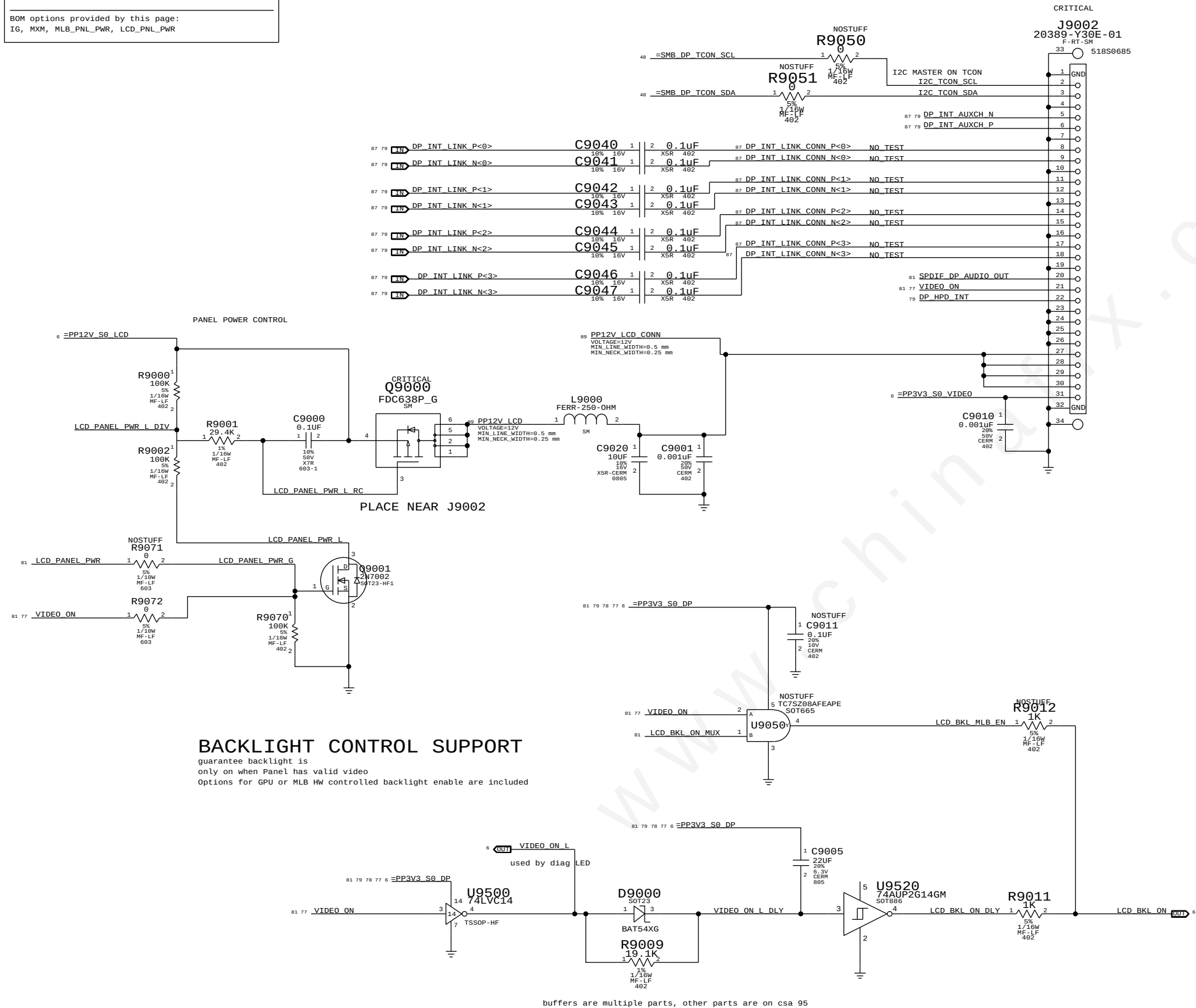
Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE



SYNC MASTER=K23 AARON SYNC DATE=07/16/2009

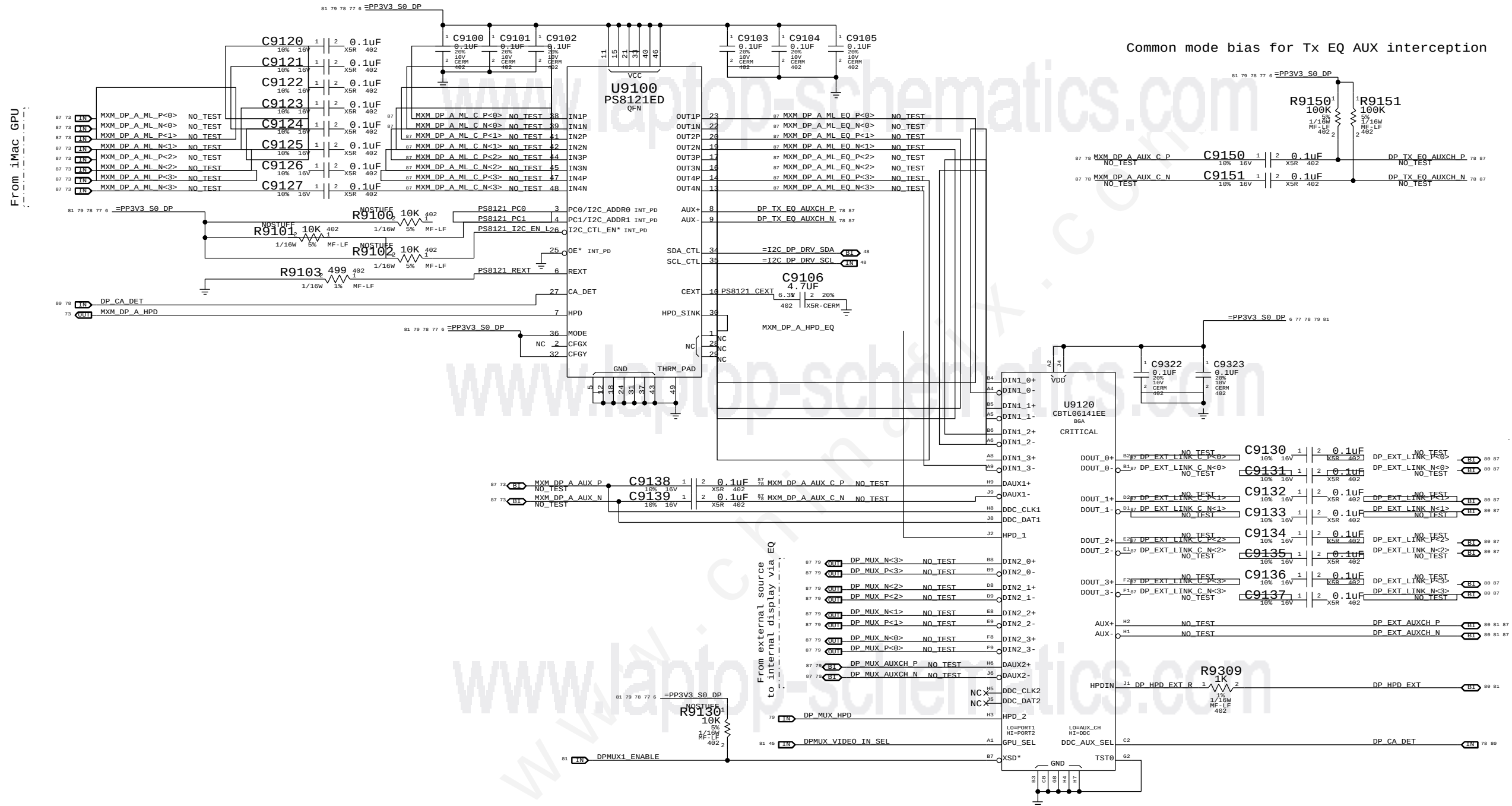
PAGE TITLE
Display: Int DP Connector

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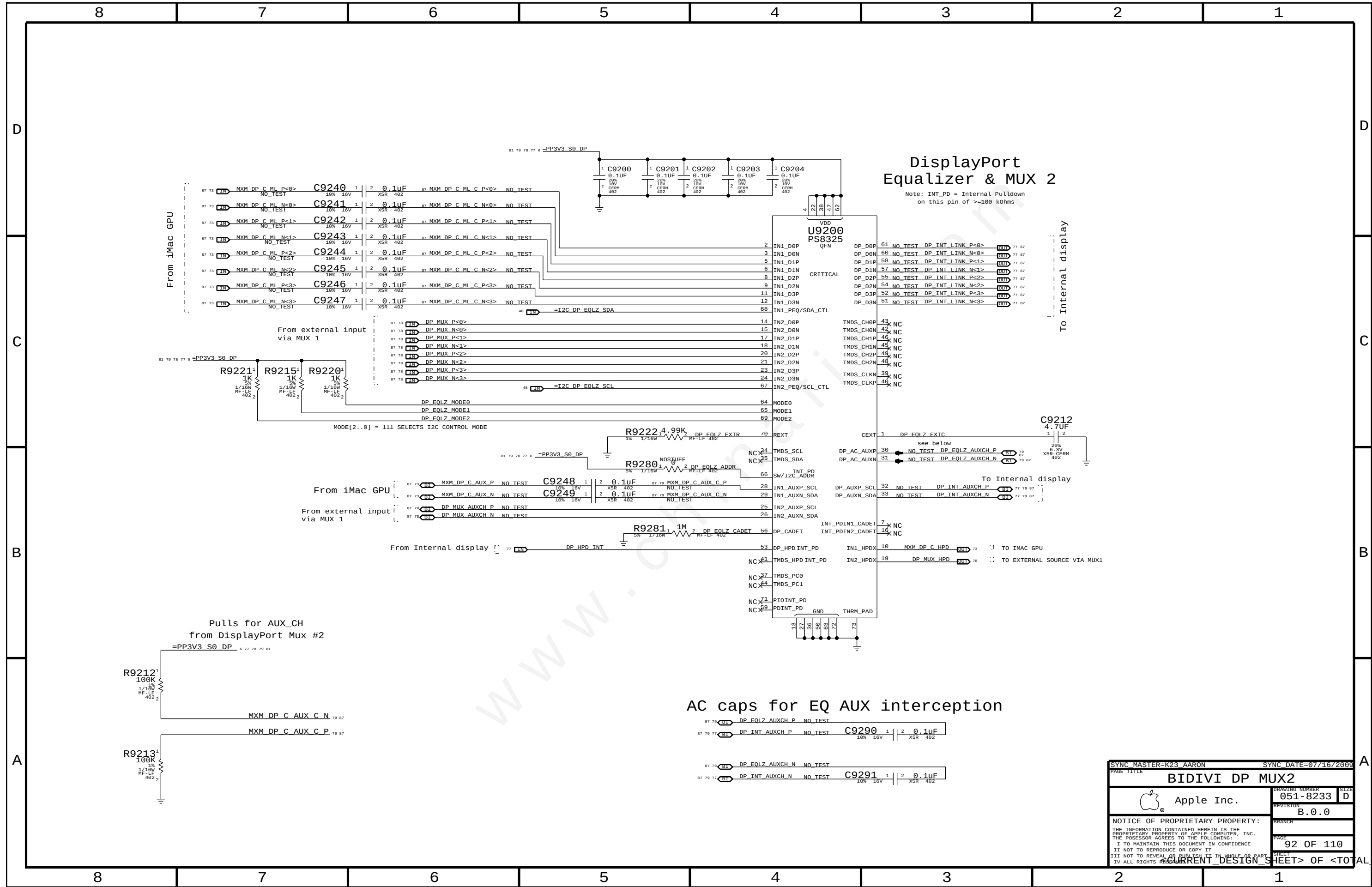
EQ & Re-Driver for DP source

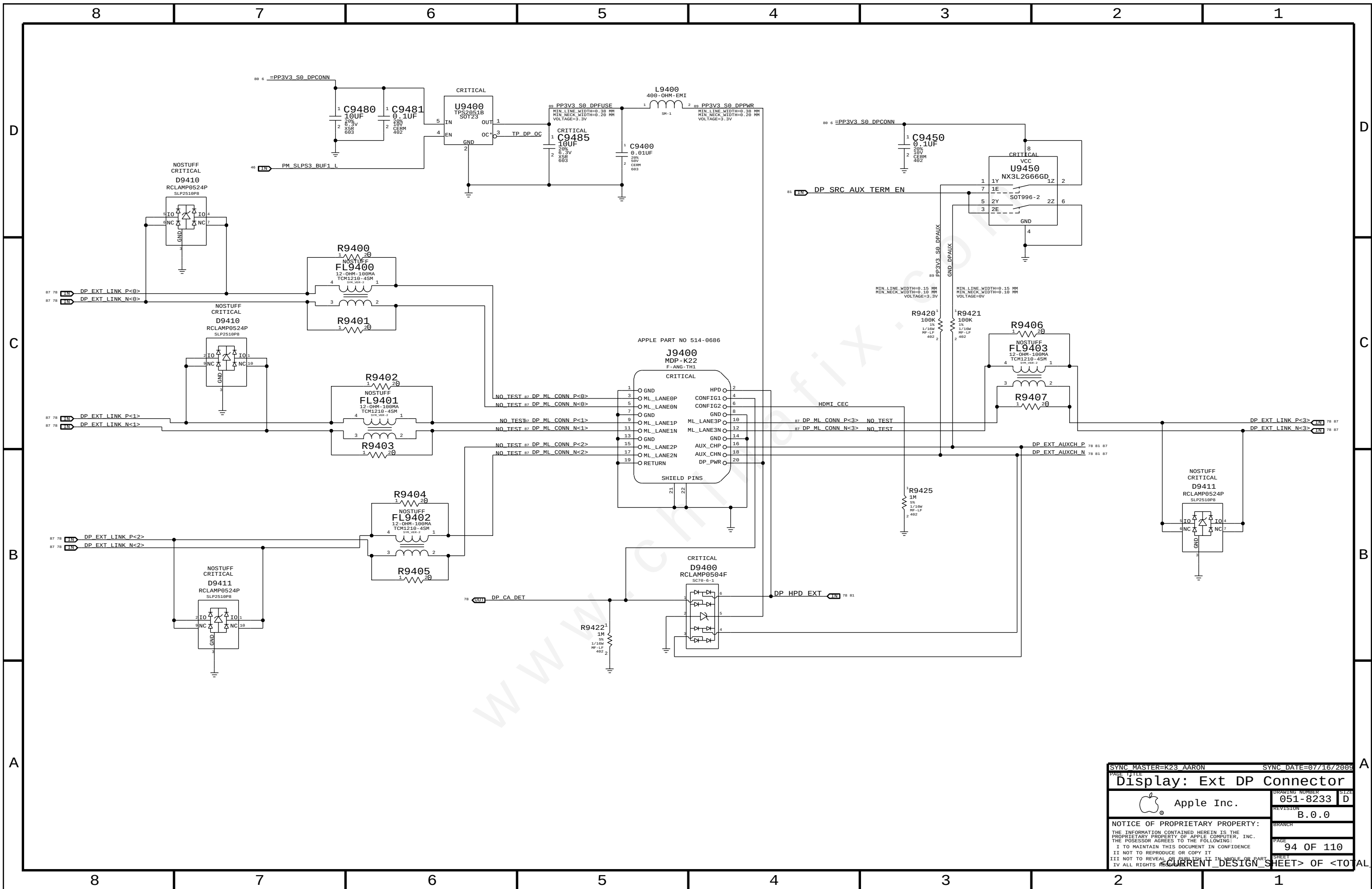
Common mode bias for Tx EQ AUX interception

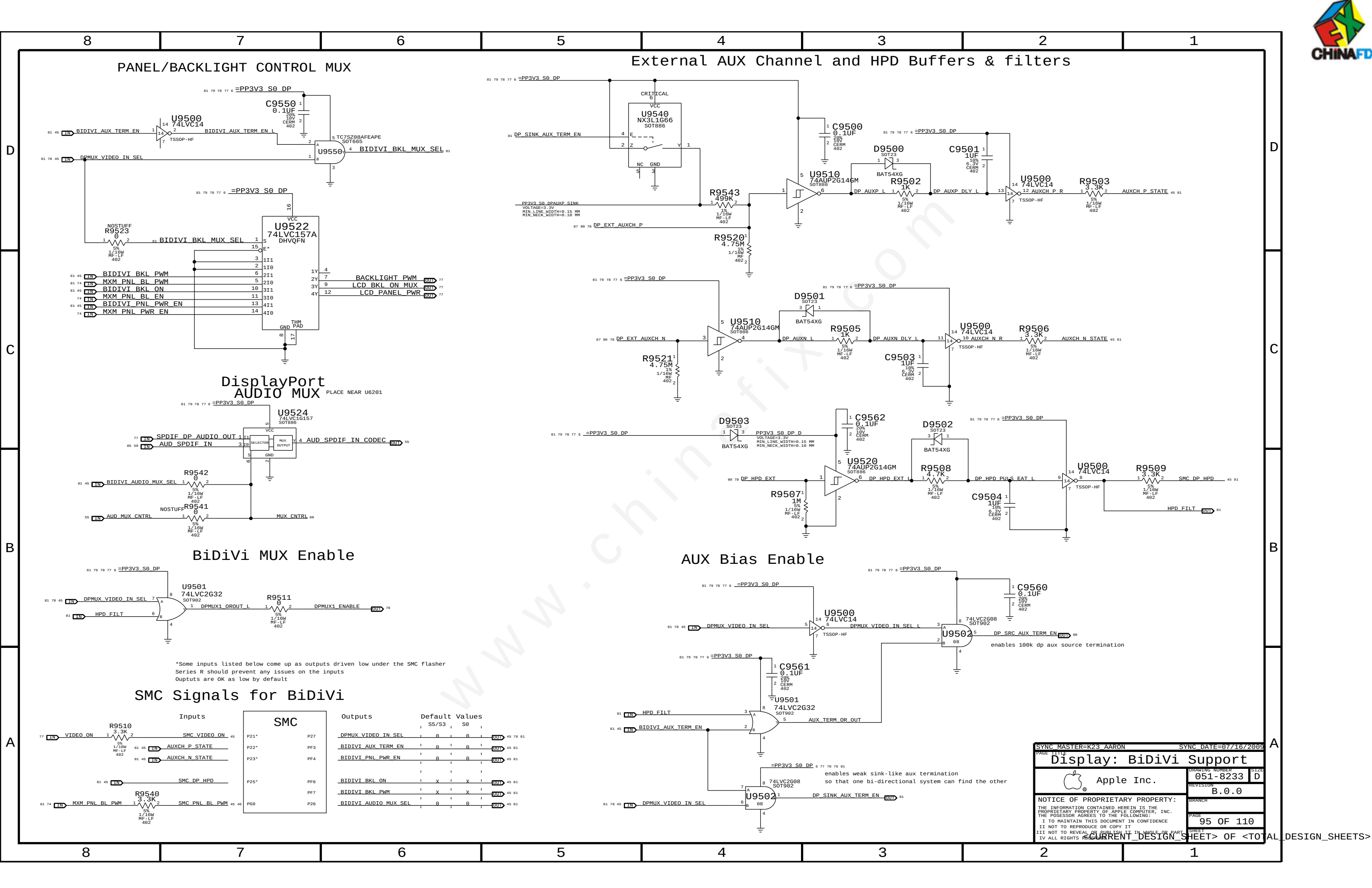


DisplayPort Mux 1
Analog mux at External Connector

SYNC MASTER=K23 AARON		SYNC DATE=07/16/2009	
PAGE TITLE		Display: BiDiVi Mux1	
DRAWING NUMBER		051-8233	
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87654321

K60/K61 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS				BOARD AREAS		BOARD UNITS (MIL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM				NO_TYPE, BGA_P1MM		MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
35_OHM_SE	TOP, BOTTOM	Y	0.21 MM	0.085 MM	=STANDARD		
35_OHM_SE	*	Y	0.19 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD		
39_OHM_SE	*	Y	0.16 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD		
45_OHM_SE	*	Y	0.12 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3, ISL6	Y	0.155 MM	0.085 MM	=STANDARD	0.135 MM	0.1 MM
70_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL6	Y	0.099 MM	0.085 MM	12 MM	0.200 MM	0.1 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL6	Y	0.081 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.085 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.150 MM	?
2X_DIELECTRIC	TOP, BOTTOM	0.160 MM	?
3X_DIELECTRIC	*	0.220 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.240 MM	?
4X_DIELECTRIC	*	0.300 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.320 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.400 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

K60/K61 RULE DEFINITIONS

SYNC_MASTER=K60_DEREK SYNC_DATE=07/01/2009

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NONE	82	92

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<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW_ROUTE_ON_LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>MEM_45S</td><td>*</td><td>=45_OHM_SE</td><td>=45_OHM_SE</td><td>=45_OHM_SE</td><td>=45_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>MEM_39S</td><td>*</td><td>=39_OHM_SE</td><td>=39_OHM_SE</td><td>=39_OHM_SE</td><td>=39_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>MEM_35S</td><td>*</td><td>=35_OHM_SE</td><td>=35_OHM_SE</td><td>=35_OHM_SE</td><td>=35_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>MEM_70D</td><td>*</td><td>=70_OHM_DIFF</td><td>=70_OHM_DIFF</td><td>=70_OHM_DIFF</td><td>=70_OHM_DIFF</td><td>=70_OHM_DIFF</td><td>=70_OHM_DIFF</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD	MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD	MEM_35S	*	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=STANDARD	=STANDARD	MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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32	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<4>	12 32	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<47..40>	12 32	MEM_45S	MEM_DQ_ODD	MEM_A_DM<5>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<55..48>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<6>	12 32	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<63..56>	12 32	MEM_45S	MEM_DQ_ODD	MEM_A_DM<7>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<0>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<0>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<1>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<1>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<2>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<2>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<3>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<3>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<4>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<4>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<5>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<5>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<6>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<6>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_P<7>	12 32	MEM_70D	MEM_DQS	MEM_A_DQS_N<7>	12 32	MEM_70D	MEM_CLK	MEM_B_CLK_P<3..0>	12 32	MEM_70D	MEM_CLK	MEM_B_CLK_N<3..0>	12 32	MEM_39S	MEM_CTRL	MEM_B_CKE<3..0>	12 31	MEM_39S	MEM_CTRL	MEM_B_CS_L<3..0>	12 31	MEM_39S	MEM_CTRL	MEM_B_ODT<3..0>	12 31	MEM_35S	MEM_CMD	MEM_B_A<15..0>	12 31	MEM_35S	MEM_CMD	MEM_B_BA<2..0>	12 31	MEM_35S	MEM_CMD	MEM_B_RAS_L	12 31	MEM_35S	MEM_CMD	MEM_B_CAS_L	12 31	MEM_35S	MEM_CMD	MEM_B_WE_L	12 31	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<7..0>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<0>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<15..8>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DM<1>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<23..16>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<2>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<31..24>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DM<3>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<39..32>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<4>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<47..40>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DM<5>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<55..48>	12 32	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<6>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<63..56>	12 32	MEM_45S	MEM_DQ_ODD	MEM_B_DM<7>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<0>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<1>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<1>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<2>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<2>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<3>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<3>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<4>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<4>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<5>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<5>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<6>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<6>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_P<7>	12 32	MEM_70D	MEM_DQS	MEM_B_DQS_N<7>	12 32	MEM_70D	MEM_DQS	CPU_SM_RCOMP0	11	MEM_70D	MEM_DQS	CPU_SM_RCOMP1	11	MEM_70D	MEM_DQS	CPU_SM_RCOMP2	11	MEM_70D	MEM_DQS	TP MEM B DQS_P<8>	8 12	MEM_70D	MEM_DQS	TP MEM B DQS_N<8>	8 12	MEM_70D	MEM_DQS	TP MEM A DQS_P<8>	8 12	MEM_70D	MEM_DQS	TP MEM A DQS_N<8>	8 12	ADD RULES TO NC_DQS<8> TO CLEAR CHECK_PLUS ERRORS								Memory Net Properties								<table><tr><th rowspan="2">ELECTRICAL_CONSTRAINT_SET</th><th 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32	MEM_70D	MEM_DQS	CPU_SM_RCOMP0	11	MEM_70D	MEM_DQS	CPU_SM_RCOMP1	11	MEM_70D	MEM_DQS	CPU_SM_RCOMP2	11	MEM_70D	MEM_DQS	TP MEM B DQS_P<8>	8 12	MEM_70D	MEM_DQS	TP MEM B DQS_N<8>	8 12	MEM_70D	MEM_DQS	TP MEM A DQS_P<8>	8 12	MEM_70D	MEM_DQS	TP MEM A DQS_N<8>	8 12	ADD RULES TO NC_DQS<8> TO CLEAR CHECK_PLUS ERRORS								Memory Net Properties								<table><tr><th rowspan="2">ELECTRICAL_CONSTRAINT_SET</th><th colspan="2">NET_TYPE</th></tr><tr><th>PHYSICAL</th><th>SPACING</th></tr><tr><td>MEM_70D</td><td>MEM_DQS</td><td>MEM_B_DQS_P<0></td><td>12 32</</td></tr></table>								ELECTRICAL_CONSTRAINT_SET	NET_TYPE		PHYSICAL	SPACING	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	12 32</
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87654321

PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_PCH	*	0.2 MM	?
COMP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
XTAL	*	=4x_DIELECTRIC	?

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
	PCI_55S	PCI	PCI_REQ0 L
	PCI_55S	PCI	PCI_REQ1 L
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN
	LPC_55S	LPC	LPC_AD<3...0>
	LPC_55S	LPC	LPC_FRAME L
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC R
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK R
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS R
	USB_90D	USB	USB_EXTN P
	USB_90D	USB	USB_EXTN N
	USB_90D	USB	USB_EXTN P
	USB_90D	USB	USB_EXTN N
	USB_90D	USB	USB_PORT1 P
	USB_90D	USB	USB_PORT1 N
	USB_90D	USB	USB_EXTN P
	USB_90D	USB	USB_EXTN N
	USB_90D	USB	USB_PORT2 P
	USB_90D	USB	USB_PORT2 N
	USB_90D	USB	USB_EXTN P
	USB_90D	USB	USB_EXTN N
	USB_90D	USB	USB_D_MUXED P
	USB_90D	USB	USB_D_MUXED N
	USB_90D	USB	USB_PORT3 P
	USB_90D	USB	USB_PORT3 N
	USB_90D	USB	USB_CAMERA P
	USB_90D	USB	USB_CAMERA N
	USB_90D	USB	USB_CAMERA L P
	USB_90D	USB	USB_CAMERA L N
	USB_90D	USB	USB_BT P
	USB_90D	USB	USB_BT N
	USB_90D	USB	USB_BT L P
	USB_90D	USB	USB_BT L N
	USB_90D	USB	USB_IR P
	USB_90D	USB	USB_IR N
	USB_90D	USB	USB_IR L P
	USB_90D	USB	USB_IR L N
	USB_90D	USB	USB_SDCARD P
	USB_90D	USB	USB_SDCARD N
	USB_90D	USB	USB_SDCARD L P
	USB_90D	USB	USB_SDCARD L N
	USB_90D	USB	USB_WM P
	USB_90D	USB	USB_WM N
	USB_90D	USB	USB_WM L P
	USB_90D	USB	USB_WM L N
	USB_90D	USB	USB_MINI P
	USB_90D	USB	USB_MINI N
	USB_90D	USB	USB_BRCRYPT P
	USB_90D	USB	USB_BRCRYPT N
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2
	CLK_XTAL	XTAL	CK505_XTAL IN
	CLK_XTAL	XTAL	CK505_XTAL OUT
	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK
	USB_90D	USB	USB_BRCRYPT L P
	USB_90D	USB	USB_BRCRYPT L N
	USB_90D	USB	USB_HUB1 UP P
	USB_90D	USB	USB_HUB1 UP N
	USB_90D	USB	USB_HUB2 UP P
	USB_90D	USB	USB_HUB2 UP N

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
	SPI_55S	SPI	SPI_CLK R
	SPI_55S	SPI	SPI_CLK
	SPI_55S	SPI	SPI_MOSI R
	SPI_55S	SPI	SPI_MOSI
	SPI_55S	SPI	SPI_MISO R
	SPI_55S	SPI	SPI_MISO
	SPI_55S	SPI	SPI_CS0 R L
	SPI_55S	SPI	SPI_CS0 L
	SPI_55S	SPI	SPI_MLB_CS L
	SPI_55S	SPI	SPI_ALT_CS L
	SPI_55S	SPI	SPIROM_USE_MLB
	SPI_55S	SPI	SPI_ALT_MOSI
	SPI_55S	SPI	SPI_ALT_MISO
	SPI_55S	SPI	SPI_ALT_CLK
	HDA_55S	HDA	HDA_BIT_CLK
	HDA_55S	HDA	HDA_BIT_CLK R
	HDA_55S	HDA	HDA_RST L
	HDA_55S	HDA	HDA_RST_R L
	HDA_55S	HDA	HDA_SDOUT
	HDA_55S	HDA	HDA_SDOUT R
	HDA_55S	HDA	HDA_SYNC
	HDA_55S	HDA	HDA_SYNC R
	HDA_55S	HDA	HDA_SPDIN
	HDA_55S	HDA	AUD_SDI_R
	HDA_55S	HDA	AUD_SPDIF_IN
	HDA_55S	HDA	AUD_SPDIF_OUT
	HDA_55S	HDA	AUD_SPDIF_CHIP
	HDA_55S	HDA	AUD_SPKR_OUTL01L_NOUT
	HDA_55S	HDA	AUD_SPKR_OUTL01L_POUT
	HDA_55S	HDA	AUD_SPKR_OUTL01R_NOUT
	HDA_55S	HDA	AUD_SPKR_OUTL01R_POUT
	HDA_55S	HDA	AUD_SPKR_OUTL02L_NOUT
	HDA_55S	HDA	AUD_SPKR_OUTL02L_POUT
	HDA_55S	HDA	AUD_SPKR_OUTL02R_NOUT
	HDA_55S	HDA	AUD_SPKR_OUTL02R_POUT
	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT
	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN
	PCH_55S	COMP_PCH	PCH_USB_RBIA5
	PCH_55S	COMP_PCH	PCH_SATAICOMP
	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP
	PCH_55S	COMP_PCH	PCH_DMI_COMP
	CLK_XTAL	XTAL	USB_HUB1_XTAL1
	CLK_XTAL	XTAL	USB_HUB1_XTAL2
	PCH_55S	COMP_PCH	USB_HUB1_RBIA5
	CLK_XTAL	XTAL	USB_HUB2_XTAL1
	CLK_XTAL	XTAL	USB_HUB2_XTAL2
	PCH_55S	COMP_PCH	USB_HUB2_RBIA5

SYNC MASTER=K60 SIJI

SYNC DATE=07/01/2009

PAGE TITLE

IBEX PEAK CONSTRAINTS

Apple Inc.

DRAWING NUMBER

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SHEET

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SMBus Interface Constraints <table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW ROUTE ON LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>SMB_55S</td><td>*</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=55_OHM_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr></table> <table><tr><th>SPACING_RULE_SET</th><th>LAYER</th><th>LINE-TO-LINE SPACING</th><th>WEIGHT</th></tr><tr><td>SMB</td><td>*</td><td>=2x_DIELECTRIC</td><td>?</td></tr></table> <table><tr><th>NET_SPACING_TYPE1</th><th>NET_SPACING_TYPE2</th><th>AREA_TYPE</th><th>SPACING_RULE_SET</th></tr><tr><td>THERMAL</td><td>*</td><td>*</td><td>4:1_SPACING</td></tr><tr><td>THERMAL</td><td>POWER</td><td>*</td><td>PWR_P2MM</td></tr><tr><td>THERMAL</td><td>GND</td><td>*</td><td>GND_P2MM</td></tr></table> <table><tr><th>NET_PHYSICAL_TYPE</th><th>AREA_TYPE</th><th>PHYSICAL_RULE_SET</th></tr><tr><td>THERM_DIFF</td><td>*</td><td>1:1_DIFFPAIR</td></tr><tr><td>SNS_DIFF</td><td>*</td><td>1:1_DIFFPAIR</td></tr></table>				PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SMB	*	=2x_DIELECTRIC	?	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	THERMAL	*	*	4:1_SPACING	THERMAL	POWER	*	PWR_P2MM	THERMAL	GND	*	GND_P2MM	NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET	THERM_DIFF	*	1:1_DIFFPAIR	SNS_DIFF	*	1:1_DIFFPAIR	SMC SMBus Net Properties <table><tr><th colspan="3">NET_TYPE</th><th></th></tr><tr><th>ELECTRICAL_CONSTRAINT_SET</th><th>PHYSICAL</th><th>SPACING</th><th></th></tr><tr><td>LE57</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC A S3_SCL 48</td></tr><tr><td>LE58</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC A S3_SDA 48</td></tr><tr><td>LE59</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC B S0_SCL 48</td></tr><tr><td>LE60</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC B S0_SDA 48</td></tr><tr><td>LE61</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC 0 S0_SCL 48</td></tr><tr><td>LE62</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC 0 S0_SDA 48</td></tr><tr><td>LE63</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC BSA_SCL 48</td></tr><tr><td>LE64</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC BSA_SDA 48</td></tr><tr><td>LE65</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC MGMT_SCL 48 88</td></tr><tr><td>LE66</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC MGMT_SDA 48 88</td></tr><tr><td>LE67</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC MGMT_SCL 48 88</td></tr><tr><td>LE68</td><td>SMB_55S</td><td>SMB</td><td>SMBUS SMC MGMT_SDA 48 88</td></tr><tr><td>LE69</td><td>SMB_55S</td><td>SMB</td><td>SMBUS PCH S0_CLK 48</td></tr><tr><td>LE70</td><td>SMB_55S</td><td>SMB</td><td>SMBUS PCH S0_DATA 48</td></tr><tr><td>LE71</td><td>SMB_55S</td><td>SMB</td><td>SMBUS PCH_CLK 18 48</td></tr><tr><td>LE72</td><td>SMB_55S</td><td>SMB</td><td>SMBUS PCH_DATA 18 48</td></tr><tr><td>LE73</td><td>SMB_55S</td><td>SMB</td><td>SML PCH 0_CLK 18 48</td></tr><tr><td>LE74</td><td>SMB_55S</td><td>SMB</td><td>SML PCH 0_DATA 18 48</td></tr><tr><td>LE75</td><td>SMB_55S</td><td>SMB</td><td>SML PCH 1_CLK 18 48</td></tr><tr><td>LE76</td><td>SMB_55S</td><td>SMB</td><td>SML PCH 1_DATA 18 48</td></tr><tr><td>LE77</td><td>CLK_XTAL</td><td>XTAL</td><td>SMC_EXTAL 45 46</td></tr><tr><td>LE78</td><td>CLK_XTAL</td><td>XTAL</td><td>SMC_XTAL 45 46</td></tr></table>				NET_TYPE				ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		LE57	SMB_55S	SMB	SMBUS SMC A S3_SCL 48	LE58	SMB_55S	SMB	SMBUS SMC A S3_SDA 48	LE59	SMB_55S	SMB	SMBUS SMC B S0_SCL 48	LE60	SMB_55S	SMB	SMBUS SMC B S0_SDA 48	LE61	SMB_55S	SMB	SMBUS SMC 0 S0_SCL 48	LE62	SMB_55S	SMB	SMBUS SMC 0 S0_SDA 48	LE63	SMB_55S	SMB	SMBUS SMC BSA_SCL 48	LE64	SMB_55S	SMB	SMBUS SMC BSA_SDA 48	LE65	SMB_55S	SMB	SMBUS SMC MGMT_SCL 48 88	LE66	SMB_55S	SMB	SMBUS SMC MGMT_SDA 48 88	LE67	SMB_55S	SMB	SMBUS SMC MGMT_SCL 48 88	LE68	SMB_55S	SMB	SMBUS SMC MGMT_SDA 48 88	LE69	SMB_55S	SMB	SMBUS PCH S0_CLK 48	LE70	SMB_55S	SMB	SMBUS PCH S0_DATA 48	LE71	SMB_55S	SMB	SMBUS PCH_CLK 18 48	LE72	SMB_55S	SMB	SMBUS PCH_DATA 18 48	LE73	SMB_55S	SMB	SML PCH 0_CLK 18 48	LE74	SMB_55S	SMB	SML PCH 0_DATA 18 48	LE75	SMB_55S	SMB	SML PCH 1_CLK 18 48	LE76	SMB_55S	SMB	SML PCH 1_DATA 18 48	LE77	CLK_XTAL	XTAL	SMC_EXTAL 45 46	LE78	CLK_XTAL	XTAL	SMC_XTAL 45 46	SMC THERMAL NET PROPERTIES <table><tr><th colspan="3">NET_TYPE</th><th></th></tr><tr><th>ELECTRICAL_CONSTRAINT_SET</th><th>PHYSICAL</th><th>SPACING</th><th></th></tr><tr><td>LE57</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DP1_DN6 51 88</td></tr><tr><td>LE58</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DN1_DP6 51 88</td></tr><tr><td>LE59</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DP2_DN3 51</td></tr><tr><td>LE60</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DN2_DP3 51</td></tr><tr><td>LE61</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DN1_DP6 51 88</td></tr><tr><td>LE62</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DP1_DN6 51 88</td></tr><tr><td>LE63</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DP4_DN5 51</td></tr><tr><td>LE64</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_T_DN4_DP5 51</td></tr><tr><td>LE65</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_LCD_P 51 92</td></tr><tr><td>LE66</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_LCD_N 51 92</td></tr><tr><td>LE67</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_ODD_P 51 92</td></tr><tr><td>LE68</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_ODD_N 51 92</td></tr><tr><td>LE69</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_CPU_H_P 51</td></tr><tr><td>LE70</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_CPU_H_N 51</td></tr><tr><td>LE71</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_SKIN_P 51 92</td></tr><tr><td>LE72</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_SKIN_N 51 92</td></tr><tr><td>LE73</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_AMB_P 51 92</td></tr><tr><td>LE74</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_AMB_N 51 92</td></tr><tr><td>LE75</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_MXM_P 51</td></tr><tr><td>LE76</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_MXM_N 51</td></tr><tr><td>LE77</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_CPU_THERMD_P 18 51</td></tr><tr><td>LE78</td><td>THERM_DIFF</td><td>THERMAL</td><td>SNS_CPU_THERMD_N 18 51</td></tr><tr><td>LE79</td><td></td><td>THERMAL</td><td>HDD_OOB_TEMP_FILT 51 92</td></tr><tr><td>LE80</td><td></td><td>THERMAL</td><td>HDD_OOB_TEMP 51</td></tr><tr><td>LE81</td><td></td><td>THERMAL</td><td>HDD_OOB_TEMP_R 51</td></tr><tr><td>LE82</td><td></td><td>THERMAL</td><td>SMC_HDD_OOB_TEMP 51</td></tr></table>				NET_TYPE				ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		LE57	THERM_DIFF	THERMAL	SNS_T_DP1_DN6 51 88	LE58	THERM_DIFF	THERMAL	SNS_T_DN1_DP6 51 88	LE59	THERM_DIFF	THERMAL	SNS_T_DP2_DN3 51	LE60	THERM_DIFF	THERMAL	SNS_T_DN2_DP3 51	LE61	THERM_DIFF	THERMAL	SNS_T_DN1_DP6 51 88	LE62	THERM_DIFF	THERMAL	SNS_T_DP1_DN6 51 88	LE63	THERM_DIFF	THERMAL	SNS_T_DP4_DN5 51	LE64	THERM_DIFF	THERMAL	SNS_T_DN4_DP5 51	LE65	THERM_DIFF	THERMAL	SNS_LCD_P 51 92	LE66	THERM_DIFF	THERMAL	SNS_LCD_N 51 92	LE67	THERM_DIFF	THERMAL	SNS_ODD_P 51 92	LE68	THERM_DIFF	THERMAL	SNS_ODD_N 51 92	LE69	THERM_DIFF	THERMAL	SNS_CPU_H_P 51	LE70	THERM_DIFF	THERMAL	SNS_CPU_H_N 51	LE71	THERM_DIFF	THERMAL	SNS_SKIN_P 51 92	LE72	THERM_DIFF	THERMAL	SNS_SKIN_N 51 92	LE73	THERM_DIFF	THERMAL	SNS_AMB_P 51 92	LE74	THERM_DIFF	THERMAL	SNS_AMB_N 51 92	LE75	THERM_DIFF	THERMAL	SNS_MXM_P 51	LE76	THERM_DIFF	THERMAL	SNS_MXM_N 51	LE77	THERM_DIFF	THERMAL	SNS_CPU_THERMD_P 18 51	LE78	THERM_DIFF	THERMAL	SNS_CPU_THERMD_N 18 51	LE79		THERMAL	HDD_OOB_TEMP_FILT 51 92	LE80		THERMAL	HDD_OOB_TEMP 51	LE81		THERMAL	HDD_OOB_TEMP_R 51	LE82		THERMAL	SMC_HDD_OOB_TEMP 51	SMC VOLTAGE/CURRENT NET PROPERTIES <table><tr><th colspan="3">NET_TYPE</th><th></th></tr><tr><th>ELECTRICAL_CONSTRAINT_SET</th><th>PHYSICAL</th><th>SPACING</th><th></th></tr><tr><td>LE57</td><td>THERM_DIFF</td><td>THERMAL</td><td>MXM_ISENSE_P 50</td></tr><tr><td>LE58</td><td>THERM_DIFF</td><td>THERMAL</td><td>MXM_ISENSE_N 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49</td></tr><tr><td>LE70</td><td>THERM_DIFF</td><td>THERMAL</td><td>SENSE_CPU_VTT2_N 49</td></tr><tr><td>LE71</td><td></td><td>THERMAL</td><td>GND_SMC_AVSS 45 46 49 50</td></tr><tr><td>LE72</td><td></td><td>THERMAL</td><td>SMC_CPU_1V5_ISENSE 46 49</td></tr><tr><td>LE73</td><td></td><td>THERMAL</td><td>SMC_CPU_1V5_ISENSE_R 49</td></tr><tr><td>LE74</td><td></td><td>THERMAL</td><td>SMC_CPU_1V5_VSENSE 46 49</td></tr><tr><td>LE75</td><td></td><td>THERMAL</td><td>SMC_CPU_VTT_ISENSE 46 49</td></tr><tr><td>LE76</td><td></td><td>THERMAL</td><td>SMC_CPU_VTT_ISENSE_R 49</td></tr><tr><td>LE77</td><td></td><td>THERMAL</td><td>SMC_CPU_VTT_VSENSE 46 49</td></tr><tr><td>LE78</td><td></td><td>THERMAL</td><td>SMC_CPU_1V8_ISENSE 46 49</td></tr><tr><td>LE79</td><td></td><td>THERMAL</td><td>SMC_CPU_1V8_ISENSE_R 46 49</td></tr><tr><td>LE80</td><td></td><td>THERMAL</td><td>SMC_CPU_1V8_VSENSE 46 49</td></tr><tr><td>LE81</td><td></td><td>THERMAL</td><td>SMC_CPU_VSENSE 46 49</td></tr><tr><td>LE82</td><td>VTD_PHY</td><td>VR_CTL</td><td>VR_CPU_IOUT 13 64</td></tr><tr><td>LE83</td><td>THERM_DIFF</td><td>THERMAL</td><td>VR_ISNS_CPU_P 49</td></tr><tr><td>LE84</td><td>THERM_DIFF</td><td>THERMAL</td><td>VR_ISNS_CPU_N 49</td></tr><tr><td>LE85</td><td></td><td>THERMAL</td><td>SNS_PS_CPU_ISNS 49</td></tr><tr><td>LE86</td><td></td><td>THERMAL</td><td>SMC_CPU_ISENSE 46 49</td></tr><tr><td>LE87</td><td></td><td>THERMAL</td><td>SMC_CPU_INPUT_ISENSE 46</td></tr><tr><td>LE88</td><td></td><td>THERMAL</td><td>SMC_CPU_INPUT_VSENSE 46</td></tr><tr><td>LE89</td><td></td><td>THERMAL</td><td>SMC_CPU_INPUT_VSENSE 46</td></tr><tr><td>LE90</td><td></td><td>THERMAL</td><td>SMC_DIMM_1V5_ISENSE 46 50</td></tr><tr><td>LE91</td><td></td><td>THERMAL</td><td>SMC_1V5_S3_ISENSE_R 46 50</td></tr><tr><td>LE92</td><td></td><td>THERMAL</td><td>SMC_DIMM_1V5_VSENSE 46 50</td></tr><tr><td>LE93</td><td></td><td>THERMAL</td><td>SMC_GPU_ISENSE 46 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49	LE75		THERMAL	SMC_CPU_VTT_ISENSE 46 49	LE76		THERMAL	SMC_CPU_VTT_ISENSE_R 49	LE77		THERMAL	SMC_CPU_VTT_VSENSE 46 49	LE78		THERMAL	SMC_CPU_1V8_ISENSE 46 49	LE79		THERMAL	SMC_CPU_1V8_ISENSE_R 46 49	LE80		THERMAL	SMC_CPU_1V8_VSENSE 46 49	LE81		THERMAL	SMC_CPU_VSENSE 46 49	LE82	VTD_PHY	VR_CTL	VR_CPU_IOUT 13 64	LE83	THERM_DIFF	THERMAL	VR_ISNS_CPU_P 49	LE84	THERM_DIFF	THERMAL	VR_ISNS_CPU_N 49	LE85		THERMAL	SNS_PS_CPU_ISNS 49	LE86		THERMAL	SMC_CPU_ISENSE 46 49	LE87		THERMAL	SMC_CPU_INPUT_ISENSE 46	LE88		THERMAL	SMC_CPU_INPUT_VSENSE 46	LE89		THERMAL	SMC_CPU_INPUT_VSENSE 46	LE90		THERMAL	SMC_DIMM_1V5_ISENSE 46 50	LE91		THERMAL	SMC_1V5_S3_ISENSE_R 46 50	LE92		THERMAL	SMC_DIMM_1V5_VSENSE 46 50	LE93		THERMAL	SMC_GPU_ISENSE 46 50	LE94		THERMAL	SMC_MXM_ISENSE_R 50	LE95		THERMAL	SMC_GPU_VSENSE 46 50
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																																																																																																																																																																													
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LE70	THERM_DIFF	THERMAL	SENSE_CPU_VTT2_N 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE71		THERMAL	GND_SMC_AVSS 45 46 49 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE72		THERMAL	SMC_CPU_1V5_ISENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE73		THERMAL	SMC_CPU_1V5_ISENSE_R 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE74		THERMAL	SMC_CPU_1V5_VSENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE75		THERMAL	SMC_CPU_VTT_ISENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE76		THERMAL	SMC_CPU_VTT_ISENSE_R 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE77		THERMAL	SMC_CPU_VTT_VSENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE78		THERMAL	SMC_CPU_1V8_ISENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE79		THERMAL	SMC_CPU_1V8_ISENSE_R 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE80		THERMAL	SMC_CPU_1V8_VSENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE81		THERMAL	SMC_CPU_VSENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE82	VTD_PHY	VR_CTL	VR_CPU_IOUT 13 64																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE83	THERM_DIFF	THERMAL	VR_ISNS_CPU_P 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE84	THERM_DIFF	THERMAL	VR_ISNS_CPU_N 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE85		THERMAL	SNS_PS_CPU_ISNS 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE86		THERMAL	SMC_CPU_ISENSE 46 49																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE87		THERMAL	SMC_CPU_INPUT_ISENSE 46																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE88		THERMAL	SMC_CPU_INPUT_VSENSE 46																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE89		THERMAL	SMC_CPU_INPUT_VSENSE 46																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE90		THERMAL	SMC_DIMM_1V5_ISENSE 46 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE91		THERMAL	SMC_1V5_S3_ISENSE_R 46 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE92		THERMAL	SMC_DIMM_1V5_VSENSE 46 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE93		THERMAL	SMC_GPU_ISENSE 46 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE94		THERMAL	SMC_MXM_ISENSE_R 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
LE95		THERMAL	SMC_GPU_VSENSE 46 50																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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SYNC DATE=07/01/2009

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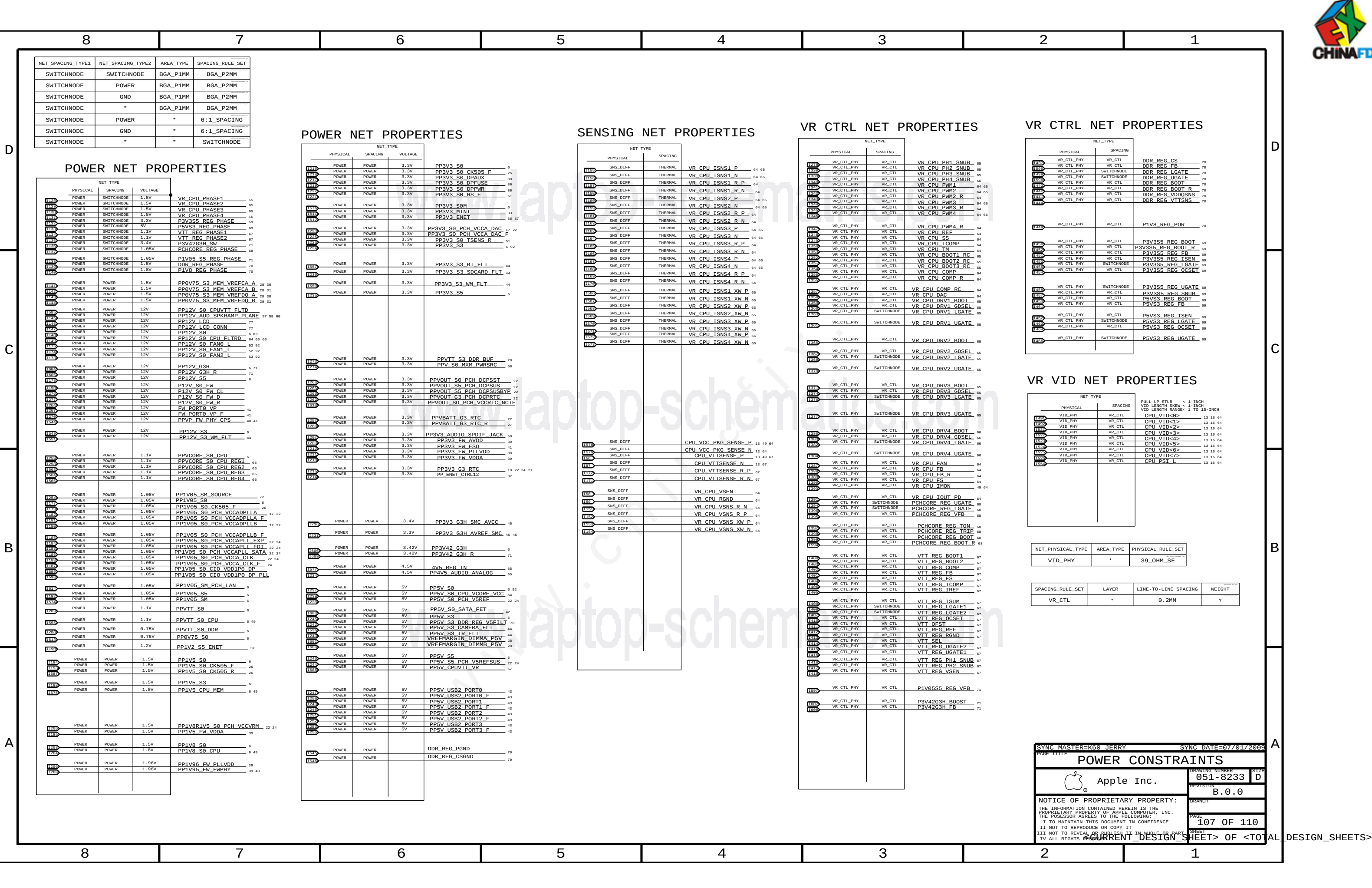
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SHEET

CURRENT DESIGN SHEET> OF <TOTAL SHEETS>



NET_TYPE				
	PHYSICAL	SPACING	VOLTAGE	
1890	POWER	SWITCHMODE	1.5V	VR_CPU_PHASE1
1891	POWER	SWITCHMODE	1.5V	VR_CPU_PHASE2
1892	POWER	SWITCHMODE	1.5V	VR_CPU_PHASE3
1893	POWER	SWITCHMODE	1.5V	VR_CPU_PHASE4
1894	POWER	SWITCHMODE	3.3V	PPV3S5_REG_PHASE
1895	POWER	SWITCHMODE	5V	PPV5S3_REG_PHASE
1896	POWER	SWITCHMODE	1.1V	VTT_REG_PHASE1
1897	POWER	SWITCHMODE	1.1V	VTT_REG_PHASE2
1898	POWER	SWITCHMODE	3.4V	P3V42G3H_SW
1899	POWER	SWITCHMODE	1.05V	PCHCORE_REG_PHASE
1900	POWER	SWITCHMODE	1.05V	PIV05_S5_REG_PHASE
1901	POWER	SWITCHMODE	1.5V	DDR_REG_PHASE
1902	POWER	SWITCHMODE	1.8V	P1V8_REG_PHASE
1903	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1904	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1905	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1906	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1907	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1908	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1909	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1910	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1911	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1912	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1913	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1914	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1915	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1916	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1917	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1918	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1919	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1920	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1921	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1922	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1923	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1924	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1925	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1926	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1927	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1928	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1929	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1930	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1931	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1932	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1933	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1934	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1935	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1936	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1937	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1938	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1939	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1940	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1941	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1942	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1943	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1944	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1945	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1946	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1947	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1948	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1949	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1950	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1951	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1952	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1953	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1954	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1955	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1956	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1957	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1958	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1959	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1960	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1961	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1962	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1963	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1964	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1965	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1966	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1967	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1968	POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA
1969	POWER	POWER	1.5V	PP0V

NET_TYPE				
PHYSICAL	SPACING	VOLTAGE		
1P30	POWER	POWER	3.3V	PP3V3 S0
1P30	POWER	POWER	3.3V	PP3V3 S0 CK505 F
1P30	POWER	POWER	3.3V	PP3V3 S0 DPAUX
1P30	POWER	POWER	3.3V	PP3V3 S0 DPFAUX
1P30	POWER	POWER	3.3V	PP3V3 S0 DDPWR
1P30	POWER	POWER	3.3V	PP3V3 S0 HS F
1P30	POWER	POWER	3.3V	PP3V3 S0M
1P30	POWER	POWER	3.3V	PP3V3 MINI
1P30	POWER	POWER	3.3V	PP3V3 ENET
1P30	POWER	POWER	3.3V	PP3V3 S0 PCH VCCA DAC
1P30	POWER	POWER	3.3V	PP3V3 S0 PCH VCCA DAC F
1P30	POWER	POWER	3.3V	PP3V3 S0 TSENS R
1P30	POWER	POWER	3.3V	PP3V3 S3
1P30	POWER	POWER	3.3V	PP3V3 S3 BT FLT
1P30	POWER	POWER	3.3V	PP3V3 S3 SDCARD FLT
1P30	POWER	POWER	3.3V	PP3V3 S3 WM FLT
1P30	POWER	POWER	3.3V	PP3V3 S5
1P30	POWER	POWER	3.3V	PPVTT S3 DDR BUF
1P30	POWER	POWER	3.3V	PPV S0 MXM PWRSRC
1P30	POWER	POWER	3.3V	PPVOUT S0 PCH DCPSS
1P30	POWER	POWER	3.3V	PPVOUT S5 PCH DCPSS
1P30	POWER	POWER	3.3V	PPVOUT S5 PCH DCPSSUBYP
1P30	POWER	POWER	3.3V	PPVOUT G3 PCH DCPRTC
1P30	POWER	POWER	3.3V	PPVOUT S0 PCH VCCRTC NCTF
1P30	POWER	POWER	3.3V	PPVBATT G3 RTC
1P30	POWER	POWER	3.3V	PPVBATT G3 RTC R
1P30	POWER	POWER	3.3V	PP3V3 AUDIO SPDIF JACK
1P30	POWER	POWER	3.3V	PP3V3 FW AVDD
1P30	POWER	POWER	3.3V	PP3V3 FW ESD
1P30	POWER	POWER	3.3V	PP3V3 FW PLLVDD
1P30	POWER	POWER	3.3V	PP3V3 FW VDDA
1P30	POWER	POWER	3.3V	PP3V3 G3 RTC
1P30	POWER	POWER	3.3V	PP ENET CTRL2
1P30	POWER	POWER	3.4V	PP3V3 G3H SMC AVCC
1P30	POWER	POWER	3.3V	PP3V3 G3H AVREF SMC
1P30	POWER	POWER	3.42V	PP3V42 G3H
1P30	POWER	POWER	3.42V	PP3V42 G3H R
1P30	POWER	POWER	4.5V	4V5 REG LN
1P30	POWER	POWER	4.5V	PP4V5 AUDIO ANALOG
1P30	POWER	POWER	5V	PP5V S0
1P30	POWER	POWER	5V	PP5V S0 CPU VCORE VCC
1P30	POWER	POWER	5V	PP5V S0 PCH V5REF
1P30	POWER	POWER	5V	PP5V S0 SATA_FET
1P30	POWER	POWER	5V	PP5V S3
1P30	POWER	POWER	5V	PP5V S3 DDR REG V5FLLT
1P30	POWER	POWER	5V	PP5V S3 CAMERA FLT
1P30	POWER	POWER	5V	PP5V S3 IR FLT
1P30	POWER	POWER	5V	VREFMARGIN DIMMA P5V
1P30	POWER	POWER	5V	VREFMARGIN DIMMB P5V
1P30	POWER	POWER	5V	PP5V S5
1P30	POWER	POWER	5V	PP5V S5 PCH V5REFSUS
1P30	POWER	POWER	5V	PP5V CPUVTT VR
1P30	POWER	POWER	5V	PP5V USB2 PORT0
1P30	POWER	POWER	5V	PP5V USB2 PORT0 F
1P30	POWER	POWER	5V	PP5V USB2 PORT1
1P30	POWER	POWER	5V	PP5V USB2 PORT2
1P30	POWER	POWER	5V	PP5V USB2 PORT2 F
1P30	POWER	POWER	5V	PP5V USB2 PORT3
1P30	POWER	POWER	5V	PP5V USB2 PORT3 F
1P30	POWER	POWER		DDR_REG_PGND
1P30	POWER	POWER		DDR_REG_CSGND

		NET_TYPE	
	PHYSICAL	SPACING	
1650A	SNS_DIFF	THERMAL	VR_CPU_TSNS1_P 64 65
1650B	SNS_DIFF	THERMAL	VR_CPU_TSNS1_N 64 65
1650C	SNS_DIFF	THERMAL	VR_CPU_TSNS1_R_P 64 65
1650D	SNS_DIFF	THERMAL	VR_CPU_TSNS1_R_N 64
1650E	SNS_DIFF	THERMAL	VR_CPU_TSNS2_P 64 65
1650F	SNS_DIFF	THERMAL	VR_CPU_TSNS2_N 64 65
1650G	SNS_DIFF	THERMAL	VR_CPU_TSNS2_R_P 64
1650H	SNS_DIFF	THERMAL	VR_CPU_TSNS2_R_N 64
1650I	SNS_DIFF	THERMAL	VR_CPU_TSNS3_P 64 65
1650J	SNS_DIFF	THERMAL	VR_CPU_TSNS3_N 64 65
1650K	SNS_DIFF	THERMAL	VR_CPU_TSNS3_R_P 64
1650L	SNS_DIFF	THERMAL	VR_CPU_TSNS3_R_N 64
1650M	SNS_DIFF	THERMAL	VR_CPU_TSNS4_P 64 66
1650N	SNS_DIFF	THERMAL	VR_CPU_TSNS4_R_P 64 66
1650O	SNS_DIFF	THERMAL	VR_CPU_TSNS4_R_N 64
1650P	SNS_DIFF	THERMAL	VR_CPU_TSNS1_XW_P 65
1650Q	SNS_DIFF	THERMAL	VR_CPU_TSNS1_XW_N 65
1650R	SNS_DIFF	THERMAL	VR_CPU_TSNS2_XW_P 65
1650S	SNS_DIFF	THERMAL	VR_CPU_TSNS2_XW_N 65
1650T	SNS_DIFF	THERMAL	VR_CPU_TSNS3_XW_P 65
1650U	SNS_DIFF	THERMAL	VR_CPU_TSNS3_XW_N 65
1650V	SNS_DIFF	THERMAL	VR_CPU_TSNS4_XW_P 66
1650W	SNS_DIFF	THERMAL	VR_CPU_TSNS4_XW_N 66
1650X	SNS_DIFF		
1650Y	SNS_DIFF		
1650Z	SNS_DIFF		
1651A	SNS_DIFF	CPU_VCC_PKG_SENSE_P 13 49 64	
1651B	SNS_DIFF	CPU_VCC_PKG_SENSE_N 13 64	
1651C	SNS_DIFF	CPU_VTTSENSE_P 13 49 67	
1651D	SNS_DIFF	CPU_VTTSENSE_N 13 67	
1651E	SNS_DIFF	CPU_VTTSENSE_R_P 67	
1651F	SNS_DIFF	CPU_VTTSENSE_R_N 67	
1651G	SNS_DIFF	VR_CPU_VSEN 64	
1651H	SNS_DIFF	VR_CPU_RGND 64	
1651I	SNS_DIFF	VR_CPU_VSNS_R_N 64	
1651J	SNS_DIFF	VR_CPU_VSNS_R_P 64	
1651K	SNS_DIFF	VR_CPU_VSNS_XW_P 64	
1651L	SNS_DIFF	VR_CPU_VSNS_XW_N 64	

NET_TYPE		
PHYSICAL	SPACING	
VR_CTL_PHY	VR_CTL	VR CPU PH1 SNUB 65
VR_CTL_PHY	VR_CTL	VR CPU PH2 SNUB 65
VR_CTL_PHY	VR_CTL	VR CPU PH3 SNUB 65
VR_CTL_PHY	VR_CTL	VR CPU PH4 SNUB 65
VR_CTL_PHY	VR_CTL	VR CPU PWM1 64 65
VR_CTL_PHY	VR_CTL	VR CPU PWM2 64 65
VR_CTL_PHY	VR_CTL	VR CPU PWM2 R 64 65
VR_CTL_PHY	VR_CTL	VR CPU PWM3 64 65
VR_CTL_PHY	VR_CTL	VR CPU PWM3 R 64 65
VR_CTL_PHY	VR_CTL	VR CPU PWM4 64 66
VR_CTL_PHY	VR_CTL	VR CPU PWM4 R 64
VR_CTL_PHY	VR_CTL	VR CPU REF 64
VR_CTL_PHY	VR_CTL	VR CPU SS 64
VR_CTL_PHY	VR_CTL	VR CPU TCOMP 64
VR_CTL_PHY	VR_CTL	VR CPU TM 64
VR_CTL_PHY	VR_CTL	VR CPU B00T1 RC 65
VR_CTL_PHY	VR_CTL	VR CPU B00T2 RC 65
VR_CTL_PHY	VR_CTL	VR CPU B00T3 RC 65
VR_CTL_PHY	VR_CTL	VR CPU COMP 64
VR_CTL_PHY	VR_CTL	VR CPU COMP R 64
VR_CTL_PHY	VR_CTL	VR CPU COMP RC 64
VR_CTL_PHY	VR_CTL	VR CPU DAC 64
VR_CTL_PHY	VR_CTL	VR CPU DRV1 B00T 65
VR_CTL_PHY	VR_CTL	VR CPU DRV1 GDSLE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV1 LGATE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV1 UGATE 65
VR_CTL_PHY	VR_CTL	VR CPU DRV2 B00T 65
VR_CTL_PHY	VR_CTL	VR CPU DRV2 GDSLE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV2 LGATE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV2 UGATE 65
VR_CTL_PHY	VR_CTL	VR CPU DRV3 B00T 65
VR_CTL_PHY	VR_CTL	VR CPU DRV3 GDSLE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV3 LGATE 65
VR_CTL_PHY	SWITCHNODE	VR CPU DRV3 UGATE 65
VR_CTL_PHY	VR_CTL	VR CPU DRV4 B00T 66
VR_CTL_PHY	VR_CTL	VR CPU DRV4 GDSLE 66
VR_CTL_PHY	SWITCHNODE	VR CPU DRV4 LGATE 66
VR_CTL_PHY	SWITCHNODE	VR CPU DRV4 UGATE 66
VR_CTL_PHY	VR_CTL	VR CPU FAN 64
VR_CTL_PHY	VR_CTL	VR CPU FB 64
VR_CTL_PHY	VR_CTL	VR CPU FB R 64
VR_CTL_PHY	VR_CTL	VR CPU FS 64
VR_CTL_PHY	VR_CTL	VR CPU IMON 49 64
VR_CTL_PHY	VR_CTL	VR CPU IOUT PD 64
VR_CTL_PHY	SWITCHNODE	PCHCORE REG UGATE 68
VR_CTL_PHY	SWITCHNODE	PCHCORE REG LGATE 68
VR_CTL_PHY	VR_CTL	PCHCORE REG VFB 68
VR_CTL_PHY	VR_CTL	PCHCORE REG TON 68
VR_CTL_PHY	VR_CTL	PCHCORE REG TRIP 68
VR_CTL_PHY	VR_CTL	PCHCORE REG BOOT 68
VR_CTL_PHY	VR_CTL	PCHCORE REG BOOT R 68
VR_CTL_PHY	VR_CTL	VTT REG B00T1 67
VR_CTL_PHY	VR_CTL	VTT REG B00T2 67
VR_CTL_PHY	VR_CTL	VTT REG COMP 67
VR_CTL_PHY	VR_CTL	VTT REG FB 67
VR_CTL_PHY	VR_CTL	VTT REG FS 67
VR_CTL_PHY	VR_CTL	VTT REG ICOMP 67
VR_CTL_PHY	VR_CTL	VTT REG IREF 67
VR_CTL_PHY	VR_CTL	VTT REG ISUM 67
VR_CTL_PHY	SWITCHNODE	VTT REG LGATE1 67
VR_CTL_PHY	SWITCHNODE	VTT REG LGATE2 67
VR_CTL_PHY	VR_CTL	VTT REG OCSET 67
VR_CTL_PHY	VR_CTL	VTT REG OFST 67
VR_CTL_PHY	VR_CTL	VTT REG REF 67
VR_CTL_PHY	VR_CTL	VTT REG REND 67
VR_CTL_PHY	VR_CTL	VTT REG SEL 67
VR_CTL_PHY	VR_CTL	VTT REG UGATE2 67
VR_CTL_PHY	VR_CTL	VTT REG UGATE1 67
VR_CTL_PHY	VR_CTL	VTT REG PH1 SNUB 67
VR_CTL_PHY	VR_CTL	VTT REG PH2 SNUB 67
VR_CTL_PHY	VR_CTL	VTT REG VSEN 67
VR_CTL_PHY	VR_CTL	P1V05S5 REG VFB 71
VR_CTL_PHY	VR_CTL	P3V42G3H B00ST 71
VR_CTL_PHY	VR_CTL	P3V42G3H FB 71

		NET_TYPE		
PHYSICAL		SPACING		
IC900	VR_CTL_PHY	VR_CTL	DDR REG CS	70
IC900	VR_CTL_PHY	VR_CTL	DDR REG FB	70
IC900	VR_CTL_PHY	SWITCHNODE	DDR REG LGATE	70
IC900	VR_CTL_PHY	SWITCHNODE	DDR REG UGATE	70
IC900	VR_CTL_PHY	VR_CTL	DDR REG BOOT	70
IC900	VR_CTL_PHY	VR_CTL	DDR REG BOOT R	70
IC900	VR_CTL_PHY	VR_CTL	DDR REG VDDQSNS	70
IC900	VR_CTL_PHY	VR_CTL	DDR REG VTTSNS	70
IC900	VR_CTL_PHY	VR_CTL	P1V8_REG_POR	70
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT	60
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT_R	60
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_FB	60
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_ISEN	60
IC900	VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_LGATE	60
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_OCSET	60
IC900	VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_UGATE	60
IC900	VR_CTL_PHY	VR_CTL	P3V3S5_REG_SNUB	60
IC900	VR_CTL_PHY	VR_CTL	P5VS3_REG_BOOT	60
IC900	VR_CTL_PHY	VR_CTL	P5VS3_REG_FB	60
IC900	VR_CTL_PHY	VR_CTL	P5VS3_REG_ISEN	60
IC900	VR_CTL_PHY	SWITCHNODE	P5VS3_REG_LGATE	60
IC900	VR_CTL_PHY	VR_CTL	P5VS3_REG_OCSET	60
IC900	VR_CTL_PHY	SWITCHNODE	P5VS3_REG_UGATE	60

VR VID NET PROPERTIES

		NET_TYPE		
PHYSICAL		SPACING		
IC900	VID_PHY	VR_CTL	CPU VID<0>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<1>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<2>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<3>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<4>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<5>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<6>	13 16 64
IC900	VID_PHY	VR_CTL	CPU VID<7>	13 16 64
IC900	VID_PHY	VR_CTL	CPU PSI L	13 16 64

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	39_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	Ø.2MM	?

SYNC MASTER=K60 JERRY

SYNC DATE=07/01/20

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NET_TYPE		PULL-UP STUB < 1-INCH	VID LENGTH SKEM < 1-INCH	VID LENGTH RANGE< 1 TO 15-INCH
PHYSICAL	SPACING			
LE800 VID_PHY	VR_CTL	CPU VID<0>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<1>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<2>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<3>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<4>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<5>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<6>		13 16 64
LE800 VID_PHY	VR_CTL	CPU VID<7>		13 16 64
LE800 VID_PHY	VR_CTL	CPU PSI L		13 16 64

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	39_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.2MM	?

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PM NET PROPERTIES
(PM, RESET, EN, PGOOD)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

PHYSICAL		NET_TYPE	SPACING
U8	PM	PLT_RESET_L	28 27
U4	PM_VTT	PLT_RESET_LS1V1_L	11
U4	PM	PM_ACDG_PS_ON	6
U3	PM	PM_BATLOW_L	19 19 45
U5	PM	PM_CLK32K_SUSCLK	9 45 85
U7	PM	PM_CLK32K_SUSCLK_R	9 19 85
U8	PM	PM_CLKRUN_L	15 19 45 47
U10	PM	PM_EXT_TS_L<0>	11 46
U9	PM	PM_EXT_TS_L<1>	11 46
U12	PM	PM_LAN_PWRGD	15 19
U13B	PM_VTT	FSB_CPURSTOUT_L	11 25
U13B	PM	USB_HUB_RESET_L	34 35
U26	PM_VTT	PM_MEM_PWRGD	11 19
U24	PM	PM_ME_PWRGD	19 63
U29	PM	PM_ME_S0_EN_G	72
U26	PM	PM_ME_S0_EN_G1	72
U26	PM	PM_ME_S0_EN_R	72
U30	PM	PM_MXM_PGOOD	63 74
U26	PM	PM_PCH_PWRGD	19 63
U33	PM	PM_PGOOD_DDRREG_S3	5 62 78
U33	PM	PM_PGOOD_PVCORE_CPU	5 26 63 64
U32	PM	PM_PWRBTN_L	19 25 45
U35	PM	PM_RSMRST_L	45 62
U24	PM	PM_RSMRST_PCH_L	19 62
U36	PM	PM_SLP_M_L	5 19 62
U36	PM	PM_SLP_M_R	
U37	PM	PM_SLP_S3_L	5 19 32 33 37 46 62 63
U36	PM	PM_SLP_S3_L_AND_S0_RDY	
U30	PM	PM_SLP_S4_1_L	19 62
U33	PM	PM_SLP_S4_2_L	19 45 46
U31	PM	PM_SLP_S4_3_L	5 19
U30	PM	PM_SLP_S4_L	19 32
U33	PM	PM_SLP_S5_L	19 45
U35	PM	PM_SUS_PWR_ACK	19
U32	PM_VTT	PM_SYNC	11 19
U32	PM	SDCARD_PLT_RST_L	27 44
U35	PM	PM_SYSRST_L	19 27 45
U35	PM	PM_SYS_PWRGD	19 32 63
U36	PM_VTT	PM_THRMTRIP_L	11 21 46
U38	PM	RSMRST_PWRGD	45 63
U37	PM	RTC_RESET_L	18 91
U36	PM_VTT	CPU_PWRGD	11 21 25
U36	PM	CPU_RESET_L	11 27
U35	PM	PGOOD_1V05ME_G1	63
U35	PM	PGOOD_1V05ME_G2	63
U37	PM	PGOOD_1V8_S0_G1	63
U34	PM	PGOOD_1V8_S0_G2	63
U36	PM	PGOOD_CPU_GFX_DDR	63
U35	PM	PGOOD_P12V_S3	
U35	PM	PGOOD_P1V05_ME_S5	
U38	PM	PGOOD_P1V5_S0	72
U37	PM	PGOOD_P1V8_S0	63
U38	PM	PGOOD_P3V3_ME	72
U36	PM	PGOOD_P3V3_S0	48 63 72
U38	PM	PGOOD_P3V3_S3	34 72
U32	PM	PGOOD_P5V_S0	62 72
U34	PM	PGOOD_PCH_AND_P1V8	63
U36	PM	PGOOD_PCH_S0	63
U36	PM	PGOOD_SYSPWROK	63
U37	PM	PGOOD_SYSPWROK_R	63
U38	PM	RTC_RESET_L	18 91
U38	PM	P12V_S3_EN	62 72
U35	PM	P1V05_ME_SM_EN	62 72
U36	PM	P1V5_S0_EN	62 72
U32	PM	P3V3ME_EN	62 72
U34	PM	P3V3S0_EN	62 72
U32	PM	P3V3S3_EN	62 72
U36	PM	P5VS0_EN	62 72
U36	PM	P5VS3_EN	62 68
U37	PM	PCHCORE_REG_EN	62 68
U37	PM	PCHCORE_REG_PGOOD	5 62 63 68
U38	PM	PEG_RESET_L	9 27
U38	PM	SDCARD_RESET	21 25 44 92

PHYSICAL		NET_TYPE	SPACING
U32	PM	4V5_REG_EN	55
U38	PM	ALL_SYS_PWRGD_R	6 25 32 63
U36	PM	ALL_SYS_PWRGD_SMC	45 63
U35	PM	CK505_27MHZ_EN	26
U35	PM	CPUVTT_REG_EN	62 67
U36	PM_VTT	CPUVTT_REG_PGOOD	11 62 63 67
U37	PM	CPU_MEM_RESET_L	11 32
U38	PM	DDRVTI_EN	32 62 78
U38	PM	DEBUG_RESET_L	27 47
U38	PM	FWPHY_RESET_L	39
U39	PM	FWXIO_SNOOP_EN	39
U39	PM	FW_RESET_L	27 39
U39	PM	GFX_VR_EN	
U34	PM	GFX_VR_PGOOD	
U36	PM	LAN_RESET_L	27 36
U36	PM	MEM_RESET_L	39 21 32
U37	PM	MINI_RESET_L	27 33
U36	PM	SMC_DELAYED_PWRGD	46 63
U36	PM	SMC_LRESET_L	27 45
U36	PM	SMC_RESET_L	45 46 47
U36	PM	T2B_RESET_L	
U36	PM_VTT	XDP_CUPWROD	11 25
U36	PM_VTT	XDP_DBRESET_L	11 25 27
U36	PM_VTT	XDP_PWRGD	25

NET PHYSICAL FOR NC NETS
REMOVE WHEN CHECKPLUS IS FIXED

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
PCIE GRAPHICS			
U38B		PCIE_R5D	PCIE NC_PCIE_CLK100M_EXCARD_N
U38B		PCIE_R5D	PCIE NC_PCIE_CLK100M_EXCARD_P
U38B		PCIE_R5D	PCIE NC_PCIE_EXCARD_D2R_N
U38B		PCIE_R5D	PCIE NC_PCIE_EXCARD_D2R_P
U38B		PCIE_R5D	PCIE NC_PCIE_EXCARD_R2D_C_N
U38B		PCIE_R5D	PCIE NC_PCIE_EXCARD_R2D_C_P
U38B	NO_TEST=TRUE	USB_90D	USB NC_USB_EXCARD_N
U38B	NO_TEST=TRUE	USB_90D	USB NC_USB_EXCARD_P
U38D	NO_TEST=TRUE	USB_90D	USB NC_USB_EXTE_N
U38D	NO_TEST=TRUE	USB_90D	USB NC_USB_EXTE_P
U38D	NO_TEST=TRUE	USB_90D	USB NC_USB_TPAD_N
U38D	NO_TEST=TRUE	USB_90D	USB NC_USB_TPAD_P

SYNC MASTER=K60 MIKE

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PM RESETS ENABLES PGOOD CONST

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SHEET

CURRENT DESIGN SHEET

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FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT							
J4700 USB CAMERA							
PP5V_S3_CAMERA FUNC_TEST=TRUE PIN_ALLOWED_TPS=1 85 44 USB_CAMERA_L_P FUNC_TEST=TRUE 85 44 USB_CAMERA_L_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4700 2 Ground Testpoints near J4700							
J4750 USB CARD READER							
85 44 USB_SDCARD_L_P FUNC_TEST=TRUE 85 44 USB_SDCARD_L_N FUNC_TEST=TRUE 91 44 25 21 SDCARD_RESET FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4750 2 Ground Testpoints near J4750							
J4720 USB BLUETOOTH							
85 44 USB_BT_L_P FUNC_TEST=TRUE 85 44 USB_BT_L_N FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4720 2 Ground Testpoints near J4720							
J4780 IR BOARD							
85 44 USB_IR_L_P FUNC_TEST=TRUE 85 44 USB_IR_L_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4780 2 Ground Testpoints near J4780							
J4520 SATA ODD (HIGH SPEED)							
84 42 SATA_ODD_R2D_P FUNC_TEST=TRUE 84 42 SATA_ODD_R2D_N FUNC_TEST=TRUE 84 42 SATA_ODD_D2R_C_N FUNC_TEST=TRUE 84 42 SATA_ODD_D2R_C_P FUNC_TEST=TRUE 45 42 SMC_ODD_DETECT FUNC_TEST=TRUE 1 PP5V_S0 Testpoint near J4520 5 Ground Testpoints near J4520							
J4510 SATA HDD (HIGH SPEED)							
84 42 SATA_HDD_R2D_P FUNC_TEST=TRUE 84 42 SATA_HDD_R2D_N FUNC_TEST=TRUE 84 42 SATA_HDD_D2R_C_N FUNC_TEST=TRUE 84 42 SATA_HDD_D2R_C_P FUNC_TEST=TRUE 3 Ground Testpoints near J4510							
J5520 ANALOG LCD TEMP SENSOR							
88 51 SNS_LCD_P FUNC_TEST=TRUE 88 51 SNS_LCD_N FUNC_TEST=TRUE							
J5521 AMBIENT TEMP SENSOR							
88 51 SNS_AMB_P FUNC_TEST=TRUE 88 51 SNS_AMB_N FUNC_TEST=TRUE							
J5551 ODD TEMP SENSOR							
88 51 SNS_ODD_P FUNC_TEST=TRUE 88 51 SNS_ODD_N FUNC_TEST=TRUE							
J5600 ODD FAN							
52 FAN_0_PWR_L FUNC_TEST=TRUE 52 FAN_TACH0_L FUNC_TEST=TRUE 89 52 PP12V_S0_FAN0_L FUNC_TEST=TRUE 52 FAN_0_GND FUNC_TEST=TRUE							
J5700 CPU FAN							
53 FAN_2_PWR_L FUNC_TEST=TRUE 53 FAN_TACH2_L FUNC_TEST=TRUE 89 53 PP12V_S0_FAN2_L FUNC_TEST=TRUE 51 FAN_2_GND FUNC_TEST=TRUE							
J5601 HD FAN							
52 FAN_1_PWR_L FUNC_TEST=TRUE 52 FAN_TACH1_L FUNC_TEST=TRUE 89 52 PP12V_S0_FAN1_L FUNC_TEST=TRUE 52 FAN_1_GND FUNC_TEST=TRUE							
J5550 HDD TEMP SENSOR							
88 51 HDD_O0B_TEMP_FILT FUNC_TEST=TRUE							
J5560 SKIN TEMP SENSOR							
88 51 SNS_SKIN_P FUNC_TEST=TRUE 88 51 SNS_SKIN_N FUNC_TEST=TRUE							
J6601 AUDIO MICROPHONE							
59 AUD_MIC_IN1_N_CONN FUNC_TEST=TRUE 59 GND_AUDIO_MIC1_CONN FUNC_TEST=TRUE 59 AUD_MIC_IN1_P_CONN FUNC_TEST=TRUE 1 Ground Testpoint near J6601							
J6602 AUDIO RIGHT SPEAKER							
85 59 58 AUD_SPKR_OUTL02R_POUT FUNC_TEST=TRUE 85 59 58 AUD_SPKR_OUTL02R_NOUT FUNC_TEST=TRUE 85 59 57 AUD_SPKR_OUTL01R_POUT FUNC_TEST=TRUE 85 59 57 AUD_SPKR_OUTL01R_NOUT FUNC_TEST=TRUE							
J6603 AUDIO LEFT SPEAKER							
85 59 58 AUD_SPKR_OUTL02L_POUT FUNC_TEST=TRUE 85 59 58 AUD_SPKR_OUTL02L_NOUT FUNC_TEST=TRUE 85 59 57 AUD_SPKR_OUTL01L_POUT FUNC_TEST=TRUE 85 59 57 AUD_SPKR_OUTL01L_NOUT FUNC_TEST=TRUE							
GND 16 TP'S							
59 PP3V3_S3 2 TP'S FUNC_TEST=TRUE PIN_ALLOWED_TPS=2							
PP5V_S3 REG 2 TP'S							
69 6 PP5V_S3_REG 2 TP'S FUNC_TEST=TRUE PIN_ALLOWED_TPS=1							
PP5V_S0							
69 6 PP5V_S0 FUNC_TEST=TRUE PIN_ALLOWED_TPS=1							